

Safety Smartwatch for Women in India: A Design Blueprint

Executive Summary

This report proposes a cutting-edge **safety-focused smartwatch** for women and girls (ages 13–35) in urban and semi-urban India. It combines wearable IoT hardware, embedded AI, and cloud services to address pressing safety concerns. The device provides one-tap distress capture (recording evidence), instant SOS alerts with live location, and hidden-camera detection in a discreet wrist-worn form factor. Key features include:

- **One-Touch Distress Recording:** A panic button immediately triggers **video and audio recording**, encrypted on-device, and streams footage to a secure personal cloud vault for evidence preservation.
- **Instant SOS & Live Location:** An SOS button sends the user's **real-time GPS location** to up to five pre-selected emergency contacts and **automatically calls them sequentially** until someone picks up.
- **Hidden Camera Detection:** An integrated module scans for **covert surveillance cameras** in private spaces (washrooms, fitting rooms, hotels) using multi-techniques (IR lens reflection, RF signal scan, etc.), alerting the wearer to potential voyeurism threats.
- **Robust and User-Centric Design:** The smartwatch is **comfortable, tamper-resistant, and durable** (e.g. waterproof), with long battery life and intuitive operation. It employs end-to-end encryption, biometric access control, and adheres to India's latest data protection laws for user privacy.

The report details the **market need** (backed by crime data), target **user personas** and scenarios, a survey of the **competitive landscape**, and comprehensive engineering specifications. A **system architecture** is presented (hardware, firmware/AI, cloud backend, and mobile app), alongside an evaluation of hidden-camera detection technologies. We outline **data security measures** (full-stack encryption, compliance with the Digital Personal Data Protection Act 2023) and the **regulatory path** (BIS certification, WPC approvals, etc.). Finally, we analyze potential **risks and mitigations**, discuss **ethical considerations**, provide a **Bill of Materials (BOM) cost forecast** from prototype to mass production, and propose a **go-to-market roadmap** (pilot programs, partnerships, and scaling strategies). Emerging technologies on the horizon (satellite SOS connectivity, AI-driven threat prediction) are also explored as future enhancements.

In summary, this blueprint serves as a comprehensive guide for designing and commercializing a “**smart safety watch**” that empowers women and girls with proactive personal security, while respecting privacy and regulatory frameworks.

Problem Statement & Market Need

Women in India continue to face alarming levels of harassment and violence, underscoring an urgent need for tech-enabled safety solutions. According to the National Crime Records Bureau (NCRB), **428,278 crimes against women** were reported in 2021 – a 15.3% increase from 2020[jurist.org](https://www.jurist.org). This included **31,677 rapes** (averaging 86 rapes daily) and over **89,000 cases** of assault intended to outrage modesty[jurist.org](https://www.jurist.org). Of particular relevance is the rise in **voyeurism (IPC Section 354C)** – e.g. clandestine filming – with **1,513 cases of voyeurism reported in 2021**, up ~20% from the prior year[endnowfoundation.org](https://www.endnowfoundation.org). Such incidents (from hidden cameras in changing rooms to videos of private acts) violate women’s privacy and safety on a large scale.

These sobering statistics highlight a clear **market need** for effective personal safety devices. While smartphone apps and helpline numbers exist, they often fall short in emergencies. In a high-stress attack scenario, a victim may not have the time or ability to unlock a phone and call for help. Moreover, critical evidence (audio/video) of the incident is usually not captured, hampering prosecutions. The NCRB data shows low conviction rates for sexual offenses, partly due to lack of real-time evidence and delayed reporting[jurist.org](https://www.jurist.org). There is a gap in the market for a **wearable solution** that can be activated instantly and discreetly to both **summon help and preserve evidence**.

The target users – urban and semi-urban women aged 13–35 – are tech-savvy and largely comfortable with wearables. Many already use fitness bands or smartwatches daily. A safety-oriented smartwatch can seamlessly integrate into their lifestyle while providing a much-needed safety net. **Use cases** range from a college student walking home at night, to a working professional in a taxi, to a teen in a shopping mall fitting room concerned about hidden cameras. Recent high-profile incidents (e.g. a hidden camera found in a Goa boutique’s trial room by a Union Minister) have made women acutely aware of such threats[opindia.com](https://www.opindia.com). Surveys indicate a significant percentage of women feel unsafe in public spaces or when traveling alone after dark, and families are seeking trusted tools to protect daughters.

In response, this project envisions a “**safety smartwatch**” that directly addresses these pain points. By **combining GPS tracking, cellular connectivity, and on-device recording**, the device can alert trusted contacts and authorities in real time while securely streaming incident data to the cloud. This both enables faster response (potentially preventing escalation) and creates a record of the event for evidence. The inclusion of **hidden camera detection** addresses the growing voyeurism problem by empowering women to scan their environments (hotel rooms, public toilets, etc.) for illicit surveillance devices.

From a market perspective, personal safety wearables form an emerging category with high social impact. Government agencies (e.g. police departments, women’s ministries) have shown interest in technology-aided safety programs, such as installing panic buttons in phones or CCTV monitoring. A dedicated safety smartwatch could complement these efforts

at the individual level. Parents of girls, educational institutions, and employers could be key market segments, especially if the device is proven effective and affordable. By leveraging **state-of-the-art IoT and AI**, this solution aims to reduce response times to incidents, deter potential perpetrators (through the knowledge that evidence is captured), and ultimately contribute to lowering the incidence of violence against women.

In summary, the confluence of **high crime rates** against women jurist.org, **technology readiness** of the target demographic, and **gaps in current solutions** creates a compelling case for a safety-focused smartwatch. The following sections detail how this device will meet the needs of at-risk women through careful design of features, technology, and safeguards.

User Personas & Threat Scenarios

To ensure the device is grounded in reality, we consider several **user personas and scenarios** that illustrate how the smartwatch would function in practice. Each scenario highlights a potential threat and how the device's features would mitigate it:

Persona 1: “Late-Night Commuter” – Nisha, 22, Urban College Student

*Nisha is walking home from her university library at 9 PM. A man begins following her down a dimly lit street. Feeling unsafe, Nisha discreetly presses the **distress-capture button** on her smartwatch twice. **Scenario:** The watch silently begins **recording video and audio** of the encounter from her wrist. Simultaneously, it sends an **SOS alert with her live GPS location** to her five emergency contacts (parents, two friends, an uncle, and a campus security number). Her location updates in real-time on the companion app her contacts have, and the watch starts calling them one by one. Her father picks up – he immediately sees an SMS with a map link to her location and hears the situation unfolding via the watch's microphone. As the follower approaches and grabs Nisha's arm, the father and friends are already alerted; one friend calls the police with Nisha's live location. The assailant, noticing the flashing LED on Nisha's watch (an optional visible deterrent), tries to snatch it – but by then the watch has **streamed the video/audio evidence to the cloud**. Police arrive within minutes (guided by GPS) and the assailant flees. Later, Nisha retrieves the encrypted recording from her personal cloud vault as evidence for the police report. *This scenario shows how rapid alert and evidence capture can **deter an assault and aid swift response**.**

Persona 2: “On a Business Trip” – Meera, 30, Software Engineer

*Meera travels to another city for work and checks into a hotel. She remembers news about hidden cameras found in hotel rooms. Using her smartwatch's **Hidden-Camera Detector**, she performs a quick sweep of her room. **Scenario:** She opens the detector mode on the watch, which uses a combination of **infrared LED scanning and RF signal detection**. As Meera slowly scans the room, the watch's sensor detects a suspicious reflection point behind the air conditioner vent. It also picks up a faint RF signal on a spycam frequency. The watch vibrates and quietly notifies her of a potential hidden camera. Meera uses her phone (via the watch app) to double-check by looking through the watch's IR filter view and indeed spots a tiny lens glint. She immediately calls hotel security and requests a room change, showing them the device's detection. In the new room, no alerts are triggered. *This scenario highlights how the device **protects privacy** by empowering users to find hidden cameras (voyeurism threats) before harm occurs.* If Meera had discovered an active spycam, she*

could also use the watch to record footage of it and initiate an SOS if she felt threatened by the situation.

Persona 3: “School Commute” – Aditi, 16, High-School Student

Aditi travels by public bus to school each day in a semi-urban town. There have been instances of teenage girls being harassed on public transport. Aditi’s parents have given her the safety smartwatch for peace of mind. **Scenario:** One morning, a stranger on the bus starts harassing Aditi, making lewd comments and inching closer. Aditi, feeling scared, presses the **SOS button** on her watch for 3 seconds. The watch quietly sends out her **live location and a distress message** to her parents and three close relatives. It then starts calling her mother. Meanwhile, it also begins **audio recording** (given privacy, maybe not video in a crowded bus) so that the threatening comments are saved. The mother receives the alert and location, and immediately calls Aditi’s phone to check. The sudden phone call and Aditi’s firm response alerts other passengers, and the harasser backs off, getting off at the next stop. Later, Aditi’s parents replay the audio clip from the cloud storage to understand what happened and decide on further action. *This scenario demonstrates a **youth-friendly use case**, where a minor can quickly call for help in a harassing situation, with the device facilitating immediate family intervention.* (Notably, for minors, the system would be set up with guardian consent and oversight, in compliance with data protection rules.)

These personas underscore the versatility of the proposed smartwatch across different threat scenarios – from physical assault to voyeurism. Common threads include the need for **immediate action**, **stealth or deterrence**, and **reliable communication**. In each case, the smartwatch provides a tailored response: recording evidence quietly, notifying loved ones and emergency services, drawing public attention with an alarm if appropriate, and detecting hidden dangers. Real-world testing with such personas would be crucial to refine the user experience (e.g. ensuring the activation method is easy under stress but not prone to false triggers). By designing with these scenarios in mind, we ensure the device remains **user-centric and effective** in the moments that matter most.

Competitive Landscape

Several products and technologies already address women’s safety, ranging from mobile apps to dedicated wearables. However, most existing solutions cover fragments of the problem, and **none integrate all the core features** (distress recording, SOS alerts, hidden camera detection) in one device. Below is an overview of the current landscape and how the proposed smartwatch compares:

- **Safety Jewelry and Panic Buttons:** A number of startups offer panic-button devices disguised as jewelry (rings, pendants, keychains). For example, the **Leaf Wearables “Safer” device** (India, ~2015) is a Bluetooth-enabled pendant that, on double-tap, sends an alert with the wearer’s GPS location to guardians via a paired smartphone designindaba.com. Safer introduced features like a 30-second cancel window to prevent false alarms and even a “SaferWalk” tracking mode designindaba.com. Similarly, devices like **Nimb Ring** and **Athena pendant** (by ROAR for Good) allow discreet SOS activation. *Limitations:* These devices **rely on a connected smartphone** for communication (they themselves don’t have cellular

connectivity or recording ability). They typically do not capture any audio/video evidence; their function is limited to sending an alert. Our smartwatch improves on this by being a **standalone communicator with built-in recording**. It does not depend on the user's phone to call for help, which is critical if the phone is inaccessible or taken by an attacker.

- **Personal Safety Alarms (Sound-based): Run Angel** (Ireland, launched 2017) is an example of a wearable focused on audible alarm + alerts. It's a wristband that emits a **120 dB siren** when activated and sends SMS/email alerts with location to pre-selected contacts via a phone app [markets.businessinsider.com](https://markets.businessinsider.com/markets.businessinsider.com). Its unique acoustic chamber design maximizes the piercing sound to attract attention [markets.businessinsider.com](https://markets.businessinsider.com/markets.businessinsider.com). Another example is **She's Safe** (a small keychain alarm) which doesn't send alerts but just makes noise. *Limitations:* Loud alarms can deter attackers in some cases, but they also **sacrifice stealth** – an attacker might quickly attempt to silence a device making noise. Additionally, an alarm alone doesn't notify specific people or record evidence. Our design offers an **optional siren mode** (for those who want that deterrence) but crucially pairs it with silent emergency communication. It can raise a racket *and* message help, or remain completely silent while still summoning help, depending on user preference.
- **Smartphones and Apps:** Most people carry smartphones with built-in emergency features (e.g. the Android and iOS Emergency SOS, which can call emergency services and share location when the power button is pressed multiple times). There are also safety apps like "bSafe" and government apps like "112 India". *Limitations:* Phones are versatile but **not wearable** – in many incidents the phone may be in a bag or pocket, making it hard to access quickly or discreetly. Attackers often target the phone first to isolate the victim. Phones also typically do not automatically record video on SOS (due to privacy concerns and technical constraints). The smartwatch complements the phone: being body-worn, it's *immediately accessible* and can work even if the phone is out of reach or dead. It also specifically features automatic evidence capture which most phone SOS apps lack.
- **Smartwatches (general-purpose):** High-end smartwatches like the **Apple Watch** and **Samsung Galaxy Watch** have some emergency features. For instance, Apple Watch can detect hard falls and call emergency contacts, and its Emergency SOS feature will call local emergency number (112 in India) if the side button is long-pressed. Some also share location with emergency contacts. *Limitations:* These are generic features not tailored to assault scenarios – they do not record video or audio, and they do not detect hidden cameras. Moreover, these watches are expensive and not designed with the sole purpose of women's safety in mind (battery life may be short when GPS/call is active, and privacy of data is not E2E encrypted by default). Our proposed device is **mission-specific** – prioritizing safety functions above all. For example, we allocate dedicated battery reserve for emergency events, and we encrypt all recorded data by design. Additionally, a specialized hidden-camera detector is a novel feature not found in Apple or Samsung devices.

- **Existing Women's Safety Wearables (IoT prototypes):** A number of academic prototypes and hackathon products exist in literature. Many use a combination of Arduino/Raspberry Pi with GSM modules and sensors (heart rate, accelerometer, etc.) to detect distress and send SMS. For example, one IoT-based women safety system uses an Arduino Uno, GPS, and GSM module to send an alert when a button is pressed [researchgate.net](https://www.researchgate.net). Another concept ("Smart Belt" or "electroshock jackets") attempted to include self-defense mechanisms. *Limitations:* These prototypes are often bulky, proof-of-concepts and not commercially available. They demonstrate interest in multi-sensor approaches (e.g. detecting elevated heart rate as sign of fear), but have not integrated cloud video streaming or camera detection. Our design builds on these ideas with a far more **integrated and wearable form factor**, leveraging modern components (e.g. NB-IoT modem, mini camera, advanced sensors) that were not readily available a few years ago.

Competitive Summary: The safety smartwatch will effectively **combine the strengths** of these existing solutions while addressing their gaps. Like Safer and Run Angel, it alerts loved ones with location – but it does so without needing a phone and also captures evidence. It offers a loud alarm option akin to personal alarms, but also a silent mode which others lack. It goes beyond pure SOS devices by including **preventive scanning for hidden cameras**, tapping into a **unique market differentiator** (privacy protection in everyday scenarios, not just emergencies). Moreover, from a privacy standpoint, our device employs **end-to-end encryption and user-controlled data**, whereas many existing solutions don't emphasize data security (for instance, some apps send data unencrypted or share data with third parties, which raises concerns).

In terms of pricing, current safety wearables like smart rings or pendants retail between ₹3,000–₹10,000 (USD \$40–\$130). The addition of a camera and cellular module in our device may raise costs, but as discussed later in the BOM section, we aim for mass-production cost efficiency (possibly subsidized by government or CSR initiatives to reach students and low-income users). Given its comprehensive feature set, the smartwatch can be positioned as a **premium safety device**, but the social need may justify innovative pricing models (e.g. installment plans or bundled telecom offers).

Finally, it's worth noting that while competition exists in pieces, **no single product currently in the Indian market offers the holistic safety net** that we envision. This presents an opportunity for first-mover advantage in a new category of "safety smartwatches." The strategy will involve not just selling a gadget but also educating users and stakeholders (families, institutions) on its life-saving potential, differentiating it clearly from generic wearables and panic buttons.

Detailed Functional Requirements

To meet the needs identified, the smartwatch must satisfy a robust set of **functional requirements**. These cover performance, usability, safety, and reliability criteria. Below is a breakdown of the key requirements:

1. Instant Distress Capture & Alerting: The device should initiate recording and alert transmission within **1–2 seconds** of the panic button press. This means the **camera** should wake from standby and start recording video (with at least 720p resolution at 30fps) almost immediately, and the first alert message (with GPS location) should be sent out **under 5 seconds** from trigger. Low-latency operation is critical so that even a sudden attack is captured and help is summoned in real-time.

2. Battery Life: The watch should function through a typical day without recharge. Target **battery life** is *at least 24 hours* in normal smartwatch use (notifications, periodic sensor use) and minimum **1 hour of active emergency use** (continuous video streaming + calling). In pure standby (watch idle with screen off), it should last **3–5 days**. This ensures the device is reliable during long outings. A **battery reserve** concept can be implemented – e.g. always keep 10% battery reserved exclusively for emergency mode (so the watch can still perform an SOS even if main battery is low). Fast charging to 50% within 30 minutes is also desirable for quick top-ups.

3. Ergonomics & Form Factor: The device must be **compact and comfortable** for all-day wear, similar to a regular smartwatch. Target weight < 50 grams, thickness ~12mm or less. It should use a **soft, adjustable strap** suitable for young girls and adults. The design should be discreet – not obviously a “safety device” to avoid tipping off an attacker. It can look like a sleek digital watch or fitness band. The **interface** for the user will be minimal: perhaps an OLED dial for time and status, and two subtle buttons (one for SOS, one for video capture). These buttons should be distinguishable by touch and easy to press even under stress (perhaps slightly protruding or textured). Haptic feedback (vibration) on button press is needed to assure the user it's triggered.

4. Durability & Environmental Resistance: Women will wear this device daily, so it needs to handle various conditions. It should be **water-resistant (IP67 or better)** so that it's safe from rain or sweat and can even survive accidental submersion (e.g. if the user is pushed into water, it should still send an SOS). It should also be **dust-proof** (for semi-urban, dusty environments) and **shock-resistant** against drops (meeting at least a 1m drop test on hard surface). Operating temperature range should cover 0°C to 45°C (since in India summers can be hot and device might be exposed to sun during outdoor use).

5. Reliable Connectivity: The watch must have **built-in cellular connectivity** (LTE-M and/or NB-IoT) to send alerts directly, independent of a smartphone. It should also have **GPS** for accurate location. In addition, **Bluetooth Low Energy (BLE)** is required to pair with the user's smartphone for secondary connectivity (and to sync data or adjust settings via the companion app). The system should intelligently choose the connectivity: e.g. if the user's phone is nearby and has internet, the watch might relay alerts via the phone (BLE to phone to internet) to save watch battery; otherwise it switches to direct cellular. The design should support a **nano-SIM or eSIM** for cellular. The device must be able to make a phone call (voice call) or at least VoIP call via LTE – this is needed for sequentially calling emergency contacts (or directly calling 112 if configured). It might use VoLTE or a voice-capable module.

6. GPS Accuracy and Tracking: The GPS receiver should be able to get a location fix within ~5 seconds outdoors and <30 seconds indoors (using AGPS or cell/Wi-Fi assist if needed). Location accuracy of ~5–10 meters is ideal. The device should continuously update

location to the cloud during an ongoing SOS situation (e.g. every 5 seconds). If GPS signal is unavailable (e.g. indoors), the system should fall back to coarse location via cell tower or last known location, and inform contacts accordingly.

7. Distress Recording Quality & Storage: The **camera** should have adequate resolution (preferably 1080p HD) with low-light capability (since many incidents happen in poor lighting). It should have a wide field-of-view (e.g. 120°) to capture surroundings even if the wrist is at an angle. The **microphone** should capture clear audio within a 5-meter radius. The device should **encrypt and store footage locally** (e.g. in internal flash memory) in case connectivity drops. A minimum of **15 minutes of video storage** in offline mode is required (e.g. 1GB flash storage for video buffer). If the incident lasts longer, the device should overwrite oldest footage (circular buffer) while trying to regain network to upload. Video should be time-stamped and maybe watermarked to prevent tampering (for evidentiary integrity). A low-power **video codec** (H.264/H.265 encoder in hardware) is needed to compress video in real-time without draining the battery or CPU too much.

8. SOS Alert Mechanism: Upon SOS activation, the watch should handle a sequence of actions: send initial text alerts (with location) to contacts, *and* start phone calls. The requirement is that at least **five contacts** can be notified. The calling logic might be: call Contact 1; if no answer in 10 seconds, call Contact 2, and so on, looping back if needed. If a contact answers, they should be able to hear the ambient audio from the watch (acting like a speakerphone) so they can assess the situation. Optionally, the system can also be configured to call the national emergency number (112) after contacting personal contacts. The device should also allow a **silent SOS** mode configuration – where it does all the above without any sound or light, versus an **audible SOS** where it also triggers a loud buzzer. Users can select their preferred mode in the app. A **cancel feature** is needed to avoid false alarms: e.g. if the button is pressed accidentally, the user can press and hold for 3 seconds within a 30-second window to cancel the alerts (and a notification would go out that it was a false alarm). This is similar to the approach used by Safer and Apple's SOS to mitigate accidental triggers designindaba.com.

9. Hidden Camera Detection: The watch should have a dedicated mode to scan for hidden cameras in a room. Functional requirements for this mode: It should detect common spycam types within a range of about **5–10 meters**. The scan should take no more than a couple of minutes for an average room. Ideally, it uses **multiple detection methods** (detailed in Section 7) – e.g. actively shining IR and looking for lens reflections, and scanning for RF signals emitting from camera transmitters. The interface could be via the mobile app for usability (e.g. user walks around with watch, and the app shows a “signal strength” or beeps faster when suspect object is near). A simpler on-watch UI might use an LED indicator or vibration patterns. The requirement is that it should reliably spot a pinhole camera or illegal CCTV with at least **90% detection rate** under typical conditions (aligning with research like CamRadar which achieved ~93% detection in tests cse.buffalo.edu). False alarm rate should be low (<5%) so that random electronics (routers, motion sensors) aren't constantly flagged – multi-sensor confirmation can help (requiring both an optical glint and an RF signal, for instance).

10. Tamper and Removal Detection: The device should detect if an attacker tries to **remove or disable** it. For example, if the strap is forcibly opened or cut during an active

SOS, the device should sense that (perhaps via a small switch or sudden loss of skin contact from the heart-rate sensor) and it could send a final “**Device removed**” alert with location. It could then continue transmitting audio/video until destroyed. Similarly, if someone tries to **power it off during SOS**, it should ignore the power-off (perhaps require a specific pattern or biometric to shut down in SOS mode). The watch could also have an **auto-SOS trigger if removed without authorization**: e.g. if someone snatches it off the user’s wrist violently, it could treat that as a distress event and send alerts on its own. These anti-tamper features ensure the device *cannot easily be silenced* by a perpetrator once activated.

11. Biometric Access & Privacy: For non-emergency interactions, sensitive functions should be locked behind a biometric. For instance, accessing the stored videos on the watch or app, or stopping an ongoing recording, could require the user’s fingerprint or a secure PIN. The watch could include a small **fingerprint sensor** (perhaps integrated into the side button) for the user to authenticate. This prevents anyone else from retrieving or deleting the data (e.g. if the watch is stolen, the footage remains inaccessible without the biometric). Additionally, pairing the watch with a new phone or disarming safety features might require biometric authentication to prevent misuse by others.

12. Data Security: All data (recordings, location logs, contact info) on the device and in transit must be **securely encrypted**. This is detailed later, but functionally the requirement is to use strong encryption (AES-256 or similar) for local storage and TLS 1.3 for any network communication. The device should generate its own cryptographic keys (with a secure element if possible) and not rely on obscure security through obscurity. Only authorized users (the wearer and explicitly allowed persons) should be able to decrypt/view the content. This ensures compliance with privacy laws and protection against hackers if the data is intercepted akitra.com.

13. Companion Mobile App (User Interface Requirements): A smartphone app (Android/iOS) will accompany the watch. Functional needs for the app:

- Onboarding: allow the user (or guardian) to configure emergency contacts, preferred SOS mode (silent/audible), and any personalized message for alerts.
- Dashboard: show device status (battery, signal, last known location), toggle features like “camera detection mode”.
- Live Tracking: in case of an SOS event, the app (for the wearer or designated guardians) should show the live location trail and possibly a low-frame-rate live video feed from the watch (if bandwidth permits).
- Data Access: allow the user to review incident recordings saved to cloud, which would be **encrypted** and require login + biometric to view. They can download and share these with police as needed.
- Health Monitoring (nice-to-have): Because it’s a smartwatch, it could also show step count, basic fitness metrics – but these are secondary to the core safety purpose. If included, ensure they do not compromise battery needed for safety.

- **Firmware Updates:** the app should deliver OTA firmware updates to the watch for improvements and security patches.
- **Security:** The app itself should be PIN/biometric protected, and communications between app and watch encrypted (e.g. BLE bonding with encryption).

14. Performance and Responsiveness: The device should feel responsive – button presses acknowledged via haptics in <0.5s. The hidden camera scan mode should give some feedback (progress or signals) in real-time to guide the user. The cloud alert system should be able to handle simultaneous alerts if an incident is triggered by multiple users (scalability). The system must be **highly available** (the cloud service especially), since a down server could mean a cry for help goes unheard – multi-region cloud deployment or failover via SMS gateway can be considered (e.g. if data network fails, fall back to sending SMS alerts directly from the device if possible).

15. Regulatory Compliance: The watch must meet regulatory standards (detailed later) as a functional requirement:

- It must cap its SAR (specific absorption rate) within permitted limits for a wearable device (since it has a cellular radio on the wrist).
- It should have proper fallbacks for emergency calling as per telecom regulations (e.g. if 112 is dialed, it should connect even without SIM balance or if the device is locked).
- Localization: default language support for English and local Indian languages for notifications (e.g. SMS to contacts could be in English with option for Hindi, etc., for accessibility).

By fulfilling these detailed requirements, the smartwatch will be **robust in real-world use**. Each requirement was chosen to either enhance **usability under duress**, **reliability of the safety functions**, or **trust in the device** (through security and compliance). For example, long battery life and waterproofing ensure the watch is *actually on and working when needed*. Instant alerting and recording ensure *no time is lost* in an emergency. Encryption and biometric locks ensure the user's data and privacy are *never compromised*, aligning with the principle that a safety device should not create new vulnerabilities.

These requirements will guide the engineering and design decisions in the system architecture, described next. We will see how hardware and software components are selected or designed to meet or exceed the above benchmarks.

System Architecture

The system architecture of the safety smartwatch encompasses multiple layers – from the hardware components in the device, to the on-device firmware (including AI algorithms), to the cloud backend and mobile app that together deliver the end-to-end functionality. This

section presents the architecture in an integrated manner, with a **block diagram** for visual reference and detailed explanations of each subsystem.

cse.buffalo.edu/markets.businessinsider.com *Figure: Conceptual system architecture of the safety smartwatch.* The diagram illustrates key hardware modules inside the watch (sensors, MCU, connectivity), the data flows (video/audio going to the cloud vault, alerts going to contacts and the mobile app), and the major software components (edge AI on the device, cloud services, and user application). This architecture is designed for scalability, low latency, and security.

Hardware Architecture (On-Device)

At the heart of the smartwatch is a **Secure Microcontroller Unit (MCU)/System-on-Chip (SoC)** that coordinates all functions. We propose using a low-power SoC that integrates both application processing and cellular connectivity – for example, the Nordic nRF9160 (which has an Arm Cortex-M33 MCU + LTE-M/NB-IoT modem + GPS) or a similar IoT-focused chipset. This MCU runs the embedded firmware/OS and handles sensor data acquisition, encryption, and communication. It should have hardware security features (ARM TrustZone or an isolated secure element) to store cryptographic keys and perform encryption.

Sensors and Modules connected to the MCU include:

- **Camera Module:** A mini camera (e.g. 8MP CMOS sensor) embedded at the watch's top or side. It's connected via a digital interface (like MIPI CSI or DCMI parallel interface) to the MCU or an image co-processor. The camera has a wide-angle lens (possibly fisheye) to capture a broad scene even when the wrist isn't pointed exactly at the target. It might be placed at an edge of the watch face to have an unobstructed view. In normal mode the camera is off (for privacy/battery), and it powers on only for distress capture or when user explicitly turns on camera-detection scanning.
- **Microphone:** A MEMS omni-directional mic, likely placed on the side of the watch face (with a water-resistant mesh). It connects to the MCU's ADC or I2S interface. It should be always listening in a low-power mode if possible (to detect a specific hotword or scream in future AI enhancements), but primarily it activates during SOS recording or two-way calls.
- **Hidden Camera Detector Module:** This may be a composite of sensors:
 - An **IR LED array** (several small infrared LEDs around the watch face) and an IR-sensitive photodiode or use of the camera sensor in IR mode. These allow an active scan for lens reflections.
 - An **RF scanner** – essentially a broadband RF receiver that can pick up transmissions in common spycam bands (1.2 GHz analog, 2.4 GHz Wi-Fi, 5.8 GHz etc.). This could be done via a small software-defined radio front-end or repurposing the LTE radio when idle to scan frequencies. Another option is a tiny **mm-wave radar** (60 GHz) module which might detect electronics by RF

reflection differences.

- A **magnetometer** (already often present in smartwatches for compass) which could serve to detect magnetic fields from devices (though small cameras emit weak fields).
- These sensors feed data to the MCU for analysis (e.g. the analog/digital readings from photodiode and RF front-end go into MCU via ADC or SPI interfaces).
- **GNSS Receiver:** For location, we use either the integrated GPS in the SoC (like nRF9160 has one) or an external GNSS module (supporting GPS and India's NavIC, plus GLONASS/Galileo for faster fixes). It connects via UART/SPI and provides latitude/longitude coordinates. It will be off most of the time to save power, activating during SOS or when user explicitly requests location (or for periodic check-ins if configured).
- **Accelerometer & Gyroscope:** A 6-axis IMU (inertial measurement unit) provides motion sensing. It's used for fall detection (if we choose to include), gesture recognition (perhaps double tapping the watch or a specific arm movement could trigger SOS if hands are free), and also tamper detection (a sudden forceful removal might register as a jolt). It also simply counts steps for fitness tracking. The accelerometer can be on to detect a strong shock (fall) and then wake the MCU.
- **Heart Rate / Skin Contact Sensor:** An optical heart rate sensor underneath could serve dual purposes: wellness (heart rate tracking) and security (detecting if the watch is on the wrist versus removed, by analyzing the signal or having a contact capacitance sensor). If the watch is torn off during an attack, the abrupt loss of heart signal or contact can prompt the MCU to flag a tamper event.
- **Buttons:** Two hardware push buttons (one dedicated to SOS, one to video/manual trigger) are wired to GPIOs on the MCU with hardware interrupts for instant wake-up. They have internal debouncing to avoid false triggers from jostling.
- **Vibrator (Haptic Motor):** For silent feedback and alerts, a small eccentric motor or linear resonant actuator is driven by the MCU (via a transistor or dedicated motor driver). It confirms user input (buzz on button press) and can silently alert the user (e.g. a subtle buzz when a hidden camera is detected or when SOS is activated in silent mode).
- **Speaker/Buzzer:** A tiny speaker can be included for the siren and for two-way audio (so the user can hear the contact's voice or an operator on a call). The speaker needs to output very loud sound for the alarm – possibly up to 90-100 dB. A piezo buzzer could be used for the siren specifically because it can be very loud and is simple to drive. For voice, a regular micro speaker is needed. They connect to the MCU's DAC / audio amp.

- **LED Indicator:** A multicolor LED can indicate device status (charging, SOS sent, etc.). In camera-detect mode, a red LED could blink to help with scanning (some detectors use a blinking red light to reflect off lenses eufy.com). The LED is optional and kept minimal to maintain stealth.
- **Power Management & Battery:** The watch will have a rechargeable Li-ion or Li-po battery (~500 mAh capacity). A Power Management IC (PMIC) handles charging (via a magnetic dock or USB-C connector), battery fuel gauging, and providing regulated voltages to components (e.g. camera at 2.8V, MCU at 3.3V, etc.). The design must isolate the power supply for the RF modules to reduce noise during scans. Also, in an SOS event, the PMIC should allow the system to draw peak currents for the cellular radio (LTE transmit bursts can draw >500mA). The PCB will include an antenna for LTE (tuned to relevant bands, likely a flexible PCB antenna around the strap) and for GPS (possibly a ceramic patch antenna on top). The Bluetooth shares the same antenna as LTE (if using a combo antenna) or a separate small antenna.

Internal Storage: The MCU/SoC will have some internal flash, but likely an external **NOR flash memory (e.g. 2GB)** is attached via SPI/QSPI for storing encrypted video/audio data and any ML models. This flash must be encrypted (the MCU can do on-the-fly encryption or store data already encrypted). If the MCU has limited RAM, an external SDRAM might be needed for buffering video frames during encoding.

In summary, the hardware is a compact assembly of a central secure MCU with **sensors for environment (camera, mic, RF)**, **sensors for context (GPS, accelerometer)**, **user interface elements (buttons, vibrator, speaker)**, and **communication modules (cellular, BLE)**. This is quite complex, but modern components and careful integration (e.g. using a highly integrated SoC) can achieve it in a smartwatch-sized device. A **system block diagram** was provided (Figure above) to visualize these connections, showing how Camera/Mic, Buttons, GPS, etc. all feed into the MCU, which then connects out to the Cellular modem and BLE module, etc., and finally to cloud/app[50†].

Firmware & Edge AI (On-Device Software)

The smartwatch runs embedded software that coordinates all safety functions with **real-time performance**. The firmware will likely be built on an RTOS (real-time operating system) such as FreeRTOS or Zephyr, or a lightweight Linux if the SoC is powerful enough (though for low-power and instant-on, an RTOS on MCU is preferred). Key components of the firmware include:

- **SOS State Machine:** Logic to handle an SOS event – reading the GPS, forming the alert message, initializing the cellular modem, and transmitting the data. It also manages the sequential calling: interfacing with the modem to dial numbers and play audio. This module will also control the recording sub-system (starting/stopping video/audio). Because multiple tasks happen on SOS (record, send location, call, etc.), the firmware uses multitasking (threads) with priority for critical tasks (e.g. sending the first alert immediately, then starting video stream).

- **Media Processing (Codec):** The firmware includes a **low-latency video encoder** to compress the camera feed. This could utilize hardware blocks if present on the SoC, or a lean codec library if using software. A likely approach is to record at moderate resolution (say 480p or 720p) to reduce data size, and encode in H.264 at a bitrate that can be sustained by LTE-M (~256 kbps perhaps). Audio from the mic is encoded (AAC or even low-bitrate OPUS) in sync. If network is available, the firmware simultaneously uploads chunks of this recording to the cloud; if not, it writes to local flash. This dual-write ensures redundancy.
- **Encryption:** Before storing or sending, data is **encrypted**. The firmware will use cryptographic libraries (like mbedTLS or similar) in a memory-optimized manner. Ideally, each recorded video segment is encrypted with a symmetric key (AES-256) that is itself protected. End-to-end encryption might mean the device encrypts with a public key of the user's account (so that only the user's app/cloud (holding the private key) can decrypt). This requires some key management: possibly when pairing the device with the app, a key exchange happens (the app gives the device a public key). The firmware then uses that for every SOS data encryption. The encryption module must be efficient (use hardware crypto accelerators if available) to avoid slowing down the recording pipeline. **Data integrity** (digital signatures or HMAC) might also be added so that any tampering of footage is detectable.
- **Edge AI & Sensing:** The device will employ on-device intelligence for two main features: *hidden camera detection* and possibly *activity/fall detection*. For hidden camera detection, the firmware can run a lightweight ML algorithm or signal processing:
 - If using the camera sensor to scan for reflections, the firmware might capture images while pulsing the IR LED and use an algorithm to spot bright spots that blink in sync (a classic approach for lens detection).
 - If using RF analysis, it might use a trained model to differentiate between normal RF noise and patterns indicative of a camera's RF (for example, using FFT of the RF signal).
 - Advanced approach: If the watch had enough processing, it could implement the CamRadar technique – analyzing electromagnetic emanations. But that requires sampling EM signals at high frequency and applying algorithms. The CamRadar research essentially found the unique EM oscillation pattern a camera produces [cse.buffalo.edu](https://cse.buffalo.edu/cse.buffalo.edu). Implementing this fully may be too heavy now, but the firmware could do simpler things like scanning for known Wi-Fi SSIDs or devices (some hidden cams create Wi-Fi APs – the watch could attempt to scan for any Wi-Fi network or Bluetooth device in the vicinity with suspicious name or just any unknown device on same network).
 - In summary, the firmware's AI might combine sensor inputs to raise an alert if criteria are met. For instance: "if photodiode sees a consistent flicker reflection AND RF scanner sees a 2.4 GHz signal, classify as hidden camera

likely”.

- The processing for this can be done on the MCU (which might be capable of simple ML like a decision tree or small neural net). Many IoT MCUs now support TinyML frameworks. If needed, we could incorporate an **AI accelerator** (some MCUs have DSP instructions or a small ML core).

For activity detection: the accelerometer data can be analyzed to detect a **fall or struggle**. A sudden fall could auto-trigger an SOS if the user doesn't cancel (like smartwatches do for falls). Struggle (rapid erratic movement plus elevated heart rate and maybe voice shouting detected from mic) could also auto-trigger – but these are complex to avoid false alarms, so likely opt-in features.

- **Power Management (Firmware):** The firmware manages power states – keeping the device mostly in sleep or low-power mode, waking instantly on button press or significant motion. It will also periodically do background tasks like sync time via NTP from cloud, or upload any buffered data. Efficient scheduling ensures we meet the battery life requirement. For example, cellular radio might be normally off; the firmware can wake it every hour briefly to check for any incoming messages or sync.
- **Self-Test & Diagnostics:** On boot, the firmware should self-test key components (camera present, mic working, sensors responding) and periodically report health to the cloud/app. This ensures any malfunction (like a broken camera) is caught before an emergency. The user could be notified to service or reset if something is wrong.
- **Firmware Security:** The device firmware should be secure against tampering. Support for **secure boot** (firmware signature verification) is needed so that malware cannot be injected. Also, firmware updates are signed by the manufacturer and verified on device.
- **Real-time OS tasks breakdown (illustrative):**
 - *High priority:* SOS Handler Task (monitors SOS button GPIO interrupt, manages emergency routine), Sensor IRQ handlers (for quick response).
 - *Medium:* Network Task (handles socket connections to cloud), Audio/Video encoding tasks (possibly running in small chunks to stream).
 - *Low:* App Communication task (BLE sync, responding to app commands), Logging tasks, etc.
 - Interrupts for things like incoming call (to pick up automatically in SOS), GPS fix events, etc., are handled promptly.

Cloud Backend (Secure Cloud Vault & Alert Server)

The cloud backend is a critical part of the architecture, acting as the **relay and storage hub**. It has two primary responsibilities: **storing encrypted evidence data** and **disseminating alerts to contacts (and possibly authorities)**. The design favors a scalable, serverless architecture to handle potentially many simultaneous incidents, with strong security controls.

Key components of the cloud backend:

- **Secure Storage (Personal Cloud Vault):** A cloud database or storage bucket where each user has a private, encrypted area for incident data. When the watch streams video/audio, it is sent to this storage. End-to-end encryption means the data might already be encrypted with the user's key on the device side, so the cloud just stores cipher blobs associated with a user ID and timestamp. Even the service provider cannot view it without user consent (zero-knowledge style) akitra.com. This could be implemented using cloud object storage (like AWS S3 or Azure Blob) with server-side encryption disabled in favor of client-side encryption. Metadata (like "SOS event at 10:05pm, 2 min video, location coords") could be stored in a database for indexing, but sensitive parts of metadata (exact coordinates or user identity) could also be encrypted to be safe. The storage should have **redundancy** (multiple copies or in multiple data centers) given the critical nature – we don't want a single server failure to lose an incident recording.
- **Streaming Architecture:** To handle live streaming of data (for example, a contact wanting to view the live video), a streaming server or a WebRTC service might be employed. However, to keep things simpler and within NB-IoT bandwidth, we might not do a live video feed to contacts, only recorded chunks. The device could upload, say, 10-second clips continuously. The cloud can make the latest clip available via a secure link to the contacts if needed. Alternatively, an **HTTPS long-polling or MQTT** pipeline can stream audio (since audio is lower bandwidth) in real time to contacts. We might use a publish-subscribe model where each incident is a "topic" and authorized contacts' phones subscribe to it to receive live updates.
- **Alert Notification Service:** When an SOS is triggered, the device might not directly SMS all contacts (especially if using data only). So the cloud has an SMS/Email/Push service. For instance, using SMS gateways (integrated with telecom) to send the emergency SMS to the contacts with the user's name and GPS link. Emails can also be sent similarly. Push notifications can be sent to those contacts who have our app installed (the app would receive an push with high priority showing the alert). We might leverage a third-party API for SMS (or even integrate with the government's 112 emergency platform if accessible via API for registered devices). This service must be **fast and reliable** – possibly using multiple gateways (if one fails, try another).
- **Sequential Calling Service:** Ideally, the watch does the phone calls itself via cellular voice. But if for some reason that's not possible (e.g., the NB-IoT module chosen doesn't support voice), an alternative is the cloud could initiate calls to contacts. This would involve a VoIP server or integration with a telephony service (like Twilio, etc.). The cloud can call Contact 1 and, when they pick up, can connect them to the audio stream from the watch (via the cloud as a bridge). However, this is complex and

dependent on telephony integration and real-time voice piping to cloud. A simpler approach is to rely on the device's phone capabilities to call, and only use cloud as backup for alerts. We will primarily assume the device calls out directly (which is more peer-to-peer and reliable if it has signal).

- **Serverless Functions & Database:** The architecture could use serverless functions that trigger on events. For example:
 - When an *SOS event message* arrives from a device (via a REST API or MQTT message), a cloud function triggers: it logs the event, sends out notifications to contacts, and allocates storage space for incoming media.
 - Another function triggers as each chunk of video arrives: it might assemble or just mark it in database.
 - A function could also handle *camera detection alerts* – e.g. if user uses the detector and finds something, perhaps the user can press a button to automatically alert hotel security or something; not core, but possible extension.
 - The database (NoSQL or SQL) keeps records of users, their contacts (with consent in place to store that info), device status (to know if device is online/offline), etc. It also can log every SOS activation for analytics (with user consent) to improve the service (e.g. areas with frequent alerts could be flagged anonymously).
- **Mobile App Integration:** The app connects to the cloud to fetch data (like to stream the location updates or download the encrypted video). The cloud should expose secure APIs. For instance, a GET endpoint `/incident/{id}/location` that returns the latest coordinates (secured by authentication). Or a websocket that the app can listen to for live updates. This likely uses standard cloud messaging (like Firebase Cloud Messaging for pushes, or AWS IoT for device messaging). We ensure that only authorized parties (the user or their chosen emergency contacts) can subscribe to these updates – enforced by authentication tokens.
- **Security & Privacy:** The cloud must implement stringent security. All endpoints enforce authentication (the watch will have an auth token or device ID with secret to send data; the user app will authenticate with user credentials). Data in transit is via HTTPS (TLS 1.3) to protect against interception [akitra.com](https://www.akitra.com). Contact info and personal data in the cloud database are encrypted at rest. Also, since this deals with personal data, compliance with India's DPDP Act is required – which means user consent for data collection, easy means to delete their data, breach notifications, etc. [techtarget.com](https://www.techtarget.com). The cloud design will include a **Data Protection by Design** approach: minimal data retention (e.g. if an incident video is older than say 6 months and the user has a copy or didn't mark it to keep, it might auto-delete to protect privacy). Also, robust access control to ensure no unauthorized viewing of

another person's incidents.

- **Scalability:** The use of serverless and decoupled microservices allows scaling up during emergencies (for example, if thousands of users trigger SOS during a citywide event, the system should handle the load). Using cloud infrastructure (like AWS Lambda, DynamoDB or equivalent) means we can scale on demand. For SMS/call service, we might have pre-provisioned capacity or multiple provider contracts to handle burst messaging.
- **Integration with Authorities (Future):** The cloud can also integrate with official emergency systems. For instance, with user permission, an SOS could also forward data to a police control room dashboard. Some cities have women safety programs where emergency devices are registered – our system could send data to those systems via an API (e.g. an authorized feed to police of live location for certain verified incidents). This is beyond initial scope but architecturally feasible.

Mobile Companion App

The mobile app serves as the user's interface to the system and a portal for emergency contacts:

For the **wearer (primary user)**, the app provides:

- **Account Setup & Device Pairing:** When first installing, user creates an account (or logs in via OTP for simplicity). They pair the watch via BLE by entering a code or scanning a QR. The app then registers the device's ID with the cloud under the user's account.
- **Emergency Contacts Management:** The user adds up to N contacts. The app obtains consent from those contacts if needed (it could send them a message that they've been added). It verifies their phone numbers/email. The user can configure the order of calling and whether to also auto-dial 112.
- **Preferences:** Toggle silent mode vs loud alarm, whether auto-fall-detect is on, etc. They can also enable/disable hidden camera detection scans from the app (maybe easier to initiate a scan via phone UI).
- **During an SOS:** If the user has their phone, the app will display a **live incident screen**: showing "Alert sent! Contacts responding." It could show a map of their current location updating. If video is being recorded and the user escapes, they might be able to watch it right away (but perhaps better to keep it uploading to cloud for safety). The app might also allow the user to press a "Call Police" button as an extra action or sound a siren from phone too.
- **Post-incident:** View the timeline and recordings. The user can add notes (useful if she wants to document what happened while fresh). She can tap an option to "Share with Police" which could produce a secure link or code she can give law enforcement

to access the evidence.

- **Routine Use:** The app might have a feature like “Safe Check-in” – e.g. the user can set a timer if she’s going somewhere and if she doesn’t cancel it, it triggers an alert. (This is an extra feature some apps have; including it could add value).
- **Device Tools:** Battery level display, toggle a “find my watch” ping, firmware update check, etc.

For **Emergency Contacts (secondary users)**: If they install the app (optional, since they can receive SMS without it), the app allows:

- **Viewing Alerts:** If a contact gets an SOS alert and has the app, a notification opens a live tracking screen. They can see the location of the person in distress on a map and any notes like “I’m being followed”. They might also see others in the contact group (if multiple contacts) acknowledging or responding (some coordination feature).
- **Communication:** The app could facilitate an audio call or a conference among contacts or directly to the wearer’s watch (if the watch supports voice).
- **History:** A contact could see past alerts if they were involved (in case needed for follow-up).
- **Privacy for User:** Note that contacts only get data during an active SOS. The user’s whereabouts are not continuously shared unless an emergency. The app should clarify that.

For both: the app will enforce login security, likely using phone’s biometrics to protect access to sensitive data.

A note on minors: Since our demographic includes 13–17, the app should have a flow for guardian consent. Possibly, a parent’s account is linked to a minor’s device and they manage contacts. This would align with DPDP Act requiring guardian consent for under 18 techtarget.com.

Technology: The app could be built cross-platform (Flutter/React Native) or native. It interacts with cloud via REST APIs and receives push notifications for SOS. It uses BLE APIs to communicate with the watch (for pairing, config, and maybe real-time transfer of small data like firmware updates).

Block Diagram Reference

(Refer to the earlier provided block diagram figure for a visual overview.) The architecture can be summarized in a layered flow:

- **Device Hardware Layer:** includes **sensors (camera, mic, GPS, accelerometer, hidden-cam sensor)** feeding into **MCU**, which connects out via **LTE-M network** to the cloud. The device also communicates locally to the **mobile app via BLE**.
- **Firmware Layer:** on MCU handles **data capture, encryption, and transmission**. It triggers cloud communications and handles incoming commands.
- **Cloud Layer:** comprises the **Alert Server** (which notifies contacts via SMS/Email and relays calls) and the **Data Vault** (which stores the encrypted recordings and provides them to authorized clients).
- **Application Layer:** the **Mobile App** for user and contacts to configure the system and respond to incidents.
- **Emergency Responders:** on the periphery, the contacts (family/friends) receive alerts on their phones, and optionally emergency services can be looped in (either through a contact or direct integration).

This end-to-end architecture ensures that from the moment a distress signal is generated on the watch to the moment help is dispatched, every component works in sync, with redundancies (device + cloud + contacts) to maximize the chances of a positive outcome. In the next section, we dive deeper into one of the unique features enabled by this architecture: the hidden-camera detection, evaluating how various technologies contribute to it.

Hidden-Camera Detection Technologies

A standout feature of this smartwatch is its ability to detect hidden cameras in sensitive spaces – addressing the menace of voyeurism (IPC 354C). Hidden cameras can be notoriously difficult to spot with the naked eye, so our device employs **multiple complementary detection techniques**. In this section, we evaluate the available technologies for hidden camera detection, including their principles, pros, and cons, and justify the choices for our implementation:

1. Radio Frequency (RF) Scanning: Many covert cameras (especially wireless nanny cams or spy cams) emit RF signals to transmit video – often using Wi-Fi, Bluetooth, or proprietary analog feeds in bands like 1.3 GHz or 2.4 GHz. An RF scanner sweeps through frequencies to find unexpected transmissions. Professional bug detectors have this functionality: they beep when a strong signal of certain modulation is detected eufy.com/eufy.com. Our watch could include a wideband RF receiver to scan for signal peaks. *Pros:* Can detect active wireless cameras even if they are well-hidden visually, and works in light or dark. It doesn't require line-of-sight – you can detect a camera transmitting from behind a wall or ceiling. *Cons:* Not all hidden cams transmit (some just record to SD cards). Also, an RF scan might flag other devices (e.g. a Wi-Fi router or a Bluetooth speaker) leading to false positives. In a hotel or office, there could be many RF sources. Additionally, scanning a broad spectrum can be time-consuming and power intensive, and might need an antenna which is challenging to fit in a small watch. Our solution is to include an **RF detection mode** focusing

on common spycam bands, and filter out known benign signals. Perhaps the user could calibrate by letting the watch know the Wi-Fi SSID of the location to ignore, etc. RF scanning will serve as one **mode of detection**, but not the sole method.

2. Lens Reflection (Optical Glint) Detection: Camera lenses, even tiny pinhole ones, are made of glass and reflect light. A proven method to find them is to shine a light and look for the characteristic “**red dot**” reflection. Many commercial detectors use this: they have a ring of red or IR LEDs and a viewfinder; as you look through, a camera lens appears as a bright spot of reflected light [eufy.com](http://eufy.com/eufy.com). Our smartwatch can replicate this by using IR LEDs around its face and using either the watch’s camera or a specialized photodiode to sense reflections. The user would move the watch around slowly in a darkened room; if a strong reflected beam returns at a consistent angle, the device signals a find. *Pros:* Works for any type of camera (wired or wireless) since it relies on physical lens presence. It’s relatively low-tech and **highly effective within line-of-sight** range (typically a few meters). *Cons:* Requires the environment to be dim or dark for best results, and requires a somewhat direct line-of-sight to the camera lens – if the lens is completely hidden behind a pinhole and not facing the detector, you might miss it. Also, in a very cluttered environment with lots of reflective surfaces (mirrors, metal objects), it might generate some false reflections (though true camera lens reflections often have a tell-tale colored shine). In our implementation, the watch will pulse IR and use the camera sensor to capture images; an algorithm will highlight tiny reflection points that blink in sync with the pulses (indicating a retroreflective lens). This optical method is a **cornerstone** of our detection as it directly targets the defining feature of cameras (the lens).

3. Magnetic/EMF Anomaly Detection: Some detection devices and smartphone apps claim to use the phone’s magnetometer to spot cameras by detecting electromagnetic fields or magnets in the camera. Tiny hidden cameras may have small motors (for autofocus or tilt) or coils that emit EMF, and often manufacturers put magnets in battery covers or mounts. The watch’s built-in magnetometer could sense stronger magnetic fields when very close to a camera (within a few centimeters). *Pros:* If the hidden device uses a speaker or motor (like a rotating security cam), those generate fields that can be picked up. Also, many common spy cams use DC adapters with transformers that emit EM. *Cons:* It’s quite limited – purely magnetic detection can miss electronics that have no significant magnetic components. It’s also range-limited; you’d have to be almost touching the hidden device. The magnetometer is more useful as a supplement – for instance, if the RF scan and optical scan are inconclusive but the magnetometer spikes at one spot, that gives additional evidence. We consider this a **secondary method**; the firmware can monitor the magnetometer during a scan and note anomalies, but it won’t be heavily relied on due to its unreliability for this purpose.

4. Computer Vision (CV) Lens Spotting: With the watch’s own camera, we could attempt to visually identify camera-like objects. For example, an algorithm could scan the environment for round glassy shapes (lenses) or the glint pattern of a camera sensor. There has been research on using deep learning (e.g. a YOLO object detection model) to find hidden camera lenses in images sciencedirect.com. *Pros:* Potentially can find lenses even without the IR reflection trick, and can work under some lighting conditions. Could also detect things like pinhole in an unusual place (like a tiny hole in a screw head) by learning their appearance. *Cons:* This is **computationally heavy** for a smartwatch MCU, and likely

requires a good quality image of the entire room which the watch's tiny camera may not capture well in one frame. Also, it may have many false positives (e.g. it might think a shiny bolt or phone camera lying around is a hidden camera). Due to resource constraints, we likely will not run a full CV model on the watch. However, in a scenario where the user has a phone, she could use the phone camera with our app for a thorough CV scan (the app could use its more powerful processor to run a model). So, the ecosystem can have the **phone app augment** hidden camera detection by using the phone's camera for visual sweep, while the watch focuses on IR and RF scans. Thus, CV lens spotting is in our toolbox but offloaded to the phone/cloud: e.g., the user could take photos of suspicious areas via the app and a cloud service could analyze them for hidden lens signatures.

5. Thermal Detection (infrared imaging): Not explicitly listed in the question, but worth noting: cameras and other electronics often emit heat. A thermal camera can reveal hotspots where a hidden device might be operating (especially if powered on for long). Our smartwatch will not have a thermal imager (too costly and power hungry), but in some advanced version, one could imagine using a small thermopile array sensor to scan for warm spots. However, this would also catch numerous false sources (warm pipes, laptops, etc.), so we put this aside.

6. mm-Wave Radar Imaging: The mention of possibly using **mm-wave sensors (60–77 GHz)** is intriguing. Texas Instruments and others make tiny mmWave radar chips that can sense objects and even create 3D maps. In theory, mmWave can penetrate certain materials (clothing, plastic) and detect metal or dielectric discontinuities. A hidden camera's circuitry or metal lens could reflect radar waves distinctively. In a research context, one might use radar to **see through** a fixture and detect a device. *Pros:* Works in darkness, doesn't require visible access, can cover a wide area with a scanning beam. *Cons:* These sensors consume significant power and data, and interpreting radar signals for something as small as a camera is non-trivial. Possibly beyond our near-term implementation, but we mention it as a futuristic approach. If in 2-3 years miniaturized radars become common in phones (some phones already have Google's Soli radar for gesture), the watch could leverage that concept. For now, mmWave might be too power-heavy for continuous scan, but perhaps could do a short sweep if a high suspicion triggers it.

After evaluating all methods, **our approach will be multi-modal:** The watch's hidden camera detection routine will utilize **IR reflection detection as the primary mechanism** (due to reliability), supplemented by **RF scanning** for wireless signals. The device will alert the user if either method finds a strong indication. We will also leverage the smartphone app to enhance scanning: for example, the app can automatically connect to the watch and display a UI that shows a bar graph of "signal strength" from the RF scanner, while also using the phone camera flash to do a lens sweep (covering more angles). By combining methods, we can greatly improve detection confidence – e.g. **CamRadar**, a recent system, cleverly utilized EM emanations to detect cameras with over 93% success cse.buffalo.edu and was not limited by camera transmission type or angle cse.buffalo.edu. Inspired by that, our system's integration of optical and RF means even if a camera isn't broadcasting (like just recording to memory), the optical method can catch it; if a camera has no direct line of sight (hidden behind something) but is broadcasting, the RF method can catch it. This redundancy is crucial.

We will implement a simple **user workflow** for scanning: say the user opens the “Scan Room” function on the app, places the watch on scan mode. They walk around the room holding up the watch. If the watch detects a suspect signal or reflection, it will vibrate and the app will show “Possible hidden camera detected!” with maybe an arrow or strength indicator to help locate it precisely (the user can then pinpoint the source). The device might also log the finding (so the user has evidence of finding a spycam, which could be reported).

It’s important to calibrate and test this feature thoroughly to avoid both false negatives (missing an actual camera) and false positives (crying wolf too often, which could erode user trust in the device). We’ll use a variety of common hidden camera models in testing – some with Wi-Fi, some standalone – to finetune sensitivity. We might also crowdsource data: the app could let users label if a detection was indeed a camera, improving the algorithm over time (keeping privacy in mind).

In conclusion, by evaluating **RF, optical, magnetic, and CV-based techniques**, we choose a **hybrid detection module** leveraging the most practical and miniaturizable technologies. This hidden-camera detector will give our smartwatch a unique capability to proactively protect privacy – a feature highly relevant given the rising voyeurism cases endnowfoundation.org. It transforms the watch from just a reactive safety tool (helping after an incident starts) to a **preventive tool** that can alert users to potential danger beforehand, aligning with the idea of creating safer environments for women through technology.

Data Security & Privacy Compliance

Given the sensitive nature of the data involved (personal videos, audio of incidents, location trails, etc.), the smartwatch ecosystem is designed with a **privacy-first approach**. We incorporate strong data security measures end-to-end, and ensure compliance with relevant data protection regulations, notably India’s **Digital Personal Data Protection Act (DPDP), 2023**. Below we outline the security architecture and privacy practices:

End-to-End Encryption (E2EE): All user data generated by the device is protected by encryption from the moment of capture until it reaches the intended recipient (user’s cloud vault or emergency contacts). The device encrypts recordings **locally before upload**, meaning even if someone intercepts the data in transit or breaches the cloud storage, they cannot decipher it akitra.com. Each SOS recording can be encrypted with a unique symmetric key that is itself encrypted with the user’s public key (established during device onboarding). Only the user’s mobile app (which holds the private key in its secure storage) can decrypt the content. Similarly, location coordinates and messages can be encrypted. For live phone calls to contacts, we use standard cellular voice which is encrypted over the air by telecom networks; if we ever route voice over IP, we’d use TLS and SRTP for encryption. **In short, the system ensures that no unauthorized party – not even the service provider – can eavesdrop on the user’s data.**

Secure Device Authentication: The watch and cloud authenticate each other to prevent impersonation. During setup, the device is provisioned with a unique device ID and secret. When connecting to the cloud, it uses this for mutual TLS authentication. Emergency communications to contacts (like SMS) are signed or include verification so the contacts

know it's genuine and not a spoof. The mobile app uses user credentials (with MFA or phone OTP) to ensure only the rightful user accesses their data.

Biometric/PIN Locks: As mentioned in requirements, the device and app both enforce **biometric or PIN unlock** for sensitive operations. The watch itself can lock its buttons in normal mode to avoid misuse – e.g. a stranger can't trigger the user's SOS or view data without permission. The user (or their guardian) would use fingerprint or phone unlock to enable/disable that. The app requires login and we encourage enabling fingerprint unlock on the app for quick yet secure access. If the watch is lost, the user can remotely deactivate it via the app (to prevent someone else pairing to it).

Data Minimization & Anonymization: In line with DPDP Act principles, we only collect data necessary for providing the safety service techtarget.com. That means:

- The watch is not continuously sending the user's location – it only does so during an SOS event or if the user explicitly uses a tracking feature. No "surveillance" of the user's daily movement is stored centrally by default.
- Audio/video is not recorded unless the distress feature is activated (no constant recording or audio monitoring, so the user's privacy in normal times is respected).
- The cloud does not store any plain personal info beyond what's needed: e.g., emergency contacts' phone numbers and maybe first names (with their consent) and the user's own name for identification in messages. Even these can be encrypted in the database, such that only the user's app can read them when needed.
- If any analytics are collected (like how often features are used or battery stats), they are aggregated or anonymized. For example, we might collect how many SOS events occurred in a region for improving police resources, but that would not include personal identifiers without consent.

Compliance with DPDP Act, 2023: The DPDP Act is India's new data protection law focusing on user consent, data security, and user rights techtarget.com. Our system is built to comply:

- **Consent:** Users provide explicit consent for processing their data when they sign up (through the app's terms/privacy policy). Emergency contacts also give consent to be notified. If the user is a minor, parental consent is obtained for collecting and processing the child's data (the app will have a flow for guardian approval) techtarget.com.
- **Purpose Limitation:** Data is used only for safety purposes. For instance, location data is used to route help during emergencies; we don't use it to serve ads or unrelated services. We clearly communicate what data we collect and why in the privacy notice (as required by DPDP).

- **Data Security Safeguards:** The law expects reasonable security measures; we likely exceed “reasonable” with end-to-end encryption, secure hardware, and rigorous testing techtarget.com makitra.com. We treat the watch much like a health device where privacy is paramount.
- **Breach Notification:** If despite all safeguards, a data breach occurs (say our cloud is compromised), we will follow DPDP requirements to notify the Data Protection Board of India and the affected users promptly techtarget.com. But since we store mostly encrypted data, the risk of sensitive info exposure is reduced (the attacker wouldn’t easily decrypt videos without the keys).
- **Data Retention & Deletion:** Users will have control over their data. Through the app, they can delete stored incident records from the cloud (e.g. after downloading evidence, they might want to wipe it). Our policy might be to auto-delete incident data after a certain period (say 1 year) unless the user marks it to keep (for an ongoing case). The DPDP Act emphasizes data should not be retained longer than needed, and our system aligns with that by not hoarding old data indefinitely. Users can also request deletion of their account and all associated data – and we will comply, after confirming no legal reason (like an ongoing criminal case) requires retention.
- **Child Data Protections:** The Act forbids tracking or behavioral monitoring of children and targeted ads towards them carnegieendowment.org. Our device does not do behavior profiling; any tracking is purely for safety initiated by the child or guardian. And obviously, we serve no ads at all. So we adhere to those provisions. Guardian controls on a child user’s device also align with the Act’s intent for parental supervision.

Secure Hardware & Firmware: We implement secure boot on the device so that unauthorized firmware (which could snoop on the user) cannot run. All debug interfaces (like JTAG) are locked down in production units to prevent extracting secrets. The use of a **secure enclave** in the MCU means cryptographic keys (for encryption, and authentication keys for cloud) are stored in hardware that even a sophisticated attacker with the device can’t extract. Firmware updates are signed and delivered over encrypted channels.

Communication Security: All network communication uses modern cryptographic protocols. For instance, when the watch connects to cloud (likely over MQTT or HTTPS), it uses TLS 1.3 ensuring forward secrecy etc. The BLE connection with the app is paired using secure pairing (LE Secure Connections with passkey) to prevent MITM; and sensitive commands over BLE are encrypted at the application layer too. We recall that wearables often had issues with Bluetooth eavesdropping; we address that by disallowing older insecure pairing modes and by keeping BLE use to non-critical data where possible.

Emergency Exceptions & Ethics: One challenge is that in an emergency, the user might want to share data with police or in court. Our encryption scheme should allow the user to do so easily (e.g. by giving them an export of the key for that incident or a secure share link). However, we will *not* build any “backdoor” for law enforcement to bypass encryption – that would undermine the security for users. Instead, if law enforcement legally needs data, they

would get it from the user (who holds the keys), preserving the user's agency. This approach is consistent with privacy rights – the DPDP Act actually penalizes unauthorized disclosure of personal data.

Privacy of Contacts: We also ensure that the emergency contacts' information isn't misused. They only get data when they need to respond. We don't show all contacts to each other except perhaps names if the user chooses (like, friends see each other's name in a group alert context). Contact details aren't shared beyond what's necessary to deliver the alert (the telecom obviously gets the number to SMS/call). We store minimal info about contacts and allow the user to manage it.

Server-Side Privacy and Auditing: Administrative access to the cloud servers will be tightly controlled. We will maintain audit logs of any administrative actions on data. Given data is encrypted, even admins can't view content – but, for example, if an admin needed to help remove a problematic account, that action is logged. This is more operational, but important for compliance and trust.

Secure Disposal: If a user stops using the device or deletes data, we securely delete encryption keys such that even backups eventually cannot be read (keys will be scrubbed or overwritten). On the hardware side, if the watch is being scrapped, the secure element ensures keys cannot be recovered (some even have self-erasure if tampered with).

Third-Party Components: We will vet any third-party SDKs (e.g. mapping API, SMS gateway) to ensure they comply with privacy standards. For instance, using Google Maps for location reverse-geocoding should be done carefully (or avoid sending personal data to external APIs by doing on-device conversion if needed).

In essence, the entire design follows “**Privacy by Design**” principles – embedding privacy into technology from the get-go, not as an afterthought [akitra.com](https://www.akitra.com). This builds user trust, which is absolutely vital for adoption of a safety device. Users (and their families) need confidence that the device will *protect them without exposing them*. By using robust encryption, secure handling of personal data, and complying with laws like DPDP (consent, breach notification, etc.), we aim to make the smartwatch as **secure as a personal safe** for one's data. And beyond compliance, these measures are simply the right thing to do ethically for a product that will often be privy to some of the most traumatic moments in a user's life.

Regulatory & Certification Path

Bringing a safety smartwatch to market in India requires navigating several regulatory and certification requirements. We must ensure the device complies with electronics safety standards, wireless spectrum regulations, telecom norms, and other relevant Indian laws. Below is the roadmap for regulatory approvals and certifications:

1. Bureau of Indian Standards (BIS) Certification: In India, electronic products, including smart wearables, are subject to mandatory certification to ensure they meet safety standards. Specifically, **smart watches have been included under BIS Compulsory Registration Scheme (CRS)** since 2018, requiring compliance with IS 13252 (Part

1):2010crsbis.in. IS 13252 (Part 1) is essentially the Indian equivalent of IEC 60950-1 / IEC 62368-1 (safety standard for information technology equipment). We will get our device tested in a BIS recognized lab for electric safety (checking for no shock hazard, fire safety of enclosure, etc.). Once tests are passed, we register the product with BIS and mark it. BIS registration ensures our device is safe for end users (no overheating, no dangerous emissions) and is legally required to sell in Indiabl-india.com. Additionally, if our device has a power adapter or charger, that charger likely needs BIS certification under the applicable standard (IS 13252 or the specific standard for power adapters).

2. Wireless Planning & Coordination (WPC) Approval: The device uses wireless communication (Bluetooth, LTE/NB-IoT, possibly Wi-Fi in testing). India's WPC, under the Dept. of Telecommunications, regulates RF emissions and spectrum use. For license-exempt bands (Bluetooth in 2.4 GHz, etc.), we need to obtain an **Equipment Type Approval (ETA)** from WPC for the RF module, unless we use a pre-approved module. Many off-the-shelf modules have ETA; otherwise, we'll submit our product RF test reports (for EMI/EMC and frequency use) to WPC. Particularly, Bluetooth LE at 2.4 GHz and any other unlicensed band sensor would be covered under ETA. Our cellular module will operate in licensed spectrum (with a carrier SIM), so as a device manufacturer we ensure it meets 3GPP specs but no separate license needed for usage – however, we must ensure the module has been approved by global bodies (like GCF/PTCRB) and by Indian operators for network compatibility.

3. Telecommunication Engineering Center (TEC) – MTCTE: India has introduced the **Mandatory Testing and Certification of Telecom Equipment (MTCTE)** for any telecom equipment to be used/sold in India. A smartwatch with a cellular radio might be classified under this (similar to how mobile phones are). We should check if our device needs TEC certification. Likely, since it's end-user equipment with a SIM, it should undergo testing for things like SAR (radiation absorption rate), EMI/EMC, safety, etc., under TEC guidelines. SAR testing ensures the device's RF emissions are within safe limits for wearing on the body. We will get SAR measured (likely it will pass since LTE-M power is low and watch is small, but it's required to document). The TEC certification would cover adherence to technical requirements for cellular devices and maybe e911/112 behavior (ensuring 112 calls go through even without SIM or with locked phone, etc., though since our device relies on SIM it just needs to allow 112 dialing by any SIM).

4. International Standards (for future export): Although initial market is India, we will design to meet CE and FCC standards too.

- **CE (Conformité Européenne):** This would require compliance with the Radio Equipment Directive (RED), which includes RF performance, EMC (electromagnetic compatibility), safety (likely EN 62368-1), and SAR for wearables. We'd run EMC tests (to ensure our device doesn't emit spurious interference and can withstand ESD, etc.). EMC compliance is also needed for BIS/TEC.
- **FCC (Federal Communications Commission, USA):** The device's transmitter must be tested for FCC Part 15 (unlicensed emissions) and Part 27 (for LTE bands) if we ever sell in the US. We'd ensure our module has FCC modular approval which

simplifies this.

5. Battery Certification & Transport Regulations: The smartwatch contains a lithium battery, which must comply with safety standards. India often references UL or IEC standards for batteries. **IEC 62133** is the international standard for safe operation of Li-ion batteries (covering overcharge, short, etc.). BIS in India has adopted IS 16046 which mirrors IEC 62133 – we should get our battery certified to IS 16046. Also, for **transportation**, lithium batteries (especially if shipping by air) must pass **UN 38.3** tests (altitude simulation, thermal, vibration, shock, short circuit, impact, etc.) mobilepowersolutions.com. We will ensure our battery supplier provides UN 38.3 certification mobilepowersolutions.com. That way, we can ship the product by air courier without issues. Airlines may also require our packaging to meet IATA packaging instructions for devices with lithium batteries (like protective packaging, markings if over certain Wh). Our battery is small (maybe ~2 Wh), likely under the limit for simplified shipping, but compliance is still necessary.

6. BIS Registration for Battery & Adapter: Besides the watch device, if we include a USB charging cable or adapter, note that adapters/power banks also require BIS under CRS. We might avoid including an AC adapter (assuming users charge via phone chargers) to simplify. But if we do, that adapter must comply with IS 13252 or IS 616 (if considered a power supply). The lithium cell itself might come under an Indian standard if above a capacity threshold, so we'll check BIS list for "Lithium-ion cell". As of now, cells used in electronics often come with UL certification but BIS might not require separate if integrated.

7. Environmental and Safety Regulations:

- **RoHS & WEEE:** Restriction of Hazardous Substances compliance ensures no lead, mercury, etc., in our device beyond permitted levels. If manufacturing in India or importing, compliance is needed (India has its own e-waste (Management) Rules that basically align with RoHS). We'll use RoHS components and label accordingly.
- **Battery Safety Transport for India:** The UN38.3 covers global, but domestically we might also follow Central Electrical Testing for batteries. Also, we'll label the battery properly with required symbols (do not dispose in fire, etc.) and provide MSDS if needed.

8. AI/Software Compliance: While not a formal certification, any AI in the device (like camera detection) should comply with ethical guidelines (no privacy violation since it runs locally). If we use cryptography, we ensure it meets Indian crypto export/import rules (generally, consumer devices with encryption are allowed; any restriction would be minimal as long as we aren't doing anything outside normal device encryption).

9. Patent/Frequency Licensing:

- We should ensure any specific technology (like CamRadar approach or certain detection methods) doesn't infringe existing patents. Do a freedom-to-operate search especially on hidden camera detection algorithms, just to be safe, or license if

needed.

- For using LTE-M/NB-IoT, our module vendor will likely handle the intellectual property licensing for cellular tech (since they pay royalties to patent pools). We'll just buy certified modules to avoid directly dealing with 3GPP patent licensing.
- If we add something like GPS, there's no license needed to receive signals.

10. Certification Timeline & Strategy:

- Early in development, we'll engage with a testing lab (maybe an NABL accredited lab in India) to do pre-compliance testing on RF, EMC. This way, we catch any issues (e.g. our PCB antenna might cause harmonic emissions) and fix them before formal certification.
- Once the product design is finalized, we proceed with BIS CRS registration (which can take a couple months including testing). We might expedite by using pre-certified modules (for Bluetooth, cellular) as that often simplifies testing – for instance, if the RF module is pre-approved, we mainly test the full product's radiated emissions and safety, not the module's transmitter from scratch.
- WPC ETA for Bluetooth is often paper-based if you have a module report (just submit module FCC/ETSI reports and get ETA). If using a custom antenna or design, we do our own testing for that band.
- TEC MTCTE can be slightly involved; we will see categories (maybe "Communication module product" or "Smart Tracking device" might be a category). If needed, we send it to TEC-accredited lab for tests which include SAR, EMI/EMC, safety, etc. Possibly our BIS testing covers a lot of those aspects, but separate submission to TEC is required with those results.
- SAR specifically: Our device worn on wrist would be considered "wearable on body". SAR limits in India follow international (1.6 W/kg averaged over 1g for head, and different for limbs maybe). Since our transmit power (23 dBm or lower) is small and the antenna is small, we expect SAR well under limits, but it must be measured and documented.

11. Packaging and Labelling: After certifications, we must label the product appropriately:

- BIS CRS logo with our registration number on device/packaging.
- WPC ETA number if applicable (some devices list the ETA number or the generic "Bluetooth use in India, not harmful" statement).
- The battery needs the Li-ion recycle logo and "CAUTION".

- If we got CE/FCC for international, their logos would be on packaging.
- Also, because it's a camera device (though a very small one), some jurisdictions might require us to indicate that device contains a camera (for import/export or privacy label, e.g. some companies label devices that can record). Not a formal law in India, but something to consider ethically (though since it's for the user's own safety, not spying, probably not an issue).
- The device will have an IMEI if cellular – we need to have the IMEI printed somewhere (like inside or on the box) and ensure each device gets a unique IMEI from GSMA range. For manufacturing, we'll obtain a TAC (Type Allocation Code) for the watch model to generate IMEIs.

12. Compliance Documentation: We will maintain a technical file with all test reports and compliances. Also, draft a user manual section on regulatory compliance (the kind that says "This device complies with BIS standards... FCC ID:..., etc." and safety instructions as required by standards).

13. BIS - Quality Control Order: In India, certain electronics also come under "Electronics and IT Goods (Requirement of Compulsory Registration) Order, 2012" which basically is the BIS CRS. We ensure our manufacturing is done under proper quality controls (BIS may audit the factory for maintaining standard). We'll also register the product on the government's CRS portal.

14. Other Indian Laws:

- **Import Duties and DGFT regulations:** If we import components or finished units, ensure no restricted substances or such. Likely fine.
- **Legal recording considerations:** Indian law (and many others) allows a person to record their own conversations or scenes they are part of, for lawful purposes like evidence. We should include a note in the user agreement that the device is intended for self-defense evidence and not to record others secretly for inappropriate reasons (to disclaim misuse).
- **Use of 112:** We might coordinate with Ministry of Home Affairs / local 112 centers if our device is widely adopted so they know how to handle automated calls from it. Currently, smartphones are mandated to have 112 on long-press of 5 or 9 or power button – our device acts similarly but in wearable form.

15. Testing for Extreme Scenarios: Though not a formal certification, we should test that the device still works under scenario conditions (like if someone is physically assaulted and the watch is banged or gets some water splashed, it still sends an alert). Perhaps get an **IEC 60529 IP test** for water resistance (IP67). And maybe some durability tests (not necessarily certified, but to ensure reliability).

By systematically obtaining these approvals, we ensure our smartwatch is **legally compliant and safe** to use. Certification may seem like a hurdle, but it is essential to protect users and validate our claims. For example, BIS testing will reassure that our device's battery won't explode and WPC ensures we don't interfere with other devices. Achieving these certifications can also be a marketing point (we can advertise that the product is certified safe, meeting national standards).

In parallel to regulatory tasks, we will also file any necessary documentation under the **India's Personal Data Protection** regime (if needed, maybe as a data fiduciary we might have to register depending on volume of data processed, and have a privacy policy that adheres to the Act). This ties into the compliance aspects described in the previous section.

Conclusion of this section: Navigating BIS, WPC, TEC, and related certifications is a non-negotiable part of our go-to-market. We have planned for it from the design phase (choosing certified modules, designing for compliance) so that there are no nasty surprises late in development. With these certifications in hand, we can instill confidence in retailers, government, and consumers that the device meets all Indian **regulatory standards** – it's safe to use, doesn't violate wireless rules, and aligns with national norms for quality.

Risk Analysis & Mitigations

Developing and deploying a safety smartwatch entails several **risks** – technical, operational, and user-related. It's crucial to identify these risks and plan mitigations to ensure the device truly serves as a reliable lifeline in critical moments. Below we discuss major risks and how we address them:

Risk 1: False Alarms (Accidental or Malicious Activation). *Scenario:* The user accidentally presses the SOS button or an overly sensitive fall detection triggers an alert, sending false alarms to contacts (or police). Too many false alarms could lead to “alert fatigue” where contacts start ignoring the device or, worse, authorities impose penalties. *Mitigations:* We have built-in a **confirmation window/cancel option** – e.g., the 5-second delay with vibration feedback where user can cancel if unintended (as described earlier, similar to smartphone SOS). Also, our algorithms for automated triggers (fall, sound detection) are tuned conservatively initially to avoid false triggers. During onboarding, we educate users on proper wear (so the SOS button isn't pressed by bending wrist ordinarily) and encourage them to test the button in training mode to get comfortable. If a false alert does go out, the message to contacts can say “If this is accidental, user will cancel in 5 sec” so they wait a moment. We also allow **user feedback**: e.g. if the watch triggers wrongly, the user can report that in the app, and the system might adjust sensitivity (machine learning on false events, if possible). By minimizing false alarms, we maintain trust in the system's credibility.

Risk 2: Failure to Trigger or Send Alert (Reliability). *Scenario:* In a real emergency, the user presses the button but nothing happens – perhaps due to device malfunction, dead battery, or network outage. This is a worst-case failure as it defeats the purpose. *Mitigations:* Multiple strategies ensure reliability:

- **Redundancy in Alerts:** The watch will try multiple channels – sending data via cellular, but also via paired phone if connected (so if one fails, the other might work). It can send both SMS and data packets as backup.
- **Device Self-Monitoring:** The device frequently monitors its battery and will *warn the user well in advance* if battery is low (via app notification or LED). We aim to have a reserve so even at “0%” on UI there is enough for one SOS. We’ll also encourage a nightly charging habit via the app.
- **Hardware Reliability:** We select automotive/industrial grade components where possible for robustness, and do stress testing. E.g., test the button 5000+ presses, test device under various temperatures. The enclosure is rugged so that it still works if thrown or impacted (to an extent).
- **Network Issues:** If there is no cellular coverage (e.g., basement), the device still records evidence offline. As soon as it finds even a 2G signal or later, it will send stored alerts (the firmware keeps trying). If the user is in a known no-signal zone (like remote area), we could consider an **offline mode** where the device at least emits a loud alarm since online help can’t be summoned. Future mitigation is satellite connectivity (see Future Work) to handle remote areas.
- We will maintain a **server heartbeat** mechanism: the device pings the server periodically, so the backend can notice if a device is offline or hasn’t checked in. If a user’s device goes offline for long (and they have a buddy system), it could notify the user or guardian to check on it.
- For minors or dependents, the guardian app could show device status (battery, last seen) so they know it’s ready.

Risk 3: Device Being Located or Destroyed by Attacker. *Scenario:* An aggressor notices the victim’s watch is recording or sending an SOS (maybe they heard a contact’s voice or saw a flashing LED) and tries to take it off and destroy it. *Mitigations:* We designed the watch to be as **discreet as possible** in silent mode (no loud sounds or obvious light unless user enabled). The attacker might not even realize an alert was sent. But if they do attempt to remove:

- We have **tamper detection** – instant alerts of “device removal” are sent if the strap is opened or cut (so contacts know something’s wrong even if data stops).
- The device uses very strong materials for the strap (e.g. a cut-resistant strap with embedded steel mesh, similar to some anti-theft bags). This could make it hard to quickly snatch or cut.
- If the device is destroyed, by that time hopefully the critical first few seconds of audio/video and location are already uploaded. Our alert protocol prioritizes sending a quick low-bandwidth alert first (just location and a distress flag) because that’s more important than video; once that’s out within seconds, even if the device is

smashed, help is on the way.

- We might also design the watch's form such that it's hard to yank off – perhaps a buckle that locks or requires two hands to remove (so an attacker can't easily do it in a struggle). The user can still remove it normally, but an attacker would be slowed.

Risk 4: Privacy Breach or Misuse of Data Collected: *Scenario:* The sensitive videos recorded by the device are leaked or someone hacks the system to snoop on **Risk 4: Data Privacy Breach or Misuse.** *Scenario:* The sensitive videos, audio, or location data collected could be accessed by unauthorized parties – either through a cyber-attack or misuse by insiders. For instance, a hacker could target the cloud storage, or someone who gains possession of the device might extract data. *Mitigations:* As detailed in the Data Security section, we apply **end-to-end encryption** so that even if cloud servers are compromised, the data remains unintelligible [\[akitra.com\]](#). Encryption keys are stored only with the user (in the app or secure element on device). Additionally, all communications use secure protocols (TLS), preventing interception. We enforce strict **access controls** on cloud databases – no one (including our admins) can arbitrarily view user data. We also follow compliance requirements of DPDP Act to handle data safely [\[techtargget.com\]](#). Regular security audits and penetration testing will be conducted on the system. On the device side, the firmware and storage are secured (encrypted flash, secure boot). If a device is lost, the user can remotely wipe it via the app to remove any cached data. By making the system **zero-trust** (even we as service provider cannot see user's private recordings), we greatly reduce the impact of any breach. As an extra measure, emergency videos stored in cloud could be set to auto-delete after a certain time (unless saved by user) to limit exposure window. Lastly, we'll be transparent with users – providing audit logs in the app (e.g., showing when their data was accessed and by whom – mostly just themselves) and promptly notifying of any security incident, thereby maintaining user trust.

Risk 5: Misuse of the Device (By Owner or Others). *Scenario:* A user (or someone who obtains the device) could misuse it to violate privacy or prank others. For example, someone could secretly record conversations using the watch (ironic misuse for voyeurism), or trigger false SOS alarms intentionally (e.g., a child "crying wolf" to test family response). *Mitigations:* We address this through **user education and app settings**. In the app's user guide and onboarding, we make clear the legal/ethical use: the camera is for emergency evidence only, not casual recording. The device's design also helps: it has no easy way to record video silently without alert (for instance, when recording, maybe a subtle indicator like a vibrating pulse every few seconds reminds the wearer it's active, preventing continuous covert recording misuse). We could also enforce that recorded videos (when not in an SOS event) are only accessible via the user's verified account, so one can't use the watch purely as a spycam and retrieve footage without trace. Regarding false SOS use: if a user repeatedly triggers false alarms, the app can issue warnings. Family contacts can give feedback through the app (e.g., mark an alert as false). In an extreme case, if someone were maliciously abusing the system (e.g., clogging public emergency lines with false SOS), we could temporarily deactivate the automated 112 dialing for that device and require reactivation after a safety check or training. Essentially, while the device empowers the user, we assume good intent. We put safeguards (both technical and policy) to deter intentional misuse, and have a **code of conduct** acceptance when setting up the device.

Risk 6: Technological Limitations or Counter-Measures by Attackers. *Scenario:* A savvy attacker might attempt to **jam the signals** (e.g., use an RF jammer to block cellular/Bluetooth) or adapt tactics (like avoiding detection by the device). Also, the hidden camera detector might not catch a very well-concealed next-gen spy device, giving a false sense of security. *Mitigations:* Jamming is a challenge – if an attacker actively jams cellular, the watch might fail to send out alerts. However, jamming is illegal and relatively rare; plus it tends to be localized. Our mitigation is multi-frequency capability: an attacker would have to jam both LTE and SMS (different bands) and possibly also the phone's network if the phone is present. If we detect something is off (e.g., the watch sees no network suddenly – possibly due to jamming), we could automatically trigger the loud siren as a backup to draw attention, since quiet alerts won't go out. Essentially, if connectivity is lost during an SOS in an area that normally has signal, the device could assume malicious interference and switch to **audible alarm mode** as a last resort. On hidden camera detection: we acknowledge no method is 100% foolproof. We mitigate by combining methods (RF + optical) to improve odds. We will clearly communicate to users that the tool greatly reduces risk but **vigilance is still advised**. Regular updates to the detection algorithm (via firmware updates) will keep pace with new types of spy devices. Also, any time a user finds a hidden camera (whether our device helped or not), we encourage reporting it so we can update our threat database (crowdsourced learning).

Risk 7: Operational and Financial Risks: *Scenario:* The project faces manufacturing delays, cost overruns, or difficulty in adoption. If the BOM cost is too high, we risk pricing out the target users; if supply chain issues occur (e.g., component shortages), product availability suffers. There's also risk in scaling customer support for a safety-critical device. *Mitigations:* We have a dedicated section on BOM and cost optimization (see next section) to ensure the device can be produced at scale for a reasonable cost, including strategies like bulk component sourcing and possibly subsidies. To avoid supply issues, we choose **proven components** available in volume (for example, using a widely used MCU/module rather than an exotic one). We maintain good relationships with suppliers and have backup options for critical components (e.g., at least two possible camera module vendors). For operations, we'll implement a rigorous QA in manufacturing (so that each unit's critical functions – SOS, calling, etc. – are tested before shipping out). Regarding user support: we plan a 24/7 helpline or chat support especially for when the device is new in market, to quickly address any malfunctions or user confusion (because in safety tech, user must be confident everything will work when needed). This risk is more business-oriented, but addressing it ensures the **solution is sustainable long-term** (a device that can't be financially sustained might end up discontinued, which would let users down).

By systematically analyzing these risks, we aim to design not just the product but also the **ecosystem** (user training, support, maintenance) to mitigate them. It's important to remember that this smartwatch is used in high-stakes situations – thus reliability and trust are paramount. Our multi-layered mitigations (technological, procedural, educational) address the risks at their root. For example, encryption and access control tackle privacy risks at a fundamental level, while UI/UX choices (cancel option, discreet design) tackle false alarms and tampering proactively. We will keep updating our risk register as the project evolves, learning from any incidents or user feedback in pilots, and iterating the design or processes accordingly.

In summary, while no system can be 100% risk-free, our approach is to **anticipate failure modes and edge cases**, and build in safeguards so that the device *fails safe*. Even in worst-case scenarios (device compromised, network down, etc.), we strive for a fallback that still offers some protection (like the audible alarm). This resilience-by-design philosophy increases the likelihood that when a real crisis occurs, the smartwatch will perform as intended and possibly save a life, which is the ultimate measure of its success.

Ethical & Societal Considerations

The development and deployment of a women's safety smartwatch must be guided not only by technical and business factors, but also by a strong ethical compass and sensitivity to societal context. Below we address the key ethical and social considerations:

Empowerment vs. Victim Blaming: One ethical concern is ensuring that technology to protect women is seen as **empowering** and not as shifting responsibility of safety solely onto women. We must frame the smartwatch as a tool that can help *in spite of* societal problems, not a solution *to* the root problem. The marketing and messaging will emphasize that women *have a right to safety* and this device is one way to assert that right – but it does not mean women are responsible for preventing crimes against them. We will actively avoid any narrative suggesting “if she wasn’t wearing it, she’s to blame for an assault.” This aligns with the broader movement that the onus is on society (and perpetrators) to stop crimes. The device is a **stop-gap protection** in an imperfect world.

Privacy of the User: It might seem paradoxical that a device with a camera and microphone worn 24/7 could raise privacy issues for the user herself. We guarantee that the watch is *not a surveillance device* on the user. By default, it does not record or transmit unless deliberately triggered, and even then the data is protected (as discussed) and only used for her safety. The user's consent and control are central – e.g., she decides who her emergency contacts are, she can turn off or on features like location tracking (maybe she doesn't want continuous tracking by family). We provide granular controls: for instance, a mode where only when she triggers SOS do others see location, vs. a mode where a parent or spouse can “follow me live” (some users may want that for certain scenarios). All such features are opt-in to respect autonomy. Additionally, since the device will likely collect sensitive health-ish data like movement or heart rate (for falls), we ensure compliance with privacy norms there too (not sharing it or selling it). We will also include a **privacy mode** where the user can disable the camera/mic except SOS – perhaps an LED indicator that confirms the camera is electrically off – to reassure those who are worried about being monitored. Essentially, we abide by the principle: the **user owns her data** and any sharing is her choice.

Privacy of Bystanders and Contacts: When recording an incident, inevitably other people (bystanders or even the attacker who is also a private person) are being recorded without their consent. Legally, this falls under exceptions (legitimate interest for safety/evidence). Ethically, we consider that these recordings are only used in justice context or personal review, not broadcast publicly without due process. We advise users that any footage should be handled responsibly – e.g., share with police, not post on social media in a way that violates privacy (unless necessary to identify a criminal at large, for example). Emergency contacts who receive audio from the scene must also use it discreetly. We might include an

ethics agreement or training snippet for contacts too (“By accepting to be an emergency contact, you agree to respect the privacy of your friend’s data and only use it to assist during emergencies.”). As for hidden camera detection – using it in someone’s property might raise issues, but if you’re in a private space you’re entitled to scan for invasive devices. It’s a defensive measure and ethically sound.

Data Use and Law Enforcement: If the device collects evidence of a crime, there’s an ethical duty to use that data for justice. We encourage (but don’t force) users to share evidence with law enforcement in prosecuting offenders. We will cooperate with law enforcement within the bounds of user consent and legal process – e.g., if police subpoena data for an investigation, we have to comply, but since it’s encrypted by user, we’d involve the user. We position ourselves as **allies to law enforcement against abuse**, possibly providing a portal (with user permission) for police to get verified evidence, thus improving conviction rates of gender-based crimes. An ethical balance is needed: we want to help catch criminals, but we also must guard against *overreach* (e.g., authorities trying to use the device to surveil users under the guise of safety). We will not provide any backdoor live access to police or anyone; everything goes through user’s control.

Inclusion and Accessibility: We must consider various groups in our user base. The target is women 13-35, but within that:

- We have **minors** (teens) – we ensure the device and app experience accounts for guardianship while still respecting the teen’s agency (some teens may not want parents tracking their every move, so an arrangement is needed – e.g., alerts only in emergencies, not constant monitoring).
- We consider women with disabilities – the device should be usable by those with visual or hearing impairments. For example, the UI should have strong haptic feedback for blind users; the app should integrate with screen readers. Deaf users might not need the loud siren but rely on the silent functions – ensure equal effectiveness without audio.
- **Socio-economic inclusion:** The cost might be a barrier for low-income women who arguably are among those who could benefit greatly. This is a societal consideration to address in go-to-market (perhaps via government subsidies or NGO programs – discussed in roadmap). We want to avoid creating a safety divide where only affluent women get this extra safety net. Pilots with subsidized devices in at-risk communities (like women in urban slums who often travel at odd hours for work) can be considered.

Cultural sensitivity: In some communities, women’s mobility is a sensitive issue. Wearing a “safety device” might be seen either as empowerment or, conversely, might attract unwanted attention (“what is that device?”). We designed it to look like a normal watch to avoid stigma. We’ll gather feedback from users in conservative areas to ensure the device doesn’t conflict with dress codes (e.g., it should be okay to wear with traditional clothing). Also, language support in the app and alerts (e.g., allowing alert messages in local language) is important to ensure effective use by non-English-speaking women.

Psychological impact: There's an emotional aspect – wearing a device that reminds one of danger could cause anxiety. We aim for the opposite: a sense of security. But we must be careful not to instill a false sense of invulnerability. Ethically, we will communicate that this watch *improves chances of help* but is not a guarantee. Users should still exercise normal safety precautions. The device should be marketed as a **confidence booster** – many women report feeling safer even just knowing they can call for help easily (womentech.net), which can encourage them to live their lives more freely. That's a positive societal effect: women not feeling as restricted from public spaces or travel due to fear. However, there is a concern: might reliance on such devices lead to reduced community focus on making streets safer? We should use some proceeds or partnerships to also contribute to broader safety education and advocacy, showing that technology is one part of a multi-faceted solution (along with better policing, street lighting, etc.).

Community and Reporting: The device could indirectly encourage more reporting of incidents. For example, if an attempted assault is recorded and the user escapes, she might be more empowered to report it since she has evidence. This could help catch repeat offenders. Ethically, this contributes to societal good (possibly preventing future crimes). We will ensure that if a user does not want to report, that choice is respected – the data remains hers. But we can provide in-app guidance on how to report and connect with support organizations (like women's helplines, legal aid) if an incident occurs. Essentially, the ecosystem should support the survivor's journey after an incident, not just the incident itself.

Male Allies and Broader Use: While the product is focused on women, it's ethically important to include male allies in the conversation. For instance, many emergency contacts will be men (fathers, friends, partners) – we view them as part of the solution and provide them with guidance on how to respond effectively when they get an alert (maybe even training material on intervening safely or coordinating with police). Also, the tech itself can be unisex – men and other genders facing safety issues (like journalists, or delivery workers at night) could use it. We market primarily for women's safety because that's a pressing need, but we **don't exclude others from benefit**. Societally, this avoids the perception that it's a "gimmick only for women" and instead a serious personal security device applicable broadly, while still addressing the unique threats women face (like voyeurism, which is gender-specific in IPC 354C).

Transparency and Accountability: We commit to **transparency** in how the device functions and what it does with data. There won't be hidden functionalities – e.g., we're clear that "audio is recorded when SOS is active" so users know what's happening. We also open ourselves to third-party audits for security. This transparency builds trust in communities that this isn't a surveillance gadget or some way to exploit users. Additionally, if there are any adverse events (like device fails in a key moment), we will analyze and publicly (to users) share what happened and how we're fixing it. Being accountable ensures ethical responsibility to our user community.

Stigma and Acceptance: In some environments, wearing a safety device might carry stigma (like implying the environment is unsafe). We want to normalize it akin to wearing a seatbelt – a prudent precaution that everyone accepts. Part of societal introduction will involve working with institutions (colleges, companies) to promote it as a positive measure,

not as an alarmist tool. If, for example, a college issues these to students, we frame it as the college caring for student safety, not that the campus is particularly dangerous.

In conclusion, the ethical and societal dimension of this project is deeply intertwined with its engineering. We measure success not just by units sold or tech performance, but by **real-world impact**: Are women feeling safer? Are potential perpetrators deterred knowing such devices exist? Is society starting to treat women's safety tech as essential (the way smoke detectors or seatbelts are)? We will gather qualitative feedback from users to gauge these aspects and will remain flexible to modify features if something inadvertently causes harm (for instance, if we found the presence of the watch in some context was escalating risk, we'd address that). Our overarching ethical stance is to **enhance safety and freedom** for women without infringing on rights or privacy, and while promoting broader societal change that, ultimately, would render the need for such devices obsolete in the long run.

Cost Breakdown & BOM Forecast

Designing an advanced safety smartwatch involves a careful balance between feature richness and cost. In this section, we break down the **Bill of Materials (BOM)** for the device and project costs for both prototyping and mass production (10k units), highlighting where economies of scale or design choices drive cost savings. All costs are approximate and in USD (with notes on INR).

Major Hardware Components and Estimated Costs (per unit):

- **Microcontroller/SoC + Cellular Module:** We plan to use an integrated solution (e.g., Nordic nRF9160 or similar) which includes the MCU, LTE-M/NB-IoT modem, and GPS. In prototype (low quantity), such modules might cost around \$20–\$25 each. In volume (10k units), price can drop to roughly **\$10–\$12*[counterpointresearch.com](https://www.counterpointresearch.com)[counterpointresearch.com](https://www.counterpointresearch.com)]. If we need a separate BLE chip (if main SoC doesn't have it), add ~\$2 (volume ~\$1). We assume one integrated SoC covers it. This component is one of the most expensive due to cellular capability.
- **Camera Module:** A tiny camera (module with sensor + lens) say 1080p capability. Low volume cost ~\$8, volume cost ~\$4. We might use a 5MP sensor which in bulk could be as low as \$3–4. Lens and IR filter included in module. If we require an IR LED array or special optics for detection, that might add \$1 or so.
- **Memory:** External flash (2GB NAND or 16Gb SPI NOR) – prototypes \$2, bulk ~\$1. If the MCU has enough internal flash for our needs, we can eliminate this, but likely we include it for video storage. RAM (if needed externally) – maybe a 16MB PSRAM at \$1. But perhaps not needed if we stream chunks.
- **Sensors:**
 - Accelerometer/gyro: \$1 prototype, \$0.5 bulk (common MEMS like BMI160).

- GPS antenna: \$0.5 bulk.
- Mic (MEMS): \$0.5 bulk.
- Vibrator motor: \$0.5 bulk.
- Heart-rate sensor (with LED): ~\$1.5 prototype, \$1 bulk.
- Magnetometer (if not in 9-axis IMU already): \$0.5.
- Hidden cam detector components: IR LEDs and photodiode ~ \$1 total; RF detector front-end ~\$3 (or if we can time-multiplex radio, maybe minimal cost).
- Speaker/Buzzer: small piezo or speaker ~\$1.
- Buttons, connectors, etc.: \$1 (volume total for all small electro-mech parts).
- **Display:** If we include a small OLED or e-paper for time/status. A 0.5–0.7” OLED could be ~\$3 in low volume, \$1.5 in bulk. Alternatively, a simple LED indicator approach could cut this cost, but having a basic display adds value (time, messages). We'll assume a simple monochrome OLED.
- **Battery:** A 500 mAh Li-poly battery (custom thin form) – prototypes \$4, bulk \$2. Batteries benefit from scale but also quality matters. We include protection circuit – often included in battery cost. Also consider a charging IC (\$0.5 bulk) and components for power management.
- **Enclosure & Strap:** Using durable polycarbonate or a blend, and a silicon/nylon strap. For prototyping (3D prints, etc.) maybe \$10–15 each (very high due to low volume). At mass production (injection molding), the casing might cost ~\$2–3 and the strap \$1–2. Additional for any steel reinforcement in strap maybe \$0.5. If we do decorative finishes (various colors), minimal cost difference at scale.
- **PCB & Assembly:** For prototypes, low volume PCB fab and assembly can be expensive (~\$20–30 per board when doing small run). For 10k units, the PCB (4-layer, small size) might be ~\$1 each, and assembly cost perhaps \$1–2 (given relatively high component count, and including testing overhead). We'll estimate around \$3 for PCB + assembly in volume. We should also factor the testing/calibration time (maybe \$0.5–\$1 per unit in mass prod for test labor/equipment amortization).

Summing up **estimated BOM cost** per unit:

- Prototype (very low quantity hand-built or small batch): MCU (\$20) + camera (\$8) + memory & sensors (\$6) + display (\$3) + battery (\$4) + misc (\$5 for PCB, etc.) = roughly **\$46**. Plus enclosure which might be expensive in prototype (maybe milled or

3D printed ~\$15), totaling ~\$60. So prototypes might be \$60–\$80 each. If we need to factor NRE (non-recurring engineering) costs like mold creation, that’s an upfront cost (a steel mold for injection could be \$5k–\$10k), but per unit it’s amortized in volume.

- 10k Unit Mass Production: MCU (\$12) + camera (\$4) + memory & sensors (\$5) + display (\$1.5) + battery & PMIC (\$2.5) + PCB & assembly (\$3) + enclosure/strap (\$4) + misc and margin for scrap/testing (\$2) = approximately **\$34** per unit BOM. Let’s say around \$30–\$35 as a target.

A BOM of \$35 means manufacturing cost (including factory margin) might be \$40. Typically, retail price might be 2–3x BOM to cover R&D, overhead, distribution, and profit. If 2x, that’s \$70 (₹5,600); 3x would be \$105 (₹8,400). We will aim for a retail price around **₹6000–₹7000** (approximately \$80–\$90) initially for early adopters or at low volume. With higher volumes, we could reduce price. For context, premium smartwatches cost much more, but basic wearables cost less – however, ours has unique capabilities which can justify a moderate price point. Additionally, if government subsidies or CSR sponsorship come in, the effective cost to end-user can be lowered. We will prepare a **BOM forecast table**:

Component	Prototype Cost (USD)	Mass Prod (10k units) Cost (USD)
Secure MCU + LTE/GPS modem	\$20	\$12
Cellular/BLE Antennas	\$1	\$0.3
Camera module	\$8	\$4
Audio (Mic + Speaker)	\$1	\$0.8
Sensors (IMU, HR, Mag, IR)	\$4	\$2
Memory (Flash/RAM)	\$2	\$1
Display (OLED)	\$3	\$1.5
Battery & PMIC	\$4	\$2.5
PCB & Assembly	\$5 (hand assembly)	\$3 (SMT automated)
Enclosure & Strap	\$15 (3D print/CNC)	\$4 (molded plastic, bulk)
Subtotal BOM	~\$63	~\$31–\$35
Testing, Certification (per unit amortized)	N/A	~\$2 (for volume, includes certification amortization)
Total per unit cost	~\$70–\$80 (proto)	~\$35–\$40 (at 10k units)

(Note: These figures are illustrative. Actual costs will vary based on final part choices and supplier negotiations.)

We see that at scale, the most significant cost drivers are the SoC and the quality sensors (camera, etc.). Fortunately, these benefit from Moore's Law and volume discounts. For example, the BOM analysis of a high-end LTE smartwatch (Google Pixel Watch) showed about a third of its ~\$123 cost was in processing and communication[counterpointresearch.com](https://www.counterpointresearch.com)]. Our design, being more focused and not having expensive OLED or fancy CPU, comes out cheaper. Also, Pixel Watch's display and sensors were a big cost chunk[counterpointresearch.com](https://www.counterpointresearch.com)]; we have a simpler display and similar sensors so our cost is lower.

BOM Reduction Strategies: As adoption grows, we can further optimize:

- Use of **custom SoC** in future (if volume hits, say, 100k units, we could consider a custom integrated chip that combines all functions which could cut costs by 20–30%).
- **Shared components:** The hidden-cam detector's RF function might be achieved by reusing the LTE antenna and radio in a sniffing mode, eliminating a separate RF module – this saves a few dollars.
- Simplifying display or removing non-essentials: If market feedback says display isn't needed, we could remove it, saving cost and battery.
- Battery size vs cost: a slightly smaller battery (400 mAh) could shave \$0.5 and still meet one-day life, but we chose a margin.
- Plastic vs metal: If we considered a metal case, cost would rise – so we stick to durable plastic for now.

Prototype and Pilot Costs: For initial 50–100 prototypes (for testing, certification, pilot users), the per-unit cost including development overhead might be very high (~\$200+ if we include engineering effort, tooling, etc.). But those are R&D expenses. We have to invest in certifications (one-time costs: BIS testing ~\$2000, TEC maybe \$1000, etc.), tooling (mold \$8000), initial app development, etc. We amortize these over units; at 10k units, those overheads might add a few dollars each. If we produce just 1000 units, overhead per unit is larger. Hence scaling to 10k significantly helps spread out NRE costs.

We plan to start with a **pilot batch (maybe 500 units)** manufactured – cost might be around \$50–\$60 each at that stage. If we price them at say ₹10,000 (\$125) for pilot customers (who often are early adopters or provided via grants), we can cover costs and gather feedback. As we streamline manufacturing, we target cost reduction and price drops.

Bill of Materials vs. Price Forecast:

- Prototype stage: BOM ~\$70, likely price n/a (not sold, just development).

- Initial small-scale production (1k units): BOM maybe ~\$45 (some volume discount, but not full 10k discount), expecting to price around ₹8,000–₹10,000 to cover costs and channel margins.
- Mass production (10k+): BOM ~\$35–\$40, enabling a retail price around **₹6,000** (approx \$75) if sold direct or maybe ₹7,500 (\$95) if through retail channels that take margin. This price range, while higher than basic fitness bands, is justified by the advanced safety features – and still relatively affordable for a high-quality smartwatch. We also expect that as volumes climb, costs could go down another 10–15%. If government subsidies a portion (for target student groups for example), the effective price to consumer could drop further.

Additional Costs to Consider:

- **Cloud service costs:** Each device will use cloud resources (data storage, SMS, etc.). We should factor an estimated cost per user per year (maybe a few dollars, especially for SMS/call notifications). We might offset this by a subscription or usage-based model if needed (or include a couple of years of service in product price).
- **Warranty and Service:** We will allocate some cost for warranty replacements (assume maybe 2% failure rate, that cost needs to be covered in margin).
- **Packaging and Logistics:** Packaging a smartwatch (box, manual, charger cable) might be ~\$1–\$2 per unit. Shipping to distributors or warehouses adds some cost. These are usually folded into either BOM or overhead.

In BOM forecasts, usually at 10k units we get good scale, but the real sweet spot might be 50k or 100k units where component pricing drops another tier. If the product sees widespread adoption (imagine being adopted in safety programs nationwide), we could see costs approach maybe \$25 BOM at 100k units (due to better pricing and maybe integrating more functions into one chip).

For now, with 10k as a reference, we have a viable cost structure. To ensure viability, we'll seek partnership with manufacturers to keep assembly costs low (maybe assembling in India under Make in India to get duty benefits). Also possibly negotiate with telecom operators for bulk SIM provisioning at low cost.

Comparative Note: A fully featured Apple Watch has a BOM of ~\$130 [counterpointresearch.com](https://www.counterpointresearch.com)], but that includes expensive AMOLED display and powerful processors for general apps. Our device is more specialized – akin to a feature phone level complexity with some extras – so its BOM is much lower. In fact, \$35–\$40 BOM at scale aligns with mid-tier smart bands. This indicates our cost target is realistic.

In conclusion, the BOM analysis suggests that the safety smartwatch can be produced at a **reasonable cost** relative to its life-saving value. With projected economies of scale and potential external funding support, we can strike a balance where the device is **affordable to**

the target audience while sustaining the business and continuous improvement. This financial viability is crucial for large-scale social impact – it means we can reach more users, iterate on the design, and perhaps even introduce lower-cost variants in the future (for instance, a version without cellular that pairs with phone at a lower price for those who always carry a phone). The cost forecast will be continuously refined as we move from prototype to production and get real supplier quotes. The goal is clear: deliver maximum safety feature punch at a price point accessible to as many women as possible.

Go-to-Market Roadmap

Developing the smartwatch is only half the battle; ensuring it reaches and effectively serves the intended users is the other half. Below is a phased **go-to-market (GTM) strategy** outlining how we plan to pilot, launch, and scale the product in India, leveraging partnerships and possibly subsidies to maximize impact:

Phase 1: Pilot Programs and User Testing (Months 1–6)

Before a full commercial launch, we will conduct **targeted pilot deployments** to validate the device in real-world conditions and incorporate feedback. Key steps in this phase:

- **Small-Scale Pilot Deployment:** Partner with a local institution, e.g., a university or a large corporate office with women employees, to recruit an initial pilot user group (perhaps 100 users). We would provide devices (possibly at a subsidized or free pilot cost) and closely monitor usage (with consent). For instance, a college in a metro city could let 50 female students and 50 staff use the watch for a semester. This will test technical performance (battery life, connectivity across the city), and user experience (how intuitive the device/app is, any false alarms, did it actually help in any incidents).
- **Collect Feedback & Refine:** Through surveys, focus groups, and analyzing device logs (anonymized), gather insights: Did users feel safer? How often did they use features like camera scanning? Any difficulties (e.g., pairing issues, button confusion)? Did emergency contacts respond effectively? This feedback loop will allow us to push firmware/app updates to fix issues before mass launch.
- **Field Test with Law Enforcement (optional):** Collaborate with a city police department (maybe via their women's safety cell) to trial integration. For example, 10 devices given to volunteer women in a high-risk area with police aware of alerts. This tests our backend's ability to forward alerts to police or how police respond to user's contacts. It also builds credibility with law enforcement early on.
- **Technical Validation:** Use pilot to finalize certification (if not already done). Also, test our cloud alert system capacity with simulated multiple simultaneous alerts to ensure scalability (like run drills where many pilot users trigger SOS at once).
- **Stakeholder Buy-In:** Document success stories from the pilot. E.g., "Student A used the watch to alert hostel guards when harassed by outsiders, preventing an incident." These real anecdotes and data will be powerful in convincing stakeholders in the

next phase.

Phase 2: Soft Launch and Partnerships (Months 6–12)

With a validated product, we proceed to a controlled launch focusing on partnerships and early adopters:

- **Corporate and Institutional Partnerships:** We approach companies (especially BPOs, IT firms with late-night shifts for women), educational institutions, and government schemes. The pitch: they can enhance their women's safety programs by adopting our device. For example, an IT services company might bulk-purchase 500 watches for their women who use late-night cab services, tying into their CSR or duty of care. We could partner with universities to offer devices to first-year female students at a discount (perhaps covered by the University's safety budget or alumni fund).
- **Government Collaboration:** Engage with bodies like the Ministry of Women and Child Development, state police departments, or the Nirbhaya Fund administrators. The **Nirbhaya Fund** (a government fund for women's safety) could subsidize projects that deploy safety devices. For instance, we could propose a city-wide pilot funded by the Nirbhaya Fund to give, say, 1000 devices to women in public transport jobs (conductors, etc.) or to college students in that city. Government backing would lend legitimacy and reduce costs to users.
- **Telecom Partnerships:** Work with telecom operators (Jio, Airtel) to bundle connectivity. For example, a **built-in eSIM with a special IoT data plan** could be bundled free for 1 year with the device. Operators benefit by attracting users and fulfilling CSR goals. They could also retail the device in their stores or online, effectively as another "connected device" offering. Additionally, telecoms could help integrate with emergency number 112 infrastructure.
- **Retail and Online Availability:** While initial distribution is partnership-focused, we also do a limited online launch (e.g., our own website and Amazon India) for those actively seeking such solutions. We list it under safety gadgets category, with clear value prop and possibly a launch promotion or EMI options (to make the ~\$80 cost more palatable). However, we might intentionally keep early consumer sales limited until we've ensured supply and support is ready for scale.
- **Awareness Campaigns:** Alongside soft launch, run awareness workshops and demos. For instance, safety workshops in colleges that not only talk about self-defense tips but also demonstrate the smartwatch. These can be done with NGOs or as part of college orientations. The idea is to educate the public that such technology exists. Press coverage from pilot success can be used: local news story "Smartwatch rescues woman – new tech in city's colleges" etc., will generate interest.
- **Customer Support Setup:** Anticipating growing user base, set up a dedicated support line (with perhaps a small team of trained support staff, including women who understand the context). Also prepare user guides in multiple languages. The support

line can help users with setup issues, feature training, and in case of any false alarm confusion (e.g., a contact might call us asking what to do if they get an alert – we should be ready to guide them).

Phase 3: Mass Market Expansion (Year 2 onward)

Once the product and processes are stable, scale up manufacturing and broaden distribution:

- **Full Consumer Launch:** Launch nationwide marketing targeting urban and semi-urban centers. This could include:
 - E-commerce push (list on Amazon, Flipkart, etc., possibly with “Fulfilled by Amazon” for logistics, and highlight in Women’s Day promotions).
 - Select physical retail partnerships, e.g., electronics retailers (Croma), or even pharmacies/health shops (some women might trust buying from places focused on well-being). For semi-urban penetration, tie-up with telecom retail outlets or post offices via government schemes for rural safety tech.
 - Utilize digital marketing with a focus on social media – for example, real stories dramatized in short videos showing how the watch works in a crisis, with a tagline like “Your Guardian on Your Wrist.” Influencer marketing could involve women safety activists or celebrities endorsing it as a must-have for women. Careful to keep tone positive (“with this device, she has one more reason to fear less” rather than fear-mongering).
- **Scaling Production to Meet Demand:** Based on early sales, ensure our supply chain scales. This may involve onboarding an EMS (electronics manufacturing services) partner in India for volume production. The BOM cost reductions discussed will kick in at this stage, possibly allowing a slight price drop or better margins that can be reinvested in scaling.
- **Subsidies & Bulk Programs:** Work with state governments or city councils to implement subsidy programs. For example, a state could announce that it will reimburse X% of cost for the first 1000 women in certain professions who buy the device. Or, integrate it with schemes like giving it free (or at nominal cost) to women who use government-run transport at night. We will lobby and present data to policymakers on how this device can complement their women’s safety initiatives (there are many, from panic buttons on buses to CCTV networks; our device fills the personal gap).
- **Evolve Business Model:** Consider a subscription for premium services (optional). The base device works standalone, but perhaps we can offer a subscription that includes, say, a 24/7 monitoring service or integration with a professional security response center. Some users (or their families) might pay a monthly fee for that extra assurance. This model is used in some senior citizen emergency pendants (connected to call centers). For general users, community-based response might

suffice (contacts and police), but offering a tier where an emergency response team is on call could attract some.

- **Community Building:** Create a community of users – perhaps via the app or social media – where tips are shared (like where hidden cameras were found – to crowd-map unsafe locations, etc.). We must moderate to keep it constructive and privacy-conscious, but a community can improve engagement and word-of-mouth.
- **Continuous Improvement (Future Models):** As we scale, gather ideas for version 2 hardware (maybe with satellite SOS) or a simpler variant (cheaper model without camera for those who mainly want SOS). But ensure the current product gets proper support and software updates. Maintaining an innovation roadmap keeps the product relevant and ahead of possible competition (others may enter the market seeing our success).

Phase 4: Partnerships for Ecosystem & Long Term (Year 3+)

- **Integration with Smart City Infrastructure:** In the long run, as cities implement command centers, our platform could feed anonymized alerts into city dashboards to help allocate patrols to hotspots. We'd pursue MoUs with city police for data-sharing (in aggregate) to improve public safety measures.
- **Educational Institutions Adoption:** Make it common for colleges to include our device in their welcome kit for female students (perhaps negotiated at a bulk academic price). Similar for companies – akin to how some give health wearables for wellness, give safety wearables for women employees.
- **International Expansion:** If proven in India, explore other markets with similar needs (South Asia, Middle East for women's safety, or even Western markets focusing on general personal safety or teen safety). Adapt to local regulations and integrate with local emergency systems. The Indian success story will help, but we approach that after solidifying home ground.
- **Metrics and Efficacy:** Work with researchers (maybe criminology or sociology departments) to study the impact of the device on user behavior and outcomes. If we can publish data like "Users of this device had X% faster emergency response and felt Y% safer according to surveys," it validates the concept and can influence policy (e.g., government grants could be more forthcoming with solid evidence).

Go-to-Market Summary: We start with **focused pilots** to refine the product, then leverage **partnerships** (corporate, institutional, governmental) for an initial rollout to those who need it most and to build credibility. Then we broaden to consumers with strong marketing emphasizing safety empowerment. We incorporate **subsidies and financing options** to overcome cost barriers. All along, we ensure robust customer support and iterate on feedback. A labeled block diagram of the *user journey flow* could be envisioned as an infographic: from **Awareness** (hearing about it via a workshop or ad) ->

Purchase/Onboarding (through a partner or online, setting up contacts) -> **Everyday Use** (wearing it, feeling secure) -> **Emergency** (device triggers, help arrives) -> **Post-event** (data used for action, user shares testimonial). Each step requires proper infrastructure (e.g., training at onboarding, reliability at emergency).

By executing this roadmap, our aim is that in a few years the safety smartwatch becomes as common and essential as carrying a mobile phone for many Indian women – an accessible, trusted companion that, alongside broader societal efforts, contributes to a tangible reduction in harm and an increase in women’s confidence to move freely.

Conclusion & Future Work

In this report, we presented a comprehensive blueprint for a women’s safety smartwatch – detailing the urgent **need**, the envisioned **features**, the underlying **architecture**, and the path to bring it to market responsibly. This device, blending wearable convenience with cutting-edge safety technology, has the potential to fill critical gaps in personal security for women and girls in India. By enabling instant distress signaling, evidence capture, and proactive threat detection (like hidden cameras), it empowers users to protect themselves and seek help in a way not previously possible.

The design is rooted in currently available technology, but we have kept an eye on the **future** throughout. Looking ahead 2–3 years, several emerging technologies and enhancements could further elevate the smartwatch’s capabilities:

1. Satellite-Based SOS Communication: One limitation of any mobile-connected device is cellular coverage. As satellite communication for consumer devices becomes more viable (we’ve already seen early adoption in some smartphones for emergency texting), a future version of our smartwatch could integrate a **satellite SOS feature**. This might involve a small L-band antenna and partnership with a satellite network (e.g., Globalstar or Iridium). In an area with no GSM/LTE signal – say a rural highway or a mountain trek – the watch could still send a basic distress signal via satellite, transmitting GPS coordinates to a central emergency service. By the time this tech is affordable for a wrist form factor (perhaps through emerging low-earth-orbit constellations offering IoT messaging), we will be ready to integrate it, ensuring the device is not bound by terrestrial network availability.

2. Advanced Anomaly Detection and Predictive Alerts: With more usage data and advancements in AI, the smartwatch could evolve from a reactive device to a more **proactive safety assistant**. For example, using the array of sensors (accelerometer, heart rate, microphone), the device could potentially detect patterns indicative of distress **even if the user hasn’t pressed SOS**. A concept for future exploration is a machine-learning model that monitors the user’s bio-signals and context – a sudden spike in heart rate and breathing, detected nervousness in voice tone, abrupt irregular motion – and if it matches a profile of an attack or panic, the device could *ask the user if they are okay* (via a vibration and gentle ping). If the user does not respond or cancels, it might trigger an SOS on their behalf. This is delicate (to avoid false positives), but with improved sensors and AI, it could add a layer of protection for cases where the user is incapacitated or unable to press the button. We dub this “**Guardian AI**” – essentially an algorithm that learns a user’s normal patterns and flags anomalies that suggest danger.

3. Integration with Public Infrastructure: In future, the device's capabilities (especially location tracking during SOS) could integrate with public safety infrastructure. For instance, cities might allow automatic forwarding of verified SOS alerts to the nearest police patrol vehicle or to CCTV control rooms (to turn cameras to that location). Also, as public panic button projects roll out (some cities put SOS buttons in streets or on public transport), the smartwatch could be integrated so that pressing the watch might also notify nearby public systems (through an API or Bluetooth beacon to a bus's system). This **connected safety net** vision could greatly reduce emergency response times.

4. Hardware Improvements: Anticipated advances in hardware can be incorporated:

- **Battery Technology:** Solid-state batteries or better energy-density cells could double the battery life without increasing size, or allow significant size reduction of the device. Alternatively, advanced energy-harvesting (solar on strap, or kinetic) could supplement battery – perhaps not to fully power it, but to extend time between charges in an eco-friendly way.
- **Miniaturized Sensors:** We might see even smaller and cheaper mmWave radar chips or thermal sensors that could be fit into the watch, enhancing features like hidden camera detection or even detecting approach of people in dark environments.
- **Biometric Authentication:** Perhaps a tiny fingerprint scanner on the watch face or side for quicker secure operations, or even continuous authentication by heartbeat pattern (research exists on using unique heart signatures to verify identity – the watch could ensure it's being worn by the owner).
- **Ruggedization:** Future iterations might tailor versions for different use cases, e.g., a more rugged version for rural/outdoor (with longer battery, satellite, and a built-in loud siren), versus a sleek urban version (with focus on discrete looks and perhaps pepper-spray integration – some concepts have put pepper spray in wearables).

5. Broadening the Safety Ecosystem: As the product gains traction, we foresee complementing it with related offerings:

- A **mobile app** that anyone (even without the watch) can use as a basic SOS (widening our user base) – with the watch offering premium capabilities when paired.
- Partnerships with ride-hailing services or public transit apps so that if a woman is taking a cab late night, her watch data could auto-alert a friend if something goes off-route or if she triggers SOS (integration via APIs).
- Data analytics service for city planners: an anonymized heat map of distress signals or detection of hidden cameras can inform where to increase security measures.
- A community-alert feature: optionally, a user in distress could allow an alert (perhaps without personal details) to be sent to other SafeWatch users in the vicinity, creating a community response (e.g., nearby volunteers coming to help, if deemed safe). This

needs careful curation to avoid vigilante issues, but it's a future concept aligning with the idea of **collective safety**.

In conclusion, the development of this safety smartwatch is not just a one-off engineering project but the establishment of a **platform for personal safety** that can continually evolve. The immediate design detailed in this report is feasible with today's technology and addresses a pressing need with a high degree of innovation (integrating video SOS and spycam detection in a wearable). The **impact** of such a device could be significant – from deterring harassment (knowing women might be recording and alerting others) to providing crucial evidence to convict perpetrators, and most importantly, giving women greater confidence and freedom in their daily lives.

As we implement this project, we will remain agile and receptive to emerging tech and user feedback, ensuring that the product roadmap aligns with both technological advancements and the real-world safety landscape. Our ultimate vision is a future where devices like this are *no longer needed* because society has become safer – but until that day, we aim to put the best of technology in the hands of women to safeguard their rights and wellbeing. The journey outlined – from concept to execution and continuous improvement – will require interdisciplinary collaboration, user-centric design, and ethical diligence, all of which we have endeavored to encapsulate in this blueprint.

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Safety Smartwatch for Women in India: A Design Blueprint

Executive Summary

This report proposes a cutting-edge **safety-focused smartwatch** for women and girls (ages 13–35) in urban and semi-urban India. It combines wearable IoT hardware, embedded AI, and cloud services to address pressing safety concerns. The device provides one-tap distress capture (recording evidence), instant SOS alerts with live location, and hidden-camera detection in a discreet wrist-worn form factor. Key features include:

- **One-Touch Distress Recording:** A panic button immediately triggers **video and audio recording**, encrypted on-device, and streams footage to a secure personal cloud vault for evidence preservation.
- **Instant SOS & Live Location:** An SOS button sends the user's **real-time GPS location** to up to five pre-selected emergency contacts and **automatically calls them sequentially** until someone picks up.
- **Hidden Camera Detection:** An integrated module scans for **covert surveillance cameras** in private spaces (washrooms, fitting rooms, hotels) using multi-techniques (IR lens reflection, RF signal scan, etc.), alerting the wearer to potential voyeurism threats.
- **Robust and User-Centric Design:** The smartwatch is **comfortable, tamper-resistant, and durable** (e.g. waterproof), with long battery life and intuitive operation. It employs end-to-end encryption, biometric access control, and adheres to India's latest data protection laws for user privacy.

The report details the **market need** (backed by crime data), target **user personas** and scenarios, a survey of the **competitive landscape**, and comprehensive engineering specifications. A **system architecture** is presented (hardware, firmware/AI, cloud backend, and mobile app), alongside an evaluation of hidden-camera detection technologies. We outline **data security measures** (full-stack encryption, compliance with the Digital Personal Data Protection Act 2023) and the **regulatory path** (BIS certification, WPC approvals, etc.). Finally, we analyze potential **risks and mitigations**, discuss **ethical considerations**,

provide a **Bill of Materials (BOM) cost forecast** from prototype to mass production, and propose a **go-to-market roadmap** (pilot programs, partnerships, and scaling strategies). Emerging technologies on the horizon (satellite SOS connectivity, AI-driven threat prediction) are also explored as future enhancements.

In summary, this blueprint serves as a comprehensive guide for designing and commercializing a “**smart safety watch**” that empowers women and girls with proactive personal security, while respecting privacy and regulatory frameworks.

Problem Statement & Market Need

Women in India continue to face alarming levels of harassment and violence, underscoring an urgent need for tech-enabled safety solutions. According to the National Crime Records Bureau (NCRB), **428,278 crimes against women** were reported in 2021 – a 15.3% increase from 2020[jurist.org](https://www.jurist.org). This included **31,677 rapes** (averaging 86 rapes daily) and over **89,000 cases** of assault intended to outrage modesty[jurist.org](https://www.jurist.org). Of particular relevance is the rise in **voyeurism (IPC Section 354C)** – e.g. clandestine filming – with **1,513 cases of voyeurism reported in 2021**, up ~20% from the prior yearendnowfoundation.org. Such incidents (from hidden cameras in changing rooms to videos of private acts) violate women’s privacy and safety on a large scale.

These sobering statistics highlight a clear **market need** for effective personal safety devices. While smartphone apps and helpline numbers exist, they often fall short in emergencies. In a high-stress attack scenario, a victim may not have the time or ability to unlock a phone and call for help. Moreover, critical evidence (audio/video) of the incident is usually not captured, hampering prosecutions. The NCRB data shows low conviction rates for sexual offenses, partly due to lack of real-time evidence and delayed reporting[jurist.org](https://www.jurist.org). There is a gap in the market for a **wearable solution** that can be activated instantly and discreetly to both **summon help and preserve evidence**.

The target users – urban and semi-urban women aged 13–35 – are tech-savvy and largely comfortable with wearables. Many already use fitness bands or smartwatches daily. A safety-oriented smartwatch can seamlessly integrate into their lifestyle while providing a much-needed safety net. **Use cases** range from a college student walking home at night, to a working professional in a taxi, to a teen in a shopping mall fitting room concerned about hidden cameras. Recent high-profile incidents (e.g. a hidden camera found in a Goa boutique’s trial room by a Union Minister) have made women acutely aware of such threats[opindia.com](https://www.opindia.com). Surveys indicate a significant percentage of women feel unsafe in public spaces or when traveling alone after dark, and families are seeking trusted tools to protect daughters.

In response, this project envisions a “**safety smartwatch**” that directly addresses these pain points. By **combining GPS tracking, cellular connectivity, and on-device recording**, the device can alert trusted contacts and authorities in real time while securely streaming incident data to the cloud. This both enables faster response (potentially preventing escalation) and creates a record of the event for evidence. The inclusion of **hidden camera detection** addresses the growing voyeurism problem by empowering women to scan their environments (hotel rooms, public toilets, etc.) for illicit surveillance devices.

From a market perspective, personal safety wearables form an emerging category with high social impact. Government agencies (e.g. police departments, women's ministries) have shown interest in technology-aided safety programs, such as installing panic buttons in phones or CCTV monitoring. A dedicated safety smartwatch could complement these efforts at the individual level. Parents of girls, educational institutions, and employers could be key market segments, especially if the device is proven effective and affordable. By leveraging **state-of-the-art IoT and AI**, this solution aims to reduce response times to incidents, deter potential perpetrators (through the knowledge that evidence is captured), and ultimately contribute to lowering the incidence of violence against women.

In summary, the confluence of **high crime rates** against women jurist.org, **technology readiness** of the target demographic, and **gaps in current solutions** creates a compelling case for a safety-focused smartwatch. The following sections detail how this device will meet the needs of at-risk women through careful design of features, technology, and safeguards.

User Personas & Threat Scenarios

To ensure the device is grounded in reality, we consider several **user personas and scenarios** that illustrate how the smartwatch would function in practice. Each scenario highlights a potential threat and how the device's features would mitigate it:

Persona 1: "Late-Night Commuter" – Nisha, 22, Urban College Student

*Nisha is walking home from her university library at 9 PM. A man begins following her down a dimly lit street. Feeling unsafe, Nisha discreetly presses the **distress-capture button** on her smartwatch twice. **Scenario:** The watch silently begins **recording video and audio** of the encounter from her wrist. Simultaneously, it sends an **SOS alert with her live GPS location** to her five emergency contacts (parents, two friends, an uncle, and a campus security number). Her location updates in real-time on the companion app her contacts have, and the watch starts calling them one by one. Her father picks up – he immediately sees an SMS with a map link to her location and hears the situation unfolding via the watch's microphone. As the follower approaches and grabs Nisha's arm, the father and friends are already alerted; one friend calls the police with Nisha's live location. The assailant, noticing the flashing LED on Nisha's watch (an optional visible deterrent), tries to snatch it – but by then the watch has **streamed the video/audio evidence to the cloud**. Police arrive within minutes (guided by GPS) and the assailant flees. Later, Nisha retrieves the encrypted recording from her personal cloud vault as evidence for the police report. *This scenario shows how rapid alert and evidence capture can **deter an assault and aid swift response**.**

Persona 2: "On a Business Trip" – Meera, 30, Software Engineer

*Meera travels to another city for work and checks into a hotel. She remembers news about hidden cameras found in hotel rooms. Using her smartwatch's **Hidden-Camera Detector**, she performs a quick sweep of her room. **Scenario:** She opens the detector mode on the watch, which uses a combination of **infrared LED scanning and RF signal detection**. As Meera slowly scans the room, the watch's sensor detects a suspicious reflection point behind the air conditioner vent. It also picks up a faint RF signal on a spycam frequency. The watch vibrates and quietly notifies her of a potential hidden camera. Meera uses her phone (via the watch app) to double-check by looking through the watch's IR filter view and indeed*

spots a tiny lens glint. She immediately calls hotel security and requests a room change, showing them the device's detection. In the new room, no alerts are triggered. *This scenario highlights how the device **protects privacy** by empowering users to find hidden cameras (voyeurism threats) before harm occurs.* If Meera had discovered an active spycam, she could also use the watch to record footage of it and initiate an SOS if she felt threatened by the situation.

Persona 3: “School Commute” – Aditi, 16, High-School Student

Aditi travels by public bus to school each day in a semi-urban town. There have been instances of teenage girls being harassed on public transport. Aditi's parents have given her the safety smartwatch for peace of mind. **Scenario:** One morning, a stranger on the bus starts harassing Aditi, making lewd comments and inching closer. Aditi, feeling scared, presses the **SOS button** on her watch for 3 seconds. The watch quietly sends out her **live location and a distress message** to her parents and three close relatives. It then starts calling her mother. Meanwhile, it also begins **audio recording** (given privacy, maybe not video in a crowded bus) so that the threatening comments are saved. The mother receives the alert and location, and immediately calls Aditi's phone to check. The sudden phone call and Aditi's firm response alerts other passengers, and the harasser backs off, getting off at the next stop. Later, Aditi's parents replay the audio clip from the cloud storage to understand what happened and decide on further action. *This scenario demonstrates a **youth-friendly use case**, where a minor can quickly call for help in a harassing situation, with the device facilitating immediate family intervention.* (Notably, for minors, the system would be set up with guardian consent and oversight, in compliance with data protection rules.)

Persona 4: “Solo Runner” – Kavya, 28, Early-Morning Jogger

Kavya goes jogging at 5 AM in a relatively quiet area. She wears the smartwatch with the **loud siren** option enabled. **Scenario:** During her run, two men on a motorcycle begin following her, catcalling. Sensing potential danger, Kavya **triggers the SOS**. In this mode, the smartwatch emits a **120 dB shrill alarm** (as loud as a rock concert speaker) to attract attention, similar to commercial personal alarms like Run Angel which use high-pitch sirens [markets.businessinsider.com](https://markets.businessinsider.com/markets.businessinsider.com). The deafening sound startles the men and likely wakes nearby residents. Simultaneously, her **location and an emergency alert** go out to her husband and running group. The men, seeing the commotion and realizing an alert has been sent, decide against approaching further and speed away. Kavya cancels the alarm once she reaches a safe location. *This scenario emphasizes the device's **deterrent effect** – by combining an audible alarm with instant alerts, it can repel threats before physical harm occurs.* It also shows user customization: Kavya chose an audible alarm, whereas others might prefer silent mode if stealth is needed.

These personas underscore the versatility of the proposed smartwatch across different threat scenarios – from physical assault to voyeurism. Common threads include the need for **immediate action, stealth or deterrence, and reliable communication**. In each case, the smartwatch provides a tailored response: recording evidence quietly, notifying loved ones and emergency services, drawing public attention with an alarm if appropriate, and detecting hidden dangers. Real-world testing with such personas would be crucial to refine the user experience (e.g. ensuring the activation method is easy under stress but not prone to false

triggers). By designing with these scenarios in mind, we ensure the device remains **user-centric and effective** in the moments that matter most.

Competitive Landscape

Several products and technologies already address women's safety, ranging from mobile apps to dedicated wearables. However, most existing solutions cover fragments of the problem, and **none integrate all the core features** (distress recording, SOS alerts, hidden camera detection) in one device. Below is an overview of the current landscape and how the proposed smartwatch compares:

- **Safety Jewelry and Panic Buttons:** A number of startups offer panic-button devices disguised as jewelry (rings, pendants, keychains). For example, the **Leaf Wearables "Safer" device** (India, ~2015) is a Bluetooth-enabled pendant that, on double-tap, sends an alert with the wearer's GPS location to guardians via a paired smartphone designindaba.com. Safer introduced features like a 30-second cancel window to prevent false alarms and even a "SaferWalk" tracking mode designindaba.com. Similarly, devices like **Nimb Ring** and **Athena pendant** (by ROAR for Good) allow discreet SOS activation. *Limitations:* These devices **rely on a connected smartphone** for communication (they themselves don't have cellular connectivity or recording ability). They typically do not capture any audio/video evidence; their function is limited to sending an alert. Our smartwatch improves on this by being a **standalone communicator with built-in recording**. It does not depend on the user's phone to call for help, which is critical if the phone is inaccessible or taken by an attacker.
- **Personal Safety Alarms (Sound-based):** **Run Angel** (Ireland, launched 2017) is an example of a wearable focused on audible alarm + alerts. It's a wristband that emits a **120 dB siren** when activated and sends SMS/email alerts with location to pre-selected contacts via a phone app markets.businessinsider.com. Its unique acoustic chamber design maximizes the piercing sound to attract attention markets.businessinsider.com. Another example is **She's Safe** (a small keychain alarm) which doesn't send alerts but just makes noise. *Limitations:* Loud alarms can deter attackers in some cases, but they also **sacrifice stealth** – an attacker might quickly attempt to silence a device making noise. Additionally, an alarm alone doesn't notify specific people or record evidence. Our design offers an **optional siren mode** (for those who want that deterrence) but crucially pairs it with silent emergency communication. It can raise a racket *and* message help, or remain completely silent while still summoning help, depending on user preference.
- **Smartphones and Apps:** Most people carry smartphones with built-in emergency features (e.g. the Android and iOS Emergency SOS, which can call emergency services and share location when the power button is pressed multiple times). There are also safety apps like "bSafe" and government apps like "112 India". *Limitations:* Phones are versatile but **not wearable** – in many incidents the phone may be in a

bag or pocket, making it hard to access quickly or discreetly. Attackers often target the phone first to isolate the victim. Phones also typically do not automatically record video on SOS (due to privacy concerns and technical constraints). The smartwatch complements the phone: being body-worn, it's *immediately accessible* and can work even if the phone is out of reach or dead. It also specifically features automatic evidence capture which most phone SOS apps lack.

- **Smartwatches (general-purpose):** High-end smartwatches like the **Apple Watch** and **Samsung Galaxy Watch** have some emergency features. For instance, Apple Watch can detect hard falls and call emergency contacts, and its Emergency SOS feature will call local emergency number (112 in India) if the side button is long-pressed. Some also share location with emergency contacts. *Limitations:* These are generic features not tailored to assault scenarios – they do not record video or audio, and they do not detect hidden cameras. Moreover, these watches are expensive and not designed with the sole purpose of women's safety in mind (battery life may be short when GPS/call is active, and privacy of data is not E2E encrypted by default). Our proposed device is **mission-specific** – prioritizing safety functions above all. For example, we allocate dedicated battery reserve for emergency events, and we encrypt all recorded data by design. Additionally, a specialized hidden-camera detector is a novel feature not found in Apple or Samsung devices.
- **Existing Women's Safety Wearables (IoT prototypes):** A number of academic prototypes and hackathon products exist in literature. Many use a combination of Arduino/Raspberry Pi with GSM modules and sensors (heart rate, accelerometer, etc.) to detect distress and send SMS. For example, one IoT-based women safety system uses an Arduino Uno, GPS, and GSM module to send an alert when a button is pressed [researchgate.net](https://www.researchgate.net). Another concept ("Smart Belt" or "electroshock jackets") attempted to include self-defense mechanisms. *Limitations:* These prototypes are often bulky, proof-of-concepts and not commercially available. They demonstrate interest in multi-sensor approaches (e.g. detecting elevated heart rate as sign of fear), but have not integrated cloud video streaming or camera detection. Our design builds on these ideas with a far more **integrated and wearable form factor**, leveraging modern components (e.g. NB-IoT modem, mini camera, advanced sensors) that were not readily available a few years ago.

Competitive Summary: The safety smartwatch will effectively **combine the strengths** of these existing solutions while addressing their gaps. Like Safer and Run Angel, it alerts loved ones with location – but it does so without needing a phone and also captures evidence. It offers a loud alarm option akin to personal alarms, but also a silent mode which others lack. It goes beyond pure SOS devices by including **preventive scanning for hidden cameras**, tapping into a **unique market differentiator** (privacy protection in everyday scenarios, not just emergencies). Moreover, from a privacy standpoint, our device employs **end-to-end encryption and user-controlled data**, whereas many existing solutions don't emphasize data security (for instance, some apps send data unencrypted or share data with third parties, which raises concerns).

In terms of pricing, current safety wearables like smart rings or pendants retail between ₹3,000–₹10,000 (USD \$40–\$130). The addition of a camera and cellular module in our device may raise costs, but as discussed later in the BOM section, we aim for mass-production cost efficiency (possibly subsidized by government or CSR initiatives to reach students and low-income users). Given its comprehensive feature set, the smartwatch can be positioned as a **premium safety device**, but the social need may justify innovative pricing models (e.g. installment plans or bundled telecom offers).

Finally, it's worth noting that while competition exists in pieces, **no single product currently in the Indian market offers the holistic safety net** that we envision. This presents an opportunity for first-mover advantage in a new category of "safety smartwatches." The strategy will involve not just selling a gadget but also educating users and stakeholders (families, institutions) on its life-saving potential, differentiating it clearly from generic wearables and panic buttons.

Detailed Functional Requirements

To meet the needs identified, the smartwatch must satisfy a robust set of **functional requirements**. These cover performance, usability, safety, and reliability criteria. Below is a breakdown of the key requirements:

1. Instant Distress Capture & Alerting: The device should initiate recording and alert transmission within **1–2 seconds** of the panic button press. This means the **camera** should wake from standby and start recording video (with at least 720p resolution at 30fps) almost immediately, and the first alert message (with GPS location) should be sent out **under 5 seconds** from trigger. Low-latency operation is critical so that even a sudden attack is captured and help is summoned in real-time.

2. Battery Life: The watch should function through a typical day without recharge. Target **battery life** is *at least 24 hours* in normal smartwatch use (notifications, periodic sensor use) and minimum **1 hour of active emergency use** (continuous video streaming + calling). In pure standby (watch idle with screen off), it should last **3–5 days**. This ensures the device is reliable during long outings. A **battery reserve** concept can be implemented – e.g. always keep 10% battery reserved exclusively for emergency mode (so the watch can still perform an SOS even if main battery is low). Fast charging to 50% within 30 minutes is also desirable for quick top-ups.

3. Ergonomics & Form Factor: The device must be **compact and comfortable** for all-day wear, similar to a regular smartwatch. Target weight < 50 grams, thickness ~12mm or less. It should use a **soft, adjustable strap** suitable for young girls and adults. The design should be discreet – not obviously a "safety device" to avoid tipping off an attacker. It can look like a sleek digital watch or fitness band. The **interface** for the user will be minimal: perhaps an OLED dial for time and status, and two subtle buttons (one for SOS, one for video capture). These buttons should be distinguishable by touch and easy to press even under stress (perhaps slightly protruding or textured). Haptic feedback (vibration) on button press is needed to assure the user it's triggered.

4. Durability & Environmental Resistance: Women will wear this device daily, so it needs to handle various conditions. It should be **water-resistant (IP67 or better)** so that it's safe from rain or sweat and can even survive accidental submersion (e.g. if the user is pushed into water, it should still send an SOS). It should also be **dust-proof** (for semi-urban, dusty environments) and **shock-resistant** against drops (meeting at least a 1m drop test on hard surface). Operating temperature range should cover 0°C to 45°C (since in India summers can be hot and device might be exposed to sun during outdoor use).

5. Reliable Connectivity: The watch must have **built-in cellular connectivity** (LTE-M and/or NB-IoT) to send alerts directly, independent of a smartphone. It should also have **GPS** for accurate location. In addition, **Bluetooth Low Energy (BLE)** is required to pair with the user's smartphone for secondary connectivity (and to sync data or adjust settings via the companion app). The system should intelligently choose the connectivity: e.g. if the user's phone is nearby and has internet, the watch might relay alerts via the phone (BLE to phone to internet) to save watch battery; otherwise it switches to direct cellular. The design should support a **nano-SIM or eSIM** for cellular. The device must be able to make a phone call (voice call) or at least VoIP call via LTE – this is needed for sequentially calling emergency contacts (or directly calling 112 if configured). It might use VoLTE or a voice-capable module.

6. GPS Accuracy and Tracking: The GPS receiver should be able to get a location fix within ~5 seconds outdoors and <30 seconds indoors (using AGPS or cell/Wi-Fi assist if needed). Location accuracy of ~5–10 meters is ideal. The device should continuously update location to the cloud during an ongoing SOS situation (e.g. every 5 seconds). If GPS signal is unavailable (e.g. indoors), the system should fall back to coarse location via cell tower or last known location, and inform contacts accordingly.

7. Distress Recording Quality & Storage: The **camera** should have adequate resolution (preferably 1080p HD) with low-light capability (since many incidents happen in poor lighting). It should have a wide field-of-view (e.g. 120°) to capture surroundings even if the wrist is at an angle. The **microphone** should capture clear audio within a 5-meter radius. The device should **encrypt and store footage locally** (e.g. in internal flash memory) in case connectivity drops. A minimum of **15 minutes of video storage** in offline mode is required (e.g. 1GB flash storage for video buffer). If the incident lasts longer, the device should overwrite oldest footage (circular buffer) while trying to regain network to upload. Video should be time-stamped and maybe watermarked to prevent tampering (for evidentiary integrity). A low-power **video codec** (H.264/H.265 encoder in hardware) is needed to compress video in real-time without draining the battery or CPU too much.

8. SOS Alert Mechanism: Upon SOS activation, the watch should handle a sequence of actions: send initial text alerts (with location) to contacts, *and* start phone calls. The requirement is that at least **five contacts** can be notified. The calling logic might be: call Contact 1; if no answer in 10 seconds, call Contact 2, and so on, looping back if needed. If a contact answers, they should be able to hear the ambient audio from the watch (acting like a speakerphone) so they can assess the situation. Optionally, the system can also be configured to call the national emergency number (112) after contacting personal contacts. The device should also allow a **silent SOS** mode configuration – where it does all the above without any sound or light, versus an **audible SOS** where it also triggers a loud buzzer. Users can select their preferred mode in the app. A **cancel feature** is needed to avoid false

alarms: e.g. if the button is pressed accidentally, the user can press and hold for 3 seconds within a 30-second window to cancel the alerts (and a notification would go out that it was a false alarm). This is similar to the approach used by Safer and Apple's SOS to mitigate accidental triggers designindaba.com.

9. Hidden Camera Detection: The watch should have a dedicated mode to scan for hidden cameras in a room. Functional requirements for this mode: It should detect common spycam types within a range of about **5–10 meters**. The scan should take no more than a couple of minutes for an average room. Ideally, it uses **multiple detection methods** (detailed in Section 7) – e.g. actively shining IR and looking for lens reflections, and scanning for RF signals emitting from camera transmitters. The interface could be via the mobile app for usability (e.g. user walks around with watch, and the app shows a “signal strength” or beeps faster when suspect object is near). A simpler on-watch UI might use an LED indicator or vibration patterns. The requirement is that it should reliably spot a pinhole camera or illegal CCTV with at least **90% detection rate** under typical conditions (aligning with research like CamRadar which achieved ~93% detection in tests cse.buffalo.edu). False alarm rate should be low (<5%) so that random electronics (routers, motion sensors) aren't constantly flagged – multi-sensor confirmation can help (requiring both an optical glint and an RF signal, for instance).

10. Tamper and Removal Detection: The device should detect if an attacker tries to **remove or disable** it. For example, if the strap is forcibly opened or cut during an active SOS, the device should sense that (perhaps via a small switch or sudden loss of skin contact from the heart-rate sensor) and it could send a final **“Device removed” alert** with location. It could then continue transmitting audio/video until destroyed. Similarly, if someone tries to **power it off during SOS**, it should ignore the power-off (perhaps require a specific pattern or biometric to shut down in SOS mode). The watch could also have an **auto-SOS trigger if removed without authorization**: e.g. if someone snatches it off the user's wrist violently, it could treat that as a distress event and send alerts on its own. These anti-tamper features ensure the device *cannot easily be silenced* by a perpetrator once activated.

11. Biometric Access & Privacy: For non-emergency interactions, sensitive functions should be locked behind a biometric. For instance, accessing the stored videos on the watch or app, or stopping an ongoing recording, could require the user's fingerprint or a secure PIN. The watch could include a small **fingerprint sensor** (perhaps integrated into the side button) for the user to authenticate. This prevents anyone else from retrieving or deleting the data (e.g. if the watch is stolen, the footage remains inaccessible without the biometric). Additionally, pairing the watch with a new phone or disarming safety features might require biometric authentication to prevent misuse by others.

12. Data Security: All data (recordings, location logs, contact info) on the device and in transit must be **securely encrypted**. This is detailed later, but functionally the requirement is to use strong encryption (AES-256 or similar) for local storage and TLS 1.3 for any network communication. The device should generate its own cryptographic keys (with a secure element if possible) and not rely on obscure security through obscurity. Only authorized users (the wearer and explicitly allowed persons) should be able to decrypt/view the content. This ensures compliance with privacy laws and protection against hackers if the data is intercepted akitra.com.

13. Companion Mobile App (User Interface Requirements): A smartphone app (Android/iOS) will accompany the watch. Functional needs for the app:

- Onboarding: allow the user (or guardian) to configure emergency contacts, preferred SOS mode (silent/audible), and any personalized message for alerts.
- Dashboard: show device status (battery, signal, last known location), toggle features like “camera detection mode”.
- Live Tracking: in case of an SOS event, the app (for the wearer or designated guardians) should show the live location trail and possibly a low-frame-rate live video feed from the watch (if bandwidth permits).
- Data Access: allow the user to review incident recordings saved to cloud, which would be **encrypted** and require login + biometric to view. They can download and share these with police as needed.
- Health Monitoring (nice-to-have): Because it’s a smartwatch, it could also show step count, basic fitness metrics – but these are secondary to the core safety purpose. If included, ensure they do not compromise battery needed for safety.
- Firmware Updates: the app should deliver OTA firmware updates to the watch for improvements and security patches.
- **Security:** The app itself should be PIN/biometric protected, and communications between app and watch encrypted (e.g. BLE bonding with encryption).

14. Performance and Responsiveness: The device should feel responsive – button presses acknowledged via haptics in <0.5s. The hidden camera scan mode should give some feedback (progress or signals) in real-time to guide the user. The cloud alert system should be able to handle simultaneous alerts if an incident is triggered by multiple users (scalability). The system must be **highly available** (the cloud service especially), since a down server could mean a cry for help goes unheard – multi-region cloud deployment or failover via SMS gateway can be considered (e.g. if data network fails, fall back to sending SMS alerts directly from the device if possible).

15. Regulatory Compliance: The watch must meet regulatory standards (detailed later) as a functional requirement:

- It must cap its SAR (specific absorption rate) within permitted limits for a wearable device (since it has a cellular radio on the wrist).
- It should have proper fallbacks for emergency calling as per telecom regulations (e.g. if 112 is dialed, it should connect even without SIM balance or if the device is locked).
- Localization: default language support for English and local Indian languages for notifications (e.g. SMS to contacts could be in English with option for Hindi, etc., for

accessibility).

By fulfilling these detailed requirements, the smartwatch will be **robust in real-world use**. Each requirement was chosen to either enhance **usability under duress**, **reliability of the safety functions**, or **trust in the device** (through security and compliance). For example, long battery life and waterproofing ensure the watch is *actually on and working when needed*. Instant alerting and recording ensure *no time is lost* in an emergency. Encryption and biometric locks ensure the user's data and privacy are *never compromised*, aligning with the principle that a safety device should not create new vulnerabilities.

These requirements will guide the engineering and design decisions in the system architecture, described next. We will see how hardware and software components are selected or designed to meet or exceed the above benchmarks.

System Architecture

The system architecture of the safety smartwatch encompasses multiple layers – from the hardware components in the device, to the on-device firmware (including AI algorithms), to the cloud backend and mobile app that together deliver the end-to-end functionality. This section presents the architecture in an integrated manner, with a **block diagram** for visual reference and detailed explanations of each subsystem.

cse.buffalo.edu/markets.businessinsider.com *Figure: Conceptual system architecture of the safety smartwatch.* The diagram illustrates key hardware modules inside the watch (sensors, MCU, connectivity), the data flows (video/audio going to the cloud vault, alerts going to contacts and the mobile app), and the major software components (edge AI on the device, cloud services, and user application). This architecture is designed for scalability, low latency, and security.

Hardware Architecture (On-Device)

At the heart of the smartwatch is a **Secure Microcontroller Unit (MCU)/System-on-Chip (SoC)** that coordinates all functions. We propose using a low-power SoC that integrates both application processing and cellular connectivity – for example, the Nordic nRF9160 (which has an Arm Cortex-M33 MCU + LTE-M/NB-IoT modem + GPS) or a similar IoT-focused chipset. This MCU runs the embedded firmware/OS and handles sensor data acquisition, encryption, and communication. It should have hardware security features (ARM TrustZone or an isolated secure element) to store cryptographic keys and perform encryption.

Sensors and Modules connected to the MCU include:

- **Camera Module:** A mini camera (e.g. 8MP CMOS sensor) embedded at the watch's top or side. It's connected via a digital interface (like MIPI CSI or DCMI parallel interface) to the MCU or an image co-processor. The camera has a wide-angle lens (possibly fisheye) to capture a broad scene even when the wrist isn't pointed exactly at the target. It might be placed at an edge of the watch face to have an unobstructed view. In normal mode the camera is off (for privacy/battery), and it powers on only for

distress capture or when user explicitly turns on camera-detection scanning.

- **Microphone:** A MEMS omni-directional mic, likely placed on the side of the watch face (with a water-resistant mesh). It connects to the MCU's ADC or I2S interface. It should be always listening in a low-power mode if possible (to detect a specific hotword or scream in future AI enhancements), but primarily it activates during SOS recording or two-way calls.
- **Hidden Camera Detector Module:** This may be a composite of sensors:
 - An **IR LED array** (several small infrared LEDs around the watch face) and an IR-sensitive photodiode or use of the camera sensor in IR mode. These allow an active scan for lens reflections.
 - An **RF scanner** – essentially a broadband RF receiver that can pick up transmissions in common spycam bands (1.2 GHz analog, 2.4 GHz Wi-Fi, 5.8 GHz etc.). This could be done via a small software-defined radio front-end or repurposing the LTE radio when idle to scan frequencies. Another option is a tiny **mm-wave radar** (60 GHz) module which might detect electronics by RF reflection differences.
 - A **magnetometer** (already often present in smartwatches for compass) which could serve to detect magnetic fields from devices (though small cameras emit weak fields).
 - These sensors feed data to the MCU for analysis (e.g. the analog/digital readings from photodiode and RF front-end go into MCU via ADC or SPI interfaces).
- **GNSS Receiver:** For location, we use either the integrated GPS in the SoC (like nRF9160 has one) or an external GNSS module (supporting GPS and India's NavIC, plus GLONASS/Galileo for faster fixes). It connects via UART/SPI and provides latitude/longitude coordinates. It will be off most of the time to save power, activating during SOS or when user explicitly requests location (or for periodic check-ins if configured).
- **Accelerometer & Gyroscope:** A 6-axis IMU (inertial measurement unit) provides motion sensing. It's used for fall detection (if we choose to include), gesture recognition (perhaps double tapping the watch or a specific arm movement could trigger SOS if hands are free), and also tamper detection (a sudden forceful removal might register as a jolt). It also simply counts steps for fitness tracking. The accelerometer can be on to detect a strong shock (fall) and then wake the MCU.
- **Heart Rate / Skin Contact Sensor:** An optical heart rate sensor underneath could serve dual purposes: wellness (heart rate tracking) and security (detecting if the watch is on the wrist versus removed, by analyzing the signal or having a contact capacitance sensor). If the watch is torn off during an attack, the abrupt loss of heart

signal or contact can prompt the MCU to flag a tamper event.

- **Buttons:** Two hardware push buttons (one dedicated to SOS, one to video/manual trigger) are wired to GPIOs on the MCU with hardware interrupts for instant wake-up. They have internal debouncing to avoid false triggers from jostling.
- **Vibrator (Haptic Motor):** For silent feedback and alerts, a small eccentric motor or linear resonant actuator is driven by the MCU (via a transistor or dedicated motor driver). It confirms user input (buzz on button press) and can silently alert the user (e.g. a subtle buzz when a hidden camera is detected or when SOS is activated in silent mode).
- **Speaker/Buzzer:** A tiny speaker can be included for the siren and for two-way audio (so the user can hear the contact's voice or an operator on a call). The speaker needs to output very loud sound for the alarm – possibly up to 90-100 dB. A piezo buzzer could be used for the siren specifically because it can be very loud and is simple to drive. For voice, a regular micro speaker is needed. They connect to the MCU's DAC / audio amp.
- **LED Indicator:** A multicolor LED can indicate device status (charging, SOS sent, etc.). In camera-detect mode, a red LED could blink to help with scanning (some detectors use a blinking red light to reflect off lenses [eufy.com](https://www.eufy.com)). The LED is optional and kept minimal to maintain stealth.
- **Power Management & Battery:** The watch will have a rechargeable Li-ion or Li-po battery (~500 mAh capacity). A Power Management IC (PMIC) handles charging (via a magnetic dock or USB-C connector), battery fuel gauging, and providing regulated voltages to components (e.g. camera at 2.8V, MCU at 3.3V, etc.). The design must isolate the power supply for the RF modules to reduce noise during scans. Also, in an SOS event, the PMIC should allow the system to draw peak currents for the cellular radio (LTE transmit bursts can draw >500mA). The PCB will include an antenna for LTE (tuned to relevant bands, likely a flexible PCB antenna around the strap) and for GPS (possibly a ceramic patch antenna on top). The Bluetooth shares the same antenna as LTE (if using a combo antenna) or a separate small antenna.

Internal Storage: The MCU/SoC will have some internal flash, but likely an external **NOR flash memory (e.g. 2GB)** is attached via SPI/QSPI for storing encrypted video/audio data and any ML models. This flash must be encrypted (the MCU can do on-the-fly encryption or store data already encrypted). If the MCU has limited RAM, an external SDRAM might be needed for buffering video frames during encoding.

In summary, the hardware is a compact assembly of a central secure MCU with **sensors for environment (camera, mic, RF), sensors for context (GPS, accelerometer), user interface elements (buttons, vibrator, speaker), and communication modules (cellular, BLE)**. This is quite complex, but modern components and careful integration (e.g. using a highly integrated SoC) can achieve it in a smartwatch-sized device. A **system block diagram** was provided (Figure above) to visualize these connections, showing how

Camera/Mic, Buttons, GPS, etc. all feed into the MCU, which then connects out to the Cellular modem and BLE module, etc., and finally to cloud/app[50†].

Firmware & Edge AI (On-Device Software)

The smartwatch runs embedded software that coordinates all safety functions with **real-time performance**. The firmware will likely be built on an RTOS (real-time operating system) such as FreeRTOS or Zephyr, or a lightweight Linux if the SoC is powerful enough (though for low-power and instant-on, an RTOS on MCU is preferred). Key components of the firmware include:

- **SOS State Machine:** Logic to handle an SOS event – reading the GPS, forming the alert message, initializing the cellular modem, and transmitting the data. It also manages the sequential calling: interfacing with the modem to dial numbers and play audio. This module will also control the recording sub-system (starting/stopping video/audio). Because multiple tasks happen on SOS (record, send location, call, etc.), the firmware uses multitasking (threads) with priority for critical tasks (e.g. sending the first alert immediately, then starting video stream).
- **Media Processing (Codec):** The firmware includes a **low-latency video encoder** to compress the camera feed. This could utilize hardware blocks if present on the SoC, or a lean codec library if using software. A likely approach is to record at moderate resolution (say 480p or 720p) to reduce data size, and encode in H.264 at a bitrate that can be sustained by LTE-M (~256 kbps perhaps). Audio from the mic is encoded (AAC or even low-bitrate OPUS) in sync. If network is available, the firmware simultaneously uploads chunks of this recording to the cloud; if not, it writes to local flash. This dual-write ensures redundancy.
- **Encryption:** Before storing or sending, data is **encrypted**. The firmware will use cryptographic libraries (like mbedTLS or similar) in a memory-optimized manner. Ideally, each recorded video segment is encrypted with a symmetric key (AES-256) that is itself protected. End-to-end encryption might mean the device encrypts with a public key of the user's account (so that only the user's app/cloud (holding the private key) can decrypt). This requires some key management: possibly when pairing the device with the app, a key exchange happens (the app gives the device a public key). The firmware then uses that for every SOS data encryption. The encryption module must be efficient (use hardware crypto accelerators if available) to avoid slowing down the recording pipeline. **Data integrity** (digital signatures or HMAC) might also be added so that any tampering of footage is detectable.
- **Edge AI & Sensing:** The device will employ on-device intelligence for two main features: *hidden camera detection* and possibly *activity/fall detection*. For hidden camera detection, the firmware can run a lightweight ML algorithm or signal processing:
 - If using the camera sensor to scan for reflections, the firmware might capture images while pulsing the IR LED and use an algorithm to spot bright spots

that blink in sync (a classic approach for lens detection).

- If using RF analysis, it might use a trained model to differentiate between normal RF noise and patterns indicative of a camera's RF (for example, using FFT of the RF signal).
 - Advanced approach: If the watch had enough processing, it could implement the CamRadar technique – analyzing electromagnetic emanations. But that requires sampling EM signals at high frequency and applying algorithms. The CamRadar research essentially found the unique EM oscillation pattern a camera produces [cse.buffalo.edu](https://cse.buffalo.edu/cse.buffalo.edu). Implementing this fully may be too heavy now, but the firmware could do simpler things like scanning for known Wi-Fi SSIDs or devices (some hidden cams create Wi-Fi APs – the watch could attempt to scan for any Wi-Fi network or Bluetooth device in the vicinity with suspicious name or just any unknown device on same network).
 - In summary, the firmware's AI might combine sensor inputs to raise an alert if criteria are met. For instance: "if photodiode sees a consistent flicker reflection AND RF scanner sees a 2.4 GHz signal, classify as hidden camera likely".
- The processing for this can be done on the MCU (which might be capable of simple ML like a decision tree or small neural net). Many IoT MCUs now support TinyML frameworks. If needed, we could incorporate an **AI accelerator** (some MCUs have DSP instructions or a small ML core).

For activity detection: the accelerometer data can be analyzed to detect a **fall or struggle**. A sudden fall could auto-trigger an SOS if the user doesn't cancel (like smartwatches do for falls). Struggle (rapid erratic movement plus elevated heart rate and maybe voice shouting detected from mic) could also auto-trigger – but these are complex to avoid false alarms, so likely opt-in features.

- **Power Management (Firmware):** The firmware manages power states – keeping the device mostly in sleep or low-power mode, waking instantly on button press or significant motion. It will also periodically do background tasks like sync time via NTP from cloud, or upload any buffered data. Efficient scheduling ensures we meet the battery life requirement. For example, cellular radio might be normally off; the firmware can wake it every hour briefly to check for any incoming messages or sync.
- **Self-Test & Diagnostics:** On boot, the firmware should self-test key components (camera present, mic working, sensors responding) and periodically report health to the cloud/app. This ensures any malfunction (like a broken camera) is caught before an emergency. The user could be notified to service or reset if something is wrong.
- **Firmware Security:** The device firmware should be secure against tampering. Support for **secure boot** (firmware signature verification) is needed so that malware cannot be injected. Also, firmware updates are signed by the manufacturer and

verified on device.

- **Real-time OS tasks breakdown (illustrative):**

- *High priority:* SOS Handler Task (monitors SOS button GPIO interrupt, manages emergency routine), Sensor IRQ handlers (for quick response).
- *Medium:* Network Task (handles socket connections to cloud), Audio/Video encoding tasks (possibly running in small chunks to stream).
- *Low:* App Communication task (BLE sync, responding to app commands), Logging tasks, etc.
- Interrupts for things like incoming call (to pick up automatically in SOS), GPS fix events, etc., are handled promptly.

Cloud Backend (Secure Cloud Vault & Alert Server)

The cloud backend is a critical part of the architecture, acting as the **relay and storage hub**. It has two primary responsibilities: **storing encrypted evidence data** and **disseminating alerts to contacts (and possibly authorities)**. The design favors a scalable, serverless architecture to handle potentially many simultaneous incidents, with strong security controls.

Key components of the cloud backend:

- **Secure Storage (Personal Cloud Vault):** A cloud database or storage bucket where each user has a private, encrypted area for incident data. When the watch streams video/audio, it is sent to this storage. End-to-end encryption means the data might already be encrypted with the user's key on the device side, so the cloud just stores cipher blobs associated with a user ID and timestamp. Even the service provider cannot view it without user consent (zero-knowledge style) akitra.com. This could be implemented using cloud object storage (like AWS S3 or Azure Blob) with server-side encryption disabled in favor of client-side encryption. Metadata (like "SOS event at 10:05pm, 2 min video, location coords") could be stored in a database for indexing, but sensitive parts of metadata (exact coordinates or user identity) could also be encrypted to be safe. The storage should have **redundancy** (multiple copies or in multiple data centers) given the critical nature – we don't want a single server failure to lose an incident recording.
- **Streaming Architecture:** To handle live streaming of data (for example, a contact wanting to view the live video), a streaming server or a WebRTC service might be employed. However, to keep things simpler and within NB-IoT bandwidth, we might not do a live video feed to contacts, only recorded chunks. The device could upload, say, 10-second clips continuously. The cloud can make the latest clip available via a secure link to the contacts if needed. Alternatively, an **HTTPS long-polling or MQTT** pipeline can stream audio (since audio is lower bandwidth) in real time to contacts. We might use a publish-subscribe model where each incident is a "topic" and

authorized contacts' phones subscribe to it to receive live updates.

- **Alert Notification Service:** When an SOS is triggered, the device might not directly SMS all contacts (especially if using data only). So the cloud has an SMS/Email/Push service. For instance, using SMS gateways (integrated with telecom) to send the emergency SMS to the contacts with the user's name and GPS link. Emails can also be sent similarly. Push notifications can be sent to those contacts who have our app installed (the app would receive a push with high priority showing the alert). We might leverage a third-party API for SMS (or even integrate with the government's 112 emergency platform if accessible via API for registered devices). This service must be **fast and reliable** – possibly using multiple gateways (if one fails, try another).
- **Sequential Calling Service:** Ideally, the watch does the phone calls itself via cellular voice. But if for some reason that's not possible (e.g., the NB-IoT module chosen doesn't support voice), an alternative is the cloud could initiate calls to contacts. This would involve a VoIP server or integration with a telephony service (like Twilio, etc.). The cloud can call Contact 1 and, when they pick up, can connect them to the audio stream from the watch (via the cloud as a bridge). However, this is complex and dependent on telephony integration and real-time voice piping to cloud. A simpler approach is to rely on the device's phone capabilities to call, and only use cloud as backup for alerts. We will primarily assume the device calls out directly (which is more peer-to-peer and reliable if it has signal).
- **Serverless Functions & Database:** The architecture could use serverless functions that trigger on events. For example:
 - When an *SOS event message* arrives from a device (via a REST API or MQTT message), a cloud function triggers: it logs the event, sends out notifications to contacts, and allocates storage space for incoming media.
 - Another function triggers as each chunk of video arrives: it might assemble or just mark it in database.
 - A function could also handle *camera detection alerts* – e.g. if user uses the detector and finds something, perhaps the user can press a button to automatically alert hotel security or something; not core, but possible extension.
 - The database (NoSQL or SQL) keeps records of users, their contacts (with consent in place to store that info), device status (to know if device is online/offline), etc. It also can log every SOS activation for analytics (with user consent) to improve the service (e.g. areas with frequent alerts could be flagged anonymously).
- **Mobile App Integration:** The app connects to the cloud to fetch data (like to stream the location updates or download the encrypted video). The cloud should expose

secure APIs. For instance, a GET endpoint `/incident/{id}/location` that returns the latest coordinates (secured by authentication). Or a websocket that the app can listen to for live updates. This likely uses standard cloud messaging (like Firebase Cloud Messaging for pushes, or AWS IoT for device messaging). We ensure that only authorized parties (the user or their chosen emergency contacts) can subscribe to these updates – enforced by authentication tokens.

- **Security & Privacy:** The cloud must implement stringent security. All endpoints enforce authentication (the watch will have an auth token or device ID with secret to send data; the user app will authenticate with user credentials). Data in transit is via HTTPS (TLS 1.3) to protect against interception akitra.com. Contact info and personal data in the cloud database are encrypted at rest. Also, since this deals with personal data, compliance with India's DPDP Act is required – which means user consent for data collection, easy means to delete their data, breach notifications, etc. techtarget.com. The cloud design will include a **Data Protection by Design** approach: minimal data retention (e.g. if an incident video is older than say 6 months and the user has a copy or didn't mark it to keep, it might auto-delete to protect privacy). Also, robust access control to ensure no unauthorized viewing of another person's incidents.
- **Scalability:** The use of serverless and decoupled microservices allows scaling up during emergencies (for example, if thousands of users trigger SOS during a citywide event, the system should handle the load). Using cloud infrastructure (like AWS Lambda, DynamoDB or equivalent) means we can scale on demand. For SMS/call service, we might have pre-provisioned capacity or multiple provider contracts to handle burst messaging.
- **Integration with Authorities (Future):** The cloud can also integrate with official emergency systems. For instance, with user permission, an SOS could also forward data to a police control room dashboard. Some cities have women safety programs where emergency devices are registered – our system could send data to those systems via an API (e.g. an authorized feed to police of live location for certain verified incidents). This is beyond initial scope but architecturally feasible.

Mobile Companion App

The mobile app serves as the user's interface to the system and a portal for emergency contacts:

For the **wearer (primary user)**, the app provides:

- **Account Setup & Device Pairing:** When first installing, user creates an account (or logs in via OTP for simplicity). They pair the watch via BLE by entering a code or scanning a QR. The app then registers the device's ID with the cloud under the user's account.

- **Emergency Contacts Management:** The user adds up to N contacts. The app obtains consent from those contacts if needed (it could send them a message that they've been added). It verifies their phone numbers/email. The user can configure the order of calling and whether to also auto-dial 112.
- **Preferences:** Toggle silent mode vs loud alarm, whether auto-fall-detect is on, etc. They can also enable/disable hidden camera detection scans from the app (maybe easier to initiate a scan via phone UI).
- **During an SOS:** If the user has their phone, the app will display a **live incident screen**: showing "Alert sent! Contacts responding." It could show a map of their current location updating. If video is being recorded and the user escapes, they might be able to watch it right away (but perhaps better to keep it uploading to cloud for safety). The app might also allow the user to press a "Call Police" button as an extra action or sound a siren from phone too.
- **Post-incident:** View the timeline and recordings. The user can add notes (useful if she wants to document what happened while fresh). She can tap an option to "Share with Police" which could produce a secure link or code she can give law enforcement to access the evidence.
- **Routine Use:** The app might have a feature like "Safe Check-in" – e.g. the user can set a timer if she's going somewhere and if she doesn't cancel it, it triggers an alert. (This is an extra feature some apps have; including it could add value).
- **Device Tools:** Battery level display, toggle a "find my watch" ping, firmware update check, etc.

For **Emergency Contacts (secondary users)**: If they install the app (optional, since they can receive SMS without it), the app allows:

- **Viewing Alerts:** If a contact gets an SOS alert and has the app, a notification opens a live tracking screen. They can see the location of the person in distress on a map and any notes like "I'm being followed". They might also see others in the contact group (if multiple contacts) acknowledging or responding (some coordination feature).
- **Communication:** The app could facilitate an audio call or a conference among contacts or directly to the wearer's watch (if the watch supports voice).
- **History:** A contact could see past alerts if they were involved (in case needed for follow-up).
- **Privacy for User:** Note that contacts only get data during an active SOS. The user's whereabouts are not continuously shared unless an emergency. The app should clarify that.

For both: the app will enforce login security, likely using phone's biometrics to protect access to sensitive data.

A note on minors: Since our demographic includes 13–17, the app should have a flow for guardian consent. Possibly, a parent's account is linked to a minor's device and they manage contacts. This would align with DPDP Act requiring guardian consent for under 18 techtarget.com.

Technology: The app could be built cross-platform (Flutter/React Native) or native. It interacts with cloud via REST APIs and receives push notifications for SOS. It uses BLE APIs to communicate with the watch (for pairing, config, and maybe real-time transfer of small data like firmware updates).

Block Diagram Reference

(Refer to the earlier provided block diagram figure for a visual overview.) The architecture can be summarized in a layered flow:

- **Device Hardware Layer:** includes **sensors (camera, mic, GPS, accelerometer, hidden-cam sensor)** feeding into **MCU**, which connects out via **LTE-M network** to the cloud. The device also communicates locally to the **mobile app via BLE**.
- **Firmware Layer:** on MCU handles **data capture, encryption, and transmission**. It triggers cloud communications and handles incoming commands.
- **Cloud Layer:** comprises the **Alert Server** (which notifies contacts via SMS/Email and relays calls) and the **Data Vault** (which stores the encrypted recordings and provides them to authorized clients).
- **Application Layer:** the **Mobile App** for user and contacts to configure the system and respond to incidents.
- **Emergency Responders:** on the periphery, the contacts (family/friends) receive alerts on their phones, and optionally emergency services can be looped in (either through a contact or direct integration).

This end-to-end architecture ensures that from the moment a distress signal is generated on the watch to the moment help is dispatched, every component works in sync, with redundancies (device + cloud + contacts) to maximize the chances of a positive outcome. In the next section, we dive deeper into one of the unique features enabled by this architecture: the hidden-camera detection, evaluating how various technologies contribute to it.

Hidden-Camera Detection Technologies

A standout feature of this smartwatch is its ability to detect hidden cameras in sensitive spaces – addressing the menace of voyeurism (IPC 354C). Hidden cameras can be notoriously difficult to spot with the naked eye, so our device employs **multiple**

complementary detection techniques. In this section, we evaluate the available technologies for hidden camera detection, including their principles, pros, and cons, and justify the choices for our implementation:

1. Radio Frequency (RF) Scanning: Many covert cameras (especially wireless nanny cams or spy cams) emit RF signals to transmit video – often using Wi-Fi, Bluetooth, or proprietary analog feeds in bands like 1.3 GHz or 2.4 GHz. An RF scanner sweeps through frequencies to find unexpected transmissions. Professional bug detectors have this functionality: they beep when a strong signal of certain modulation is detected eufy.comeufy.com. Our watch could include a wideband RF receiver to scan for signal peaks. *Pros:* Can detect active wireless cameras even if they are well-hidden visually, and works in light or dark. It doesn't require line-of-sight – you can detect a camera transmitting from behind a wall or ceiling. *Cons:* Not all hidden cams transmit (some just record to SD cards). Also, an RF scan might flag other devices (e.g. a Wi-Fi router or a Bluetooth speaker) leading to false positives. In a hotel or office, there could be many RF sources. Additionally, scanning a broad spectrum can be time-consuming and power intensive, and might need an antenna which is challenging to fit in a small watch. Our solution is to include an **RF detection mode** focusing on common spycam bands, and filter out known benign signals. Perhaps the user could calibrate by letting the watch know the Wi-Fi SSID of the location to ignore, etc. RF scanning will serve as one **mode of detection**, but not the sole method.

2. Lens Reflection (Optical Glint) Detection: Camera lenses, even tiny pinhole ones, are made of glass and reflect light. A proven method to find them is to shine a light and look for the characteristic “**red dot**” reflection. Many commercial detectors use this: they have a ring of red or IR LEDs and a viewfinder; as you look through, a camera lens appears as a bright spot of reflected light eufy.comeufy.com. Our smartwatch can replicate this by using IR LEDs around its face and using either the watch's camera or a specialized photodiode to sense reflections. The user would move the watch around slowly in a darkened room; if a strong reflected beam returns at a consistent angle, the device signals a find. *Pros:* Works for any type of camera (wired or wireless) since it relies on physical lens presence. It's relatively low-tech and **highly effective within line-of-sight** range (typically a few meters). *Cons:* Requires the environment to be dim or dark for best results, and requires a somewhat direct line-of-sight to the camera lens – if the lens is completely hidden behind a pinhole and not facing the detector, you might miss it. Also, in a very cluttered environment with lots of reflective surfaces (mirrors, metal objects), it might generate some false reflections (though true camera lens reflections often have a tell-tale colored shine). In our implementation, the watch will pulse IR and use the camera sensor to capture images; an algorithm will highlight tiny reflection points that blink in sync with the pulses (indicating a retroreflective lens). This optical method is a **cornerstone** of our detection as it directly targets the defining feature of cameras (the lens).

3. Magnetic/EMF Anomaly Detection: Some detection devices and smartphone apps claim to use the phone's magnetometer to spot cameras by detecting electromagnetic fields or magnets in the camera. Tiny hidden cameras may have small motors (for autofocus or tilt) or coils that emit EMF, and often manufacturers put magnets in battery covers or mounts. The watch's built-in magnetometer could sense stronger magnetic fields when very close to a camera (within a few centimeters). *Pros:* If the hidden device uses a speaker or motor (like a rotating security cam), those generate fields that can be picked up. Also, many common spy

cams use DC adapters with transformers that emit EM. *Cons:* It's quite limited – purely magnetic detection can miss electronics that have no significant magnetic components. It's also range-limited; you'd have to be almost touching the hidden device. The magnetometer is more useful as a supplement – for instance, if the RF scan and optical scan are inconclusive but the magnetometer spikes at one spot, that gives additional evidence. We consider this a **secondary method**; the firmware can monitor the magnetometer during a scan and note anomalies, but it won't be heavily relied on due to its unreliability for this purpose.

4. Computer Vision (CV) Lens Spotting: With the watch's own camera, we could attempt to visually identify camera-like objects. For example, an algorithm could scan the environment for round glassy shapes (lenses) or the glint pattern of a camera sensor. There has been research on using deep learning (e.g. a YOLO object detection model) to find hidden camera lenses in images [sciencedirect.com](https://www.sciencedirect.com). *Pros:* Potentially can find lenses even without the IR reflection trick, and can work under some lighting conditions. Could also detect things like pinhole in an unusual place (like a tiny hole in a screw head) by learning their appearance. *Cons:* This is **computationally heavy** for a smartwatch MCU, and likely requires a good quality image of the entire room which the watch's tiny camera may not capture well in one frame. Also, it may have many false positives (e.g. it might think a shiny bolt or phone camera lying around is a hidden camera). Due to resource constraints, we likely will not run a full CV model on the watch. However, in a scenario where the user has a phone, she could use the phone camera with our app for a thorough CV scan (the app could use its more powerful processor to run a model). So, the ecosystem can have the **phone app augment** hidden camera detection by using the phone's camera for visual sweep, while the watch focuses on IR and RF scans. Thus, CV lens spotting is in our toolbox but offloaded to the phone/cloud: e.g., the user could take photos of suspicious areas via the app and a cloud service could analyze them for hidden lens signatures.

5. Thermal Detection (infrared imaging): Not explicitly listed in the question, but worth noting: cameras and other electronics often emit heat. A thermal camera can reveal hotspots where a hidden device might be operating (especially if powered on for long). Our smartwatch will not have a thermal imager (too costly and power hungry), but in some advanced version, one could imagine using a small thermopile array sensor to scan for warm spots. However, this would also catch numerous false sources (warm pipes, laptops, etc.), so we put this aside.

6. mm-Wave Radar Imaging: The mention of possibly using **mm-wave sensors (60–77 GHz)** is intriguing. Texas Instruments and others make tiny mmWave radar chips that can sense objects and even create 3D maps. In theory, mmWave can penetrate certain materials (clothing, plastic) and detect metal or dielectric discontinuities. A hidden camera's circuitry or metal lens could reflect radar waves distinctively. In a research context, one might use radar to **see through** a fixture and detect a device. *Pros:* Works in darkness, doesn't require visible access, can cover a wide area with a scanning beam. *Cons:* These sensors consume significant power and data, and interpreting radar signals for something as small as a camera is non-trivial. Possibly beyond our near-term implementation, but we mention it as a futuristic approach. If in 2-3 years miniaturized radars become common in phones (some phones already have Google's Soli radar for gesture), the watch could leverage that

concept. For now, mmWave might be too power-heavy for continuous scan, but perhaps could do a short sweep if a high suspicion triggers it.

After evaluating all methods, **our approach will be multi-modal**: The watch's hidden camera detection routine will utilize **IR reflection detection as the primary mechanism** (due to reliability), supplemented by **RF scanning** for wireless signals. The device will alert the user if either method finds a strong indication. We will also leverage the smartphone app to enhance scanning: for example, the app can automatically connect to the watch and display a UI that shows a bar graph of "signal strength" from the RF scanner, while also using the phone camera flash to do a lens sweep (covering more angles). By combining methods, we can greatly improve detection confidence – e.g. **CamRadar**, a recent system, cleverly utilized EM emanations to detect cameras with over 93% successcse.buffalo.edu and was not limited by camera transmission type or anglecse.buffalo.edu. Inspired by that, our system's integration of optical and RF means even if a camera isn't broadcasting (like just recording to memory), the optical method can catch it; if a camera has no direct line of sight (hidden behind something) but is broadcasting, the RF method can catch it. This redundancy is crucial.

We will implement a simple **user workflow** for scanning: say the user opens the "Scan Room" function on the app, places the watch on scan mode. They walk around the room holding up the watch. If the watch detects a suspect signal or reflection, it will vibrate and the app will show "Possible hidden camera detected!" with maybe an arrow or strength indicator to help locate it precisely (the user can then pinpoint the source). The device might also log the finding (so the user has evidence of finding a spycam, which could be reported).

It's important to calibrate and test this feature thoroughly to avoid both false negatives (missing an actual camera) and false positives (crying wolf too often, which could erode user trust in the device). We'll use a variety of common hidden camera models in testing – some with Wi-Fi, some standalone – to finetune sensitivity. We might also crowdsource data: the app could let users label if a detection was indeed a camera, improving the algorithm over time (keeping privacy in mind).

In conclusion, by evaluating **RF, optical, magnetic, and CV-based techniques**, we choose a **hybrid detection module** leveraging the most practical and miniaturizable technologies. This hidden-camera detector will give our smartwatch a unique capability to proactively protect privacy – a feature highly relevant given the rising voyeurism casesendnowfoundation.org. It transforms the watch from just a reactive safety tool (helping after an incident starts) to a **preventive tool** that can alert users to potential danger beforehand, aligning with the idea of creating safer environments for women through technology.

Data Security & Privacy Compliance

Given the sensitive nature of the data involved (personal videos, audio of incidents, location trails, etc.), the smartwatch ecosystem is designed with a **privacy-first approach**. We incorporate strong data security measures end-to-end, and ensure compliance with relevant data protection regulations, notably India's **Digital Personal Data Protection Act (DPDP), 2023**. Below we outline the security architecture and privacy practices:

End-to-End Encryption (E2EE): All user data generated by the device is protected by encryption from the moment of capture until it reaches the intended recipient (user's cloud vault or emergency contacts). The device encrypts recordings **locally before upload**, meaning even if someone intercepts the data in transit or breaches the cloud storage, they cannot decipher it [akitra.com](https://www.akitra.com). Each SOS recording can be encrypted with a unique symmetric key that is itself encrypted with the user's public key (established during device onboarding). Only the user's mobile app (which holds the private key in its secure storage) can decrypt the content. Similarly, location coordinates and messages can be encrypted. For live phone calls to contacts, we use standard cellular voice which is encrypted over the air by telecom networks; if we ever route voice over IP, we'd use TLS and SRTP for encryption. **In short, the system ensures that no unauthorized party – not even the service provider – can eavesdrop on the user's data.**

Secure Device Authentication: The watch and cloud authenticate each other to prevent impersonation. During setup, the device is provisioned with a unique device ID and secret. When connecting to the cloud, it uses this for mutual TLS authentication. Emergency communications to contacts (like SMS) are signed or include verification so the contacts know it's genuine and not a spoof. The mobile app uses user credentials (with MFA or phone OTP) to ensure only the rightful user accesses their data.

Biometric/PIN Locks: As mentioned in requirements, the device and app both enforce **biometric or PIN unlock** for sensitive operations. The watch itself can lock its buttons in normal mode to avoid misuse – e.g. a stranger can't trigger the user's SOS or view data without permission. The user (or their guardian) would use fingerprint or phone unlock to enable/disable that. The app requires login and we encourage enabling fingerprint unlock on the app for quick yet secure access. If the watch is lost, the user can remotely deactivate it via the app (to prevent someone else pairing to it).

Data Minimization & Anonymization: In line with DPDP Act principles, we only collect data necessary for providing the safety service [techtarget.com](https://www.techtarget.com). That means:

- The watch is not continuously sending the user's location – it only does so during an SOS event or if the user explicitly uses a tracking feature. No "surveillance" of the user's daily movement is stored centrally by default.
- Audio/video is not recorded unless the distress feature is activated (no constant recording or audio monitoring, so the user's privacy in normal times is respected).
- The cloud does not store any plain personal info beyond what's needed: e.g., emergency contacts' phone numbers and maybe first names (with their consent) and the user's own name for identification in messages. Even these can be encrypted in the database, such that only the user's app can read them when needed.
- If any analytics are collected (like how often features are used or battery stats), they are aggregated or anonymized. For example, we might collect how many SOS events occurred in a region for improving police resources, but that would not include personal identifiers without consent.

Compliance with DPDP Act, 2023: The DPDP Act is India's new data protection law focusing on user consent, data security, and user rightstechtarget.comtechtarget.com. Our system is built to comply:

- **Consent:** Users provide explicit consent for processing their data when they sign up (through the app's terms/privacy policy). Emergency contacts also give consent to be notified. If the user is a minor, parental consent is obtained for collecting and processing the child's data (the app will have a flow for guardian approval)techtarget.com.
- **Purpose Limitation:** Data is used only for safety purposes. For instance, location data is used to route help during emergencies; we don't use it to serve ads or unrelated services. We clearly communicate what data we collect and why in the privacy notice (as required by DPDP).
- **Data Security Safeguards:** The law expects reasonable security measures; we likely exceed "reasonable" with end-to-end encryption, secure hardware, and rigorous testingtechtarget.commakitra.com. We treat the watch much like a health device where privacy is paramount.
- **Breach Notification:** If despite all safeguards, a data breach occurs (say our cloud is compromised), we will follow DPDP requirements to notify the Data Protection Board of India and the affected users promptlytechtarget.com. But since we store mostly encrypted data, the risk of sensitive info exposure is reduced (the attacker wouldn't easily decrypt videos without the keys).
- **Data Retention & Deletion:** Users will have control over their data. Through the app, they can delete stored incident records from the cloud (e.g. after downloading evidence, they might want to wipe it). Our policy might be to auto-delete incident data after a certain period (say 1 year) unless the user marks it to keep (for an ongoing case). The DPDP Act emphasizes data should not be retained longer than needed, and our system aligns with that by not hoarding old data indefinitely. Users can also request deletion of their account and all associated data – and we will comply, after confirming no legal reason (like an ongoing criminal case) requires retention.
- **Child Data Protections:** The Act forbids tracking or behavioral monitoring of children and targeted ads towards themcarnegieendowment.org. Our device does not do behavior profiling; any tracking is purely for safety initiated by the child or guardian. And obviously, we serve no ads at all. So we adhere to those provisions. Guardian controls on a child user's device also align with the Act's intent for parental supervision.

Secure Hardware & Firmware: We implement secure boot on the device so that unauthorized firmware (which could snoop on the user) cannot run. All debug interfaces (like JTAG) are locked down in production units to prevent extracting secrets. The use of a **secure enclave** in the MCU means cryptographic keys (for encryption, and authentication

keys for cloud) are stored in hardware that even a sophisticated attacker with the device can't extract. Firmware updates are signed and delivered over encrypted channels.

Communication Security: All network communication uses modern cryptographic protocols. For instance, when the watch connects to cloud (likely over MQTT or HTTPS), it uses TLS 1.3 ensuring forward secrecy etc. The BLE connection with the app is paired using secure pairing (LE Secure Connections with passkey) to prevent MITM; and sensitive commands over BLE are encrypted at the application layer too. We recall that wearables often had issues with Bluetooth eavesdropping; we address that by disallowing older insecure pairing modes and by keeping BLE use to non-critical data where possible.

Emergency Exceptions & Ethics: One challenge is that in an emergency, the user might want to share data with police or in court. Our encryption scheme should allow the user to do so easily (e.g. by giving them an export of the key for that incident or a secure share link). However, we will *not* build any "backdoor" for law enforcement to bypass encryption – that would undermine the security for users. Instead, if law enforcement legally needs data, they would get it from the user (who holds the keys), preserving the user's agency. This approach is consistent with privacy rights – the DPDP Act actually penalizes unauthorized disclosure of personal data.

Privacy of Contacts: We also ensure that the emergency contacts' information isn't misused. They only get data when they need to respond. We don't show all contacts to each other except perhaps names if the user chooses (like, friends see each other's name in a group alert context). Contact details aren't shared beyond what's necessary to deliver the alert (the telecom obviously gets the number to SMS/call). We store minimal info about contacts and allow the user to manage it.

Server-Side Privacy and Auditing: Administrative access to the cloud servers will be tightly controlled. We will maintain audit logs of any administrative actions on data. Given data is encrypted, even admins can't view content – but, for example, if an admin needed to help remove a problematic account, that action is logged. This is more operational, but important for compliance and trust.

Secure Disposal: If a user stops using the device or deletes data, we securely delete encryption keys such that even backups eventually cannot be read (keys will be scrubbed or overwritten). On the hardware side, if the watch is being scrapped, the secure element ensures keys cannot be recovered (some even have self-erasure if tampered with).

Third-Party Components: We will vet any third-party SDKs (e.g. mapping API, SMS gateway) to ensure they comply with privacy standards. For instance, using Google Maps for location reverse-geocoding should be done carefully (or avoid sending personal data to external APIs by doing on-device conversion if needed).

In essence, the entire design follows "**Privacy by Design**" principles – embedding privacy into technology from the get-go, not as an afterthought [akitra.com](https://www.akitra.com). This builds user trust, which is absolutely vital for adoption of a safety device. Users (and their families) need confidence that the device will *protect them without exposing them*. By using robust encryption, secure handling of personal data, and complying with laws like DPDP (consent, breach notification, etc.), we aim to make the smartwatch as **secure as a personal safe** for

one's data. And beyond compliance, these measures are simply the right thing to do ethically for a product that will often be privy to some of the most traumatic moments in a user's life.

Regulatory & Certification Path

Bringing a safety smartwatch to market in India requires navigating several regulatory and certification requirements. We must ensure the device complies with electronics safety standards, wireless spectrum regulations, telecom norms, and other relevant Indian laws. Below is the roadmap for regulatory approvals and certifications:

1. Bureau of Indian Standards (BIS) Certification: In India, electronic products, including smart wearables, are subject to mandatory certification to ensure they meet safety standards. Specifically, **smart watches have been included under BIS Compulsory Registration Scheme (CRS)** since 2018, requiring compliance with IS 13252 (Part 1):2010 crsbis.in. IS 13252 (Part 1) is essentially the Indian equivalent of IEC 60950-1 / IEC 62368-1 (safety standard for information technology equipment). We will get our device tested in a BIS recognized lab for electric safety (checking for no shock hazard, fire safety of enclosure, etc.). Once tests are passed, we register the product with BIS and mark it. BIS registration ensures our device is safe for end users (no overheating, no dangerous emissions) and is legally required to sell in India bl-india.com. Additionally, if our device has a power adapter or charger, that charger likely needs BIS certification under the applicable standard (IS 13252 or the specific standard for power adapters).

2. Wireless Planning & Coordination (WPC) Approval: The device uses wireless communication (Bluetooth, LTE/NB-IoT, possibly Wi-Fi in testing). India's WPC, under the Dept. of Telecommunications, regulates RF emissions and spectrum use. For license-exempt bands (Bluetooth in 2.4 GHz, etc.), we need to obtain an **Equipment Type Approval (ETA)** from WPC for the RF module, unless we use a pre-approved module. Many off-the-shelf modules have ETA; otherwise, we'll submit our product RF test reports (for EMI/EMC and frequency use) to WPC. Particularly, Bluetooth LE at 2.4 GHz and any other unlicensed band sensor would be covered under ETA. Our cellular module will operate in licensed spectrum (with a carrier SIM), so as a device manufacturer we ensure it meets 3GPP specs but no separate license needed for usage – however, we must ensure the module has been approved by global bodies (like GCF/PTCRB) and by Indian operators for network compatibility.

3. Telecommunication Engineering Center (TEC) – MTCTE: India has introduced the **Mandatory Testing and Certification of Telecom Equipment (MTCTE)** for any telecom equipment to be used/sold in India. A smartwatch with a cellular radio might be classified under this (similar to how mobile phones are). We should check if our device needs TEC certification. Likely, since it's end-user equipment with a SIM, it should undergo testing for things like SAR (radiation absorption rate), EMI/EMC, safety, etc., under TEC guidelines. SAR testing ensures the device's RF emissions are within safe limits for wearing on the body. We will get SAR measured (likely it will pass since LTE-M power is low and watch is small, but it's required to document). The TEC certification would cover adherence to technical requirements for cellular devices and maybe e911/112 behavior (ensuring 112 calls

go through even without SIM or with locked phone, etc., though since our device relies on SIM it just needs to allow 112 dialing by any SIM).

4. International Standards (for future export): Although initial market is India, we will design to meet CE and FCC standards too.

- **CE (Conformité Européenne):** This would require compliance with the Radio Equipment Directive (RED), which includes RF performance, EMC (electromagnetic compatibility), safety (likely EN 62368-1), and SAR for wearables. We'd run EMC tests (to ensure our device doesn't emit spurious interference and can withstand ESD, etc.). EMC compliance is also needed for BIS/TEC.
- **FCC (Federal Communications Commission, USA):** The device's transmitter must be tested for FCC Part 15 (unlicensed emissions) and Part 27 (for LTE bands) if we ever sell in the US. We'd ensure our module has FCC modular approval which simplifies this.

5. Battery Certification & Transport Regulations: The smartwatch contains a lithium battery, which must comply with safety standards. India often references UL or IEC standards for batteries. **IEC 62133** is the international standard for safe operation of Li-ion batteries (covering overcharge, short, etc.). BIS in India has adopted IS 16046 which mirrors IEC 62133 – we should get our battery certified to IS 16046. Also, for **transportation**, lithium batteries (especially if shipping by air) must pass **UN 38.3** tests (altitude simulation, thermal, vibration, shock, short circuit, impact, etc.)mobilepowersolutions.com. We will ensure our battery supplier provides UN 38.3 certificationmobilepowersolutions.com. That way, we can ship the product by air courier without issues. Airlines may also require our packaging to meet IATA packaging instructions for devices with lithium batteries (like protective packaging, markings if over certain Wh). Our battery is small (maybe ~2 Wh), likely under the limit for simplified shipping, but compliance is still necessary.

6. BIS Registration for Battery & Adapter: Besides the watch device, if we include a USB charging cable or adapter, note that adapters/power banks also require BIS under CRS. We might avoid including an AC adapter (assuming users charge via phone chargers) to simplify. But if we do, that adapter must comply with IS 13252 or IS 616 (if considered a power supply). The lithium cell itself might come under an Indian standard if above a capacity threshold, so we'll check BIS list for "Lithium-ion cell". As of now, cells used in electronics often come with UL certification but BIS might not require separate if integrated.

7. Environmental and Safety Regulations:

- **RoHS & WEEE:** Restriction of Hazardous Substances compliance ensures no lead, mercury, etc., in our device beyond permitted levels. If manufacturing in India or importing, compliance is needed (India has its own e-waste (Management) Rules that basically align with RoHS). We'll use RoHS components and label accordingly.
- **Battery Safety Transport for India:** The UN38.3 covers global, but domestically we might also follow Central Electrical Testing for batteries. Also, we'll label the battery properly with required symbols (do not dispose in fire, etc.) and provide MSDS if

needed.

8. AI/Software Compliance: While not a formal certification, any AI in the device (like camera detection) should comply with ethical guidelines (no privacy violation since it runs locally). If we use cryptography, we ensure it meets Indian crypto export/import rules (generally, consumer devices with encryption are allowed; any restriction would be minimal as long as we aren't doing anything outside normal device encryption).

9. Patent/Frequency Licensing:

- We should ensure any specific technology (like CamRadar approach or certain detection methods) doesn't infringe existing patents. Do a freedom-to-operate search especially on hidden camera detection algorithms, just to be safe, or license if needed.
- For using LTE-M/NB-IoT, our module vendor will likely handle the intellectual property licensing for cellular tech (since they pay royalties to patent pools). We'll just buy certified modules to avoid directly dealing with 3GPP patent licensing.
- If we add something like GPS, there's no license needed to receive signals.

10. Certification Timeline & Strategy:

- Early in development, we'll engage with a testing lab (maybe an NABL accredited lab in India) to do pre-compliance testing on RF, EMC. This way, we catch any issues (e.g. our PCB antenna might cause harmonic emissions) and fix them before formal certification.
- Once the product design is finalized, we proceed with BIS CRS registration (which can take a couple months including testing). We might expedite by using pre-certified modules (for Bluetooth, cellular) as that often simplifies testing – for instance, if the RF module is pre-approved, we mainly test the full product's radiated emissions and safety, not the module's transmitter from scratch.
- WPC ETA for Bluetooth is often paper-based if you have a module report (just submit module FCC/ETSI reports and get ETA). If using a custom antenna or design, we do our own testing for that band.
- TEC MTCTE can be slightly involved; we will see categories (maybe "Communication module product" or "Smart Tracking device" might be a category). If needed, we send it to TEC-accredited lab for tests which include SAR, EMI/EMC, safety, etc. Possibly our BIS testing covers a lot of those aspects, but separate submission to TEC is required with those results.
- SAR specifically: Our device worn on wrist would be considered "wearable on body". SAR limits in India follow international (1.6 W/kg averaged over 1g for head, and

different for limbs maybe). Since our transmit power (23 dBm or lower) is small and the antenna is small, we expect SAR well under limits, but it must be measured and documented.

11. Packaging and Labelling: After certifications, we must label the product appropriately:

- BIS CRS logo with our registration number on device/packaging.
- WPC ETA number if applicable (some devices list the ETA number or the generic “Bluetooth use in India, not harmful” statement).
- The battery needs the Li-ion recycle logo and “CAUTION”.
- If we got CE/FCC for international, their logos would be on packaging.
- Also, because it’s a camera device (though a very small one), some jurisdictions might require us to indicate that device contains a camera (for import/export or privacy label, e.g. some companies label devices that can record). Not a formal law in India, but something to consider ethically (though since it’s for the user’s own safety, not spying, probably not an issue).
- The device will have an IMEI if cellular – we need to have the IMEI printed somewhere (like inside or on the box) and ensure each device gets a unique IMEI from GSMA range. For manufacturing, we’ll obtain a TAC (Type Allocation Code) for the watch model to generate IMEIs.

12. Compliance Documentation: We will maintain a technical file with all test reports and compliances. Also, draft a user manual section on regulatory compliance (the kind that says “This device complies with BIS standards... FCC ID:..., etc.” and safety instructions as required by standards).

13. BIS - Quality Control Order: In India, certain electronics also come under “Electronics and IT Goods (Requirement of Compulsory Registration) Order, 2012” which basically is the BIS CRS. We ensure our manufacturing is done under proper quality controls (BIS may audit the factory for maintaining standard). We’ll also register the product on the government’s CRS portal.

14. Other Indian Laws:

- **Import Duties and DGFT regulations:** If we import components or finished units, ensure no restricted substances or such. Likely fine.
- **Legal recording considerations:** Indian law (and many others) allows a person to record their own conversations or scenes they are part of, for lawful purposes like evidence. We should include a note in the user agreement that the device is intended for self-defense evidence and not to record others secretly for inappropriate reasons

(to disclaim misuse).

- **Use of 112:** We might coordinate with Ministry of Home Affairs / local 112 centers if our device is widely adopted so they know how to handle automated calls from it. Currently, smartphones are mandated to have 112 on long-press of 5 or 9 or power button – our device acts similarly but in wearable form.

15. Testing for Extreme Scenarios: Though not a formal certification, we should test that the device still works under scenario conditions (like if someone is physically assaulted and the watch is banged or gets some water splashed, it still sends an alert). Perhaps get an **IEC 60529 IP test** for water resistance (IP67). And maybe some durability tests (not necessarily certified, but to ensure reliability).

By systematically obtaining these approvals, we ensure our smartwatch is **legally compliant and safe** to use. Certification may seem like a hurdle, but it is essential to protect users and validate our claims. For example, BIS testing will reassure that our device's battery won't explode and WPC ensures we don't interfere with other devices. Achieving these certifications can also be a marketing point (we can advertise that the product is certified safe, meeting national standards).

In parallel to regulatory tasks, we will also file any necessary documentation under the **India's Personal Data Protection** regime (if needed, maybe as a data fiduciary we might have to register depending on volume of data processed, and have a privacy policy that adheres to the Act). This ties into the compliance aspects described in the previous section.

Conclusion of this section: Navigating BIS, WPC, TEC, and related certifications is a non-negotiable part of our go-to-market. We have planned for it from the design phase (choosing certified modules, designing for compliance) so that there are no nasty surprises late in development. With these certifications in hand, we can instill confidence in retailers, government, and consumers that the device meets all Indian **regulatory standards** – it's safe to use, doesn't violate wireless rules, and aligns with national norms for quality.

Risk Analysis & Mitigations

Developing and deploying a safety smartwatch entails several **risks** – technical, operational, and user-related. It's crucial to identify these risks and plan mitigations to ensure the device truly serves as a reliable lifeline in critical moments. Below we discuss major risks and how we address them:

Risk 1: False Alarms (Accidental or Malicious Activation). *Scenario:* The user accidentally presses the SOS button or an overly sensitive fall detection triggers an alert, sending false alarms to contacts (or police). Too many false alarms could lead to “alert fatigue” where contacts start ignoring the device or, worse, authorities impose penalties. *Mitigations:* We have built-in a **confirmation window/cancel option** – e.g., the 5-second delay with vibration feedback where user can cancel if unintended (as described earlier, similar to smartphone SOS). Also, our algorithms for automated triggers (fall, sound detection) are tuned conservatively initially to avoid false triggers. During onboarding, we

educate users on proper wear (so the SOS button isn't pressed by bending wrist ordinarily) and encourage them to test the button in training mode to get comfortable. If a false alert does go out, the message to contacts can say "If this is accidental, user will cancel in 5 sec" so they wait a moment. We also allow **user feedback**: e.g. if the watch triggers wrongly, the user can report that in the app, and the system might adjust sensitivity (machine learning on false events, if possible). By minimizing false alarms, we maintain trust in the system's credibility.

Risk 2: Failure to Trigger or Send Alert (Reliability). *Scenario:* In a real emergency, the user presses the button but nothing happens – perhaps due to device malfunction, dead battery, or network outage. This is a worst-case failure as it defeats the purpose. *Mitigations:* Multiple strategies ensure reliability:

- **Redundancy in Alerts:** The watch will try multiple channels – sending data via cellular, but also via paired phone if connected (so if one fails, the other might work). It can send both SMS and data packets as backup.
- **Device Self-Monitoring:** The device frequently monitors its battery and will *warn the user well in advance* if battery is low (via app notification or LED). We aim to have a reserve so even at "0%" on UI there is enough for one SOS. We'll also encourage a nightly charging habit via the app.
- **Hardware Reliability:** We select automotive/industrial grade components where possible for robustness, and do stress testing. E.g., test the button 5000+ presses, test device under various temperatures. The enclosure is rugged so that it still works if thrown or impacted (to an extent).
- **Network Issues:** If there is no cellular coverage (e.g., basement), the device still records evidence offline. As soon as it finds even a 2G signal or later, it will send stored alerts (the firmware keeps trying). If the user is in a known no-signal zone (like remote area), we could consider an **offline mode** where the device at least emits a loud alarm since online help can't be summoned. Future mitigation is satellite connectivity (see Future Work) to handle remote areas.
- We will maintain a **server heartbeat** mechanism: the device pings the server periodically, so the backend can notice if a device is offline or hasn't checked in. If a user's device goes offline for long (and they have a buddy system), it could notify the user or guardian to check on it.
- For minors or dependents, the guardian app could show device status (battery, last seen) so they know it's ready.

Risk 3: Device Being Located or Destroyed by Attacker. *Scenario:* An aggressor notices the victim's watch is recording or sending an SOS (maybe they heard a contact's voice or saw a flashing LED) and tries to take it off and destroy it. *Mitigations:* We designed the watch to be as **discreet as possible** in silent mode (no loud sounds or obvious light unless user

enabled). The attacker might not even realize an alert was sent. But if they do attempt to remove:

- We have **tamper detection** – instant alerts of “device removal” are sent if the strap is opened or cut (so contacts know something’s wrong even if data stops).
- The device uses very strong materials for the strap (e.g. a cut-resistant strap with embedded steel mesh, similar to some anti-theft bags). This could make it hard to quickly snatch or cut.
- If the device is destroyed, by that time hopefully the critical first few seconds of audio/video and location are already uploaded. Our alert protocol prioritizes sending a quick low-bandwidth alert first (just location and a distress flag) because that’s more important than video; once that’s out within seconds, even if the device is smashed, help is on the way.
- We might also design the watch’s form such that it’s hard to yank off – perhaps a buckle that locks or requires two hands to remove (so an attacker can’t easily do it in a struggle). The user can still remove it normally, but an attacker would be slowed.

Risk 4: Privacy Breach or Misuse of Data Collected: *Scenario:* The sensitive videos recorded by the device are leaked or someone hacks the system to snoop on **Risk 4: Data Privacy Breach or Misuse.** *Scenario:* The sensitive videos, audio, or location data collected could be accessed by unauthorized parties – either through a cyber-attack or misuse by insiders. For instance, a hacker could target the cloud storage, or someone who gains possession of the device might extract data. *Mitigations:* As detailed in the Data Security section, we apply **end-to-end encryption** so that even if cloud servers are compromised, the data remains unintelligible [\[akitra.com\]](#). Encryption keys are stored only with the user (in the app or secure element on device). Additionally, all communications use secure protocols (TLS), preventing interception. We enforce strict **access controls** on cloud databases – no one (including our admins) can arbitrarily view user data. We also follow compliance requirements of DPDP Act to handle data safely [\[techtargget.comtechtargget.com\]](#). Regular security audits and penetration testing will be conducted on the system. On the device side, the firmware and storage are secured (encrypted flash, secure boot). If a device is lost, the user can remotely wipe it via the app to remove any cached data. By making the system **zero-trust** (even we as service provider cannot see user’s private recordings), we greatly reduce the impact of any breach. As an extra measure, emergency videos stored in cloud could be set to auto-delete after a certain time (unless saved by user) to limit exposure window. Lastly, we’ll be transparent with users – providing audit logs in the app (e.g., showing when their data was accessed and by whom – mostly just themselves) and promptly notifying of any security incident, thereby maintaining user trust.

Risk 5: Misuse of the Device (By Owner or Others). *Scenario:* A user (or someone who obtains the device) could misuse it to violate privacy or prank others. For example, someone could secretly record conversations using the watch (ironic misuse for voyeurism), or trigger false SOS alarms intentionally (e.g., a child “crying wolf” to test family response). *Mitigations:* We address this through **user education and app settings**. In the app’s user guide and

onboarding, we make clear the legal/ethical use: the camera is for emergency evidence only, not casual recording. The device's design also helps: it has no easy way to record video silently without alert (for instance, when recording, maybe a subtle indicator like a vibrating pulse every few seconds reminds the wearer it's active, preventing continuous covert recording misuse). We could also enforce that recorded videos (when not in an SOS event) are only accessible via the user's verified account, so one can't use the watch purely as a spycam and retrieve footage without trace. Regarding false SOS use: if a user repeatedly triggers false alarms, the app can issue warnings. Family contacts can give feedback through the app (e.g., mark an alert as false). In an extreme case, if someone were maliciously abusing the system (e.g., clogging public emergency lines with false SOS), we could temporarily deactivate the automated 112 dialing for that device and require reactivation after a safety check or training. Essentially, while the device empowers the user, we assume good intent. We put safeguards (both technical and policy) to deter intentional misuse, and have a **code of conduct** acceptance when setting up the device.

Risk 6: Technological Limitations or Counter-Measures by Attackers. *Scenario:* A savvy attacker might attempt to **jam the signals** (e.g., use an RF jammer to block cellular/Bluetooth) or adapt tactics (like avoiding detection by the device). Also, the hidden camera detector might not catch a very well-concealed next-gen spy device, giving a false sense of security. *Mitigations:* Jamming is a challenge – if an attacker actively jams cellular, the watch might fail to send out alerts. However, jamming is illegal and relatively rare; plus it tends to be localized. Our mitigation is multi-frequency capability: an attacker would have to jam both LTE and SMS (different bands) and possibly also the phone's network if the phone is present. If we detect something is off (e.g., the watch sees no network suddenly – possibly due to jamming), we could automatically trigger the loud siren as a backup to draw attention, since quiet alerts won't go out. Essentially, if connectivity is lost during an SOS in an area that normally has signal, the device could assume malicious interference and switch to **audible alarm mode** as a last resort. On hidden camera detection: we acknowledge no method is 100% foolproof. We mitigate by combining methods (RF + optical) to improve odds. We will clearly communicate to users that the tool greatly reduces risk but **vigilance is still advised**. Regular updates to the detection algorithm (via firmware updates) will keep pace with new types of spy devices. Also, any time a user finds a hidden camera (whether our device helped or not), we encourage reporting it so we can update our threat database (crowdsourced learning).

Risk 7: Operational and Financial Risks: *Scenario:* The project faces manufacturing delays, cost overruns, or difficulty in adoption. If the BOM cost is too high, we risk pricing out the target users; if supply chain issues occur (e.g., component shortages), product availability suffers. There's also risk in scaling customer support for a safety-critical device. *Mitigations:* We have a dedicated section on BOM and cost optimization (see next section) to ensure the device can be produced at scale for a reasonable cost, including strategies like bulk component sourcing and possibly subsidies. To avoid supply issues, we choose **proven components** available in volume (for example, using a widely used MCU/module rather than an exotic one). We maintain good relationships with suppliers and have backup options for critical components (e.g., at least two possible camera module vendors). For operations, we'll implement a rigorous QA in manufacturing (so that each unit's critical functions – SOS, calling, etc. – are tested before shipping out). Regarding user support: we plan a 24/7 helpline or chat support especially for when the device is new in market, to quickly address

any malfunctions or user confusion (because in safety tech, user must be confident everything will work when needed). This risk is more business-oriented, but addressing it ensures the **solution is sustainable long-term** (a device that can't be financially sustained might end up discontinued, which would let users down).

By systematically analyzing these risks, we aim to design not just the product but also the **ecosystem** (user training, support, maintenance) to mitigate them. It's important to remember that this smartwatch is used in high-stakes situations – thus reliability and trust are paramount. Our multi-layered mitigations (technological, procedural, educational) address the risks at their root. For example, encryption and access control tackle privacy risks at a fundamental level, while UI/UX choices (cancel option, discreet design) tackle false alarms and tampering proactively. We will keep updating our risk register as the project evolves, learning from any incidents or user feedback in pilots, and iterating the design or processes accordingly.

In summary, while no system can be 100% risk-free, our approach is to **anticipate failure modes and edge cases**, and build in safeguards so that the device *fails safe*. Even in worst-case scenarios (device compromised, network down, etc.), we strive for a fallback that still offers some protection (like the audible alarm). This resilience-by-design philosophy increases the likelihood that when a real crisis occurs, the smartwatch will perform as intended and possibly save a life, which is the ultimate measure of its success.

Ethical & Societal Considerations

The development and deployment of a women's safety smartwatch must be guided not only by technical and business factors, but also by a strong ethical compass and sensitivity to societal context. Below we address the key ethical and social considerations:

Empowerment vs. Victim Blaming: One ethical concern is ensuring that technology to protect women is seen as **empowering** and not as shifting responsibility of safety solely onto women. We must frame the smartwatch as a tool that can help *in spite of* societal problems, not a solution *to* the root problem. The marketing and messaging will emphasize that women *have a right to safety* and this device is one way to assert that right – but it does not mean women are responsible for preventing crimes against them. We will actively avoid any narrative suggesting “if she wasn't wearing it, she's to blame for an assault.” This aligns with the broader movement that the onus is on society (and perpetrators) to stop crimes. The device is a **stop-gap protection** in an imperfect world.

Privacy of the User: It might seem paradoxical that a device with a camera and microphone worn 24/7 could raise privacy issues for the user herself. We guarantee that the watch is *not a surveillance device* on the user. By default, it does not record or transmit unless deliberately triggered, and even then the data is protected (as discussed) and only used for her safety. The user's consent and control are central – e.g., she decides who her emergency contacts are, she can turn off or on features like location tracking (maybe she doesn't want continuous tracking by family). We provide granular controls: for instance, a mode where only when she triggers SOS do others see location, vs. a mode where a parent or spouse can “follow me live” (some users may want that for certain scenarios). All such features are opt-in to respect autonomy. Additionally, since the device will likely collect

sensitive health-ish data like movement or heart rate (for falls), we ensure compliance with privacy norms there too (not sharing it or selling it). We will also include a **privacy mode** where the user can disable the camera/mic except SOS – perhaps an LED indicator that confirms the camera is electrically off – to reassure those who are worried about being monitored. Essentially, we abide by the principle: the **user owns her data** and any sharing is her choice.

Privacy of Bystanders and Contacts: When recording an incident, inevitably other people (bystanders or even the attacker who is also a private person) are being recorded without their consent. Legally, this falls under exceptions (legitimate interest for safety/evidence). Ethically, we consider that these recordings are only used in justice context or personal review, not broadcast publicly without due process. We advise users that any footage should be handled responsibly – e.g., share with police, not post on social media in a way that violates privacy (unless necessary to identify a criminal at large, for example). Emergency contacts who receive audio from the scene must also use it discreetly. We might include an **ethics agreement** or training snippet for contacts too (“By accepting to be an emergency contact, you agree to respect the privacy of your friend’s data and only use it to assist during emergencies.”). As for hidden camera detection – using it in someone’s property might raise issues, but if you’re in a private space you’re entitled to scan for invasive devices. It’s a defensive measure and ethically sound.

Data Use and Law Enforcement: If the device collects evidence of a crime, there’s an ethical duty to use that data for justice. We encourage (but don’t force) users to share evidence with law enforcement in prosecuting offenders. We will cooperate with law enforcement within the bounds of user consent and legal process – e.g., if police subpoena data for an investigation, we have to comply, but since it’s encrypted by user, we’d involve the user. We position ourselves as **allies to law enforcement against abuse**, possibly providing a portal (with user permission) for police to get verified evidence, thus improving conviction rates of gender-based crimes. An ethical balance is needed: we want to help catch criminals, but we also must guard against *overreach* (e.g., authorities trying to use the device to surveil users under the guise of safety). We will not provide any backdoor live access to police or anyone; everything goes through user’s control.

Inclusion and Accessibility: We must consider various groups in our user base. The target is women 13-35, but within that:

- We have **minors** (teens) – we ensure the device and app experience accounts for guardianship while still respecting the teen’s agency (some teens may not want parents tracking their every move, so an arrangement is needed – e.g., alerts only in emergencies, not constant monitoring).
- We consider women with disabilities – the device should be usable by those with visual or hearing impairments. For example, the UI should have strong haptic feedback for blind users; the app should integrate with screen readers. Deaf users might not need the loud siren but rely on the silent functions – ensure equal effectiveness without audio.

- **Socio-economic inclusion:** The cost might be a barrier for low-income women who arguably are among those who could benefit greatly. This is a societal consideration to address in go-to-market (perhaps via government subsidies or NGO programs – discussed in roadmap). We want to avoid creating a safety divide where only affluent women get this extra safety net. Pilots with subsidized devices in at-risk communities (like women in urban slums who often travel at odd hours for work) can be considered.

Cultural sensitivity: In some communities, women’s mobility is a sensitive issue. Wearing a “safety device” might be seen either as empowerment or, conversely, might attract unwanted attention (“what is that device?”). We designed it to look like a normal watch to avoid stigma. We’ll gather feedback from users in conservative areas to ensure the device doesn’t conflict with dress codes (e.g., it should be okay to wear with traditional clothing). Also, language support in the app and alerts (e.g., allowing alert messages in local language) is important to ensure effective use by non-English-speaking women.

Psychological impact: There’s an emotional aspect – wearing a device that reminds one of danger could cause anxiety. We aim for the opposite: a sense of security. But we must be careful not to instill a false sense of invulnerability. Ethically, we will communicate that this watch *improves chances of help* but is not a guarantee. Users should still exercise normal safety precautions. The device should be marketed as a **confidence booster** – many women report feeling safer even just knowing they can call for help easily womentech.net], which can encourage them to live their lives more freely. That’s a positive societal effect: women not feeling as restricted from public spaces or travel due to fear. However, there is a concern: might reliance on such devices lead to reduced community focus on making streets safer? We should use some proceeds or partnerships to also contribute to broader safety education and advocacy, showing that technology is one part of a multi-faceted solution (along with better policing, street lighting, etc.).

Community and Reporting: The device could indirectly encourage more reporting of incidents. For example, if an attempted assault is recorded and the user escapes, she might be more empowered to report it since she has evidence. This could help catch repeat offenders. Ethically, this contributes to societal good (possibly preventing future crimes). We will ensure that if a user does not want to report, that choice is respected – the data remains hers. But we can provide in-app guidance on how to report and connect with support organizations (like women’s helplines, legal aid) if an incident occurs. Essentially, the ecosystem should support the survivor’s journey after an incident, not just the incident itself.

Male Allies and Broader Use: While the product is focused on women, it’s ethically important to include male allies in the conversation. For instance, many emergency contacts will be men (fathers, friends, partners) – we view them as part of the solution and provide them with guidance on how to respond effectively when they get an alert (maybe even training material on intervening safely or coordinating with police). Also, the tech itself can be unisex – men and other genders facing safety issues (like journalists, or delivery workers at night) could use it. We market primarily for women’s safety because that’s a pressing need, but we **don’t exclude others from benefit**. Societally, this avoids the perception that it’s a “gimmick only for women” and instead a serious personal security device applicable broadly,

while still addressing the unique threats women face (like voyeurism, which is gender-specific in IPC 354C).

Transparency and Accountability: We commit to **transparency** in how the device functions and what it does with data. There won't be hidden functionalities – e.g., we're clear that "audio is recorded when SOS is active" so users know what's happening. We also open ourselves to third-party audits for security. This transparency builds trust in communities that this isn't a surveillance gadget or some way to exploit users. Additionally, if there are any adverse events (like device fails in a key moment), we will analyze and publicly (to users) share what happened and how we're fixing it. Being accountable ensures ethical responsibility to our user community.

Stigma and Acceptance: In some environments, wearing a safety device might carry stigma (like implying the environment is unsafe). We want to normalize it akin to wearing a seatbelt – a prudent precaution that everyone accepts. Part of societal introduction will involve working with institutions (colleges, companies) to promote it as a positive measure, not as an alarmist tool. If, for example, a college issues these to students, we frame it as the college caring for student safety, not that the campus is particularly dangerous.

In conclusion, the ethical and societal dimension of this project is deeply intertwined with its engineering. We measure success not just by units sold or tech performance, but by **real-world impact**: Are women feeling safer? Are potential perpetrators deterred knowing such devices exist? Is society starting to treat women's safety tech as essential (the way smoke detectors or seatbelts are)? We will gather qualitative feedback from users to gauge these aspects and will remain flexible to modify features if something inadvertently causes harm (for instance, if we found the presence of the watch in some context was escalating risk, we'd address that). Our overarching ethical stance is to **enhance safety and freedom** for women without infringing on rights or privacy, and while promoting broader societal change that, ultimately, would render the need for such devices obsolete in the long run.

Cost Breakdown & BOM Forecast

Designing an advanced safety smartwatch involves a careful balance between feature richness and cost. In this section, we break down the **Bill of Materials (BOM)** for the device and project costs for both prototyping and mass production (10k units), highlighting where economies of scale or design choices drive cost savings. All costs are approximate and in USD (with notes on INR).

Major Hardware Components and Estimated Costs (per unit):

- **Microcontroller/SoC + Cellular Module:** We plan to use an integrated solution (e.g., Nordic nRF9160 or similar) which includes the MCU, LTE-M/NB-IoT modem, and GPS. In prototype (low quantity), such modules might cost around \$20–\$25 each. In volume (10k units), price can drop to roughly **\$10–\$12*[counterpointresearch.com](https://www.counterpointresearch.com)[counterpointresearch.com](https://www.counterpointresearch.com)]. If we need a separate BLE chip (if main SoC doesn't have it), add ~\$2 (volume ~\$1). We assume one integrated SoC covers it. This component is one of the most expensive due to

cellular capability.

- **Camera Module:** A tiny camera (module with sensor + lens) say 1080p capability. Low volume cost ~\$8, volume cost ~\$4. We might use a 5MP sensor which in bulk could be as low as \$3–4. Lens and IR filter included in module. If we require an IR LED array or special optics for detection, that might add \$1 or so.
- **Memory:** External flash (2GB NAND or 16Gb SPI NOR) – prototypes \$2, bulk ~\$1. If the MCU has enough internal flash for our needs, we can eliminate this, but likely we include it for video storage. RAM (if needed externally) – maybe a 16MB PSRAM at \$1. But perhaps not needed if we stream chunks.
- **Sensors:**
 - Accelerometer/gyro: \$1 prototype, \$0.5 bulk (common MEMS like BMI160).
 - GPS antenna: \$0.5 bulk.
 - Mic (MEMS): \$0.5 bulk.
 - Vibrator motor: \$0.5 bulk.
 - Heart-rate sensor (with LED): ~\$1.5 prototype, \$1 bulk.
 - Magnetometer (if not in 9-axis IMU already): \$0.5.
 - Hidden cam detector components: IR LEDs and photodiode ~ \$1 total; RF detector front-end ~\$3 (or if we can time-multiplex radio, maybe minimal cost).
 - Speaker/Buzzer: small piezo or speaker ~\$1.
 - Buttons, connectors, etc.: \$1 (volume total for all small electro-mech parts).
- **Display:** If we include a small OLED or e-paper for time/status. A 0.5–0.7” OLED could be ~\$3 in low volume, \$1.5 in bulk. Alternatively, a simple LED indicator approach could cut this cost, but having a basic display adds value (time, messages). We'll assume a simple monochrome OLED.
- **Battery:** A 500 mAh Li-poly battery (custom thin form) – prototypes \$4, bulk \$2. Batteries benefit from scale but also quality matters. We include protection circuit – often included in battery cost. Also consider a charging IC (\$0.5 bulk) and components for power management.
- **Enclosure & Strap:** Using durable polycarbonate or a blend, and a silicon/nylon strap. For prototyping (3D prints, etc.) maybe \$10–15 each (very high due to low volume). At mass production (injection molding), the casing might cost ~\$2–3 and the strap \$1–2. Additional for any steel reinforcement in strap maybe \$0.5. If we do

decorative finishes (various colors), minimal cost difference at scale.

- **PCB & Assembly:** For prototypes, low volume PCB fab and assembly can be expensive (~\$20–30 per board when doing small run). For 10k units, the PCB (4-layer, small size) might be ~\$1 each, and assembly cost perhaps \$1–2 (given relatively high component count, and including testing overhead). We'll estimate around \$3 for PCB + assembly in volume. We should also factor the testing/calibration time (maybe \$0.5–\$1 per unit in mass prod for test labor/equipment amortization).

Summing up **estimated BOM cost** per unit:

- Prototype (very low quantity hand-built or small batch): MCU (\$20) + camera (\$8) + memory & sensors (\$6) + display (\$3) + battery (\$4) + misc (\$5 for PCB, etc.) = roughly **\$46**. Plus enclosure which might be expensive in prototype (maybe milled or 3D printed ~\$15), totaling ~\$60. So prototypes might be \$60–\$80 each. If we need to factor NRE (non-recurring engineering) costs like mold creation, that's an upfront cost (a steel mold for injection could be \$5k–\$10k), but per unit it's amortized in volume.
- 10k Unit Mass Production: MCU (\$12) + camera (\$4) + memory & sensors (\$5) + display (\$1.5) + battery & PMIC (\$2.5) + PCB & assembly (\$3) + enclosure/strap (\$4) + misc and margin for scrap/testing (\$2) = approximately **\$34** per unit BOM. Let's say around \$30–\$35 as a target.

A BOM of \$35 means manufacturing cost (including factory margin) might be \$40. Typically, retail price might be 2–3x BOM to cover R&D, overhead, distribution, and profit. If 2x, that's \$70 (₹5,600); 3x would be \$105 (₹8,400). We will aim for a retail price around **₹6000–₹7000** (approximately \$80–\$90) initially for early adopters or at low volume. With higher volumes, we could reduce price. For context, premium smartwatches cost much more, but basic wearables cost less – however, ours has unique capabilities which can justify a moderate price point. Additionally, if government subsidies or CSR sponsorship come in, the effective cost to end-user can be lowered. We will prepare a **BOM forecast table**:

Component	Prototype Cost (USD)	Mass Prod (10k units) Cost (USD)
Secure MCU + LTE/GPS modem	\$20	\$12
Cellular/BLE Antennas	\$1	\$0.3
Camera module	\$8	\$4
Audio (Mic + Speaker)	\$1	\$0.8
Sensors (IMU, HR, Mag, IR)	\$4	\$2

Memory (Flash/RAM)	\$2	\$1
Display (OLED)	\$3	\$1.5
Battery & PMIC	\$4	\$2.5
PCB & Assembly	\$5 (hand assembly)	\$3 (SMT automated)
Enclosure & Strap	\$15 (3D print/CNC)	\$4 (molded plastic, bulk)
Subtotal BOM	~\$63	~\$31–\$35
Testing, Certification (per unit amortized)	N/A	~\$2 (for volume, includes certification amortization)
Total per unit cost	~\$70–\$80 (proto)	~\$35–\$40 (at 10k units)

(Note: These figures are illustrative. Actual costs will vary based on final part choices and supplier negotiations.)

We see that at scale, the most significant cost drivers are the SoC and the quality sensors (camera, etc.). Fortunately, these benefit from Moore's Law and volume discounts. For example, the BOM analysis of a high-end LTE smartwatch (Google Pixel Watch) showed about a third of its ~\$123 cost was in processing and communication [counterpointresearch.com](https://www.counterpointresearch.com)]. Our design, being more focused and not having expensive OLED or fancy CPU, comes out cheaper. Also, Pixel Watch's display and sensors were a big cost chunk [counterpointresearch.com](https://www.counterpointresearch.com)]; we have a simpler display and similar sensors so our cost is lower.

BOM Reduction Strategies: As adoption grows, we can further optimize:

- Use of **custom SoC** in future (if volume hits, say, 100k units, we could consider a custom integrated chip that combines all functions which could cut costs by 20–30%).
- **Shared components:** The hidden-cam detector's RF function might be achieved by reusing the LTE antenna and radio in a sniffing mode, eliminating a separate RF module – this saves a few dollars.
- Simplifying display or removing non-essentials: If market feedback says display isn't needed, we could remove it, saving cost and battery.
- Battery size vs cost: a slightly smaller battery (400 mAh) could shave \$0.5 and still meet one-day life, but we chose a margin.
- Plastic vs metal: If we considered a metal case, cost would rise – so we stick to durable plastic for now.

Prototype and Pilot Costs: For initial 50–100 prototypes (for testing, certification, pilot users), the per-unit cost including development overhead might be very high (~\$200+ if we include engineering effort, tooling, etc.). But those are R&D expenses. We have to invest in certifications (one-time costs: BIS testing ~\$2000, TEC maybe \$1000, etc.), tooling (mold \$8000), initial app development, etc. We amortize these over units; at 10k units, those overheads might add a few dollars each. If we produce just 1000 units, overhead per unit is larger. Hence scaling to 10k significantly helps spread out NRE costs.

We plan to start with a **pilot batch (maybe 500 units)** manufactured – cost might be around \$50–\$60 each at that stage. If we price them at say ₹10,000 (\$125) for pilot customers (who often are early adopters or provided via grants), we can cover costs and gather feedback. As we streamline manufacturing, we target cost reduction and price drops.

Bill of Materials vs. Price Forecast:

- Prototype stage: BOM ~\$70, likely price n/a (not sold, just development).
- Initial small-scale production (1k units): BOM maybe ~\$45 (some volume discount, but not full 10k discount), expecting to price around ₹8,000–₹10,000 to cover costs and channel margins.
- Mass production (10k+): BOM ~\$35–\$40, enabling a retail price around **₹6,000** (approx \$75) if sold direct or maybe ₹7,500 (\$95) if through retail channels that take margin. This price range, while higher than basic fitness bands, is justified by the advanced safety features – and still relatively affordable for a high-quality smartwatch. We also expect that as volumes climb, costs could go down another 10–15%. If government subsidies a portion (for target student groups for example), the effective price to consumer could drop further.

Additional Costs to Consider:

- **Cloud service costs:** Each device will use cloud resources (data storage, SMS, etc.). We should factor an estimated cost per user per year (maybe a few dollars, especially for SMS/call notifications). We might offset this by a subscription or usage-based model if needed (or include a couple of years of service in product price).
- **Warranty and Service:** We will allocate some cost for warranty replacements (assume maybe 2% failure rate, that cost needs to be covered in margin).
- **Packaging and Logistics:** Packaging a smartwatch (box, manual, charger cable) might be ~\$1–\$2 per unit. Shipping to distributors or warehouses adds some cost. These are usually folded into either BOM or overhead.

In BOM forecasts, usually at 10k units we get good scale, but the real sweet spot might be 50k or 100k units where component pricing drops another tier. If the product sees widespread adoption (imagine being adopted in safety programs nationwide), we could see

costs approach maybe \$25 BOM at 100k units (due to better pricing and maybe integrating more functions into one chip).

For now, with 10k as a reference, we have a viable cost structure. To ensure viability, we'll seek partnership with manufacturers to keep assembly costs low (maybe assembling in India under Make in India to get duty benefits). Also possibly negotiate with telecom operators for bulk SIM provisioning at low cost.

Comparative Note: A fully featured Apple Watch has a BOM of ~\$130 [counterpointresearch.com](https://www.counterpointresearch.com)], but that includes expensive AMOLED display and powerful processors for general apps. Our device is more specialized – akin to a feature phone level complexity with some extras – so its BOM is much lower. In fact, \$35–\$40 BOM at scale aligns with mid-tier smart bands. This indicates our cost target is realistic.

In conclusion, the BOM analysis suggests that the safety smartwatch can be produced at a **reasonable cost** relative to its life-saving value. With projected economies of scale and potential external funding support, we can strike a balance where the device is **affordable to the target audience** while sustaining the business and continuous improvement. This financial viability is crucial for large-scale social impact – it means we can reach more users, iterate on the design, and perhaps even introduce lower-cost variants in the future (for instance, a version without cellular that pairs with phone at a lower price for those who always carry a phone). The cost forecast will be continuously refined as we move from prototype to production and get real supplier quotes. The goal is clear: deliver maximum safety feature punch at a price point accessible to as many women as possible.

Go-to-Market Roadmap

Developing the smartwatch is only half the battle; ensuring it reaches and effectively serves the intended users is the other half. Below is a phased **go-to-market (GTM) strategy** outlining how we plan to pilot, launch, and scale the product in India, leveraging partnerships and possibly subsidies to maximize impact:

Phase 1: Pilot Programs and User Testing (Months 1–6)

Before a full commercial launch, we will conduct **targeted pilot deployments** to validate the device in real-world conditions and incorporate feedback. Key steps in this phase:

- **Small-Scale Pilot Deployment:** Partner with a local institution, e.g., a university or a large corporate office with women employees, to recruit an initial pilot user group (perhaps 100 users). We would provide devices (possibly at a subsidized or free pilot cost) and closely monitor usage (with consent). For instance, a college in a metro city could let 50 female students and 50 staff use the watch for a semester. This will test technical performance (battery life, connectivity across the city), and user experience (how intuitive the device/app is, any false alarms, did it actually help in any incidents).
- **Collect Feedback & Refine:** Through surveys, focus groups, and analyzing device logs (anonymized), gather insights: Did users feel safer? How often did they use features like camera scanning? Any difficulties (e.g., pairing issues, button confusion)? Did emergency contacts respond effectively? This feedback loop will

allow us to push firmware/app updates to fix issues before mass launch.

- **Field Test with Law Enforcement (optional):** Collaborate with a city police department (maybe via their women's safety cell) to trial integration. For example, 10 devices given to volunteer women in a high-risk area with police aware of alerts. This tests our backend's ability to forward alerts to police or how police respond to user's contacts. It also builds credibility with law enforcement early on.
- **Technical Validation:** Use pilot to finalize certification (if not already done). Also, test our cloud alert system capacity with simulated multiple simultaneous alerts to ensure scalability (like run drills where many pilot users trigger SOS at once).
- **Stakeholder Buy-In:** Document success stories from the pilot. E.g., "Student A used the watch to alert hostel guards when harassed by outsiders, preventing an incident." These real anecdotes and data will be powerful in convincing stakeholders in the next phase.

Phase 2: Soft Launch and Partnerships (Months 6–12)

With a validated product, we proceed to a controlled launch focusing on partnerships and early adopters:

- **Corporate and Institutional Partnerships:** We approach companies (especially BPOs, IT firms with late-night shifts for women), educational institutions, and government schemes. The pitch: they can enhance their women's safety programs by adopting our device. For example, an IT services company might bulk-purchase 500 watches for their women who use late-night cab services, tying into their CSR or duty of care. We could partner with universities to offer devices to first-year female students at a discount (perhaps covered by the University's safety budget or alumni fund).
- **Government Collaboration:** Engage with bodies like the Ministry of Women and Child Development, state police departments, or the Nirbhaya Fund administrators. The **Nirbhaya Fund** (a government fund for women's safety) could subsidize projects that deploy safety devices. For instance, we could propose a city-wide pilot funded by the Nirbhaya Fund to give, say, 1000 devices to women in public transport jobs (conductors, etc.) or to college students in that city. Government backing would lend legitimacy and reduce costs to users.
- **Telecom Partnerships:** Work with telecom operators (Jio, Airtel) to bundle connectivity. For example, a **built-in eSIM with a special IoT data plan** could be bundled free for 1 year with the device. Operators benefit by attracting users and fulfilling CSR goals. They could also retail the device in their stores or online, effectively as another "connected device" offering. Additionally, telecoms could help integrate with emergency number 112 infrastructure.
- **Retail and Online Availability:** While initial distribution is partnership-focused, we also do a limited online launch (e.g., our own website and Amazon India) for those

actively seeking such solutions. We list it under safety gadgets category, with clear value prop and possibly a launch promotion or EMI options (to make the ~\$80 cost more palatable). However, we might intentionally keep early consumer sales limited until we've ensured supply and support is ready for scale.

- **Awareness Campaigns:** Alongside soft launch, run awareness workshops and demos. For instance, safety workshops in colleges that not only talk about self-defense tips but also demonstrate the smartwatch. These can be done with NGOs or as part of college orientations. The idea is to educate the public that such technology exists. Press coverage from pilot success can be used: local news story "Smartwatch rescues woman – new tech in city's colleges" etc., will generate interest.
- **Customer Support Setup:** Anticipating growing user base, set up a dedicated support line (with perhaps a small team of trained support staff, including women who understand the context). Also prepare user guides in multiple languages. The support line can help users with setup issues, feature training, and in case of any false alarm confusion (e.g., a contact might call us asking what to do if they get an alert – we should be ready to guide them).

Phase 3: Mass Market Expansion (Year 2 onward)

Once the product and processes are stable, scale up manufacturing and broaden distribution:

- **Full Consumer Launch:** Launch nationwide marketing targeting urban and semi-urban centers. This could include:
 - E-commerce push (list on Amazon, Flipkart, etc., possibly with "Fulfilled by Amazon" for logistics, and highlight in Women's Day promotions).
 - Select physical retail partnerships, e.g., electronics retailers (Croma), or even pharmacies/health shops (some women might trust buying from places focused on well-being). For semi-urban penetration, tie-up with telecom retail outlets or post offices via government schemes for rural safety tech.
 - Utilize digital marketing with a focus on social media – for example, real stories dramatized in short videos showing how the watch works in a crisis, with a tagline like "Your Guardian on Your Wrist." Influencer marketing could involve women safety activists or celebrities endorsing it as a must-have for women. Careful to keep tone positive ("with this device, she has one more reason to fear less" rather than fear-mongering).
- **Scaling Production to Meet Demand:** Based on early sales, ensure our supply chain scales. This may involve onboarding an EMS (electronics manufacturing services) partner in India for volume production. The BOM cost reductions discussed will kick in at this stage, possibly allowing a slight price drop or better margins that can be reinvested in scaling.

- **Subsidies & Bulk Programs:** Work with state governments or city councils to implement subsidy programs. For example, a state could announce that it will reimburse X% of cost for the first 1000 women in certain professions who buy the device. Or, integrate it with schemes like giving it free (or at nominal cost) to women who use government-run transport at night. We will lobby and present data to policymakers on how this device can complement their women's safety initiatives (there are many, from panic buttons on buses to CCTV networks; our device fills the personal gap).
- **Evolve Business Model:** Consider a subscription for premium services (optional). The base device works standalone, but perhaps we can offer a subscription that includes, say, a 24/7 monitoring service or integration with a professional security response center. Some users (or their families) might pay a monthly fee for that extra assurance. This model is used in some senior citizen emergency pendants (connected to call centers). For general users, community-based response might suffice (contacts and police), but offering a tier where an emergency response team is on call could attract some.
- **Community Building:** Create a community of users – perhaps via the app or social media – where tips are shared (like where hidden cameras were found – to crowd-map unsafe locations, etc.). We must moderate to keep it constructive and privacy-conscious, but a community can improve engagement and word-of-mouth.
- **Continuous Improvement (Future Models):** As we scale, gather ideas for version 2 hardware (maybe with satellite SOS) or a simpler variant (cheaper model without camera for those who mainly want SOS). But ensure the current product gets proper support and software updates. Maintaining an innovation roadmap keeps the product relevant and ahead of possible competition (others may enter the market seeing our success).

Phase 4: Partnerships for Ecosystem & Long Term (Year 3+)

- **Integration with Smart City Infrastructure:** In the long run, as cities implement command centers, our platform could feed anonymized alerts into city dashboards to help allocate patrols to hotspots. We'd pursue MoUs with city police for data-sharing (in aggregate) to improve public safety measures.
- **Educational Institutions Adoption:** Make it common for colleges to include our device in their welcome kit for female students (perhaps negotiated at a bulk academic price). Similar for companies – akin to how some give health wearables for wellness, give safety wearables for women employees.
- **International Expansion:** If proven in India, explore other markets with similar needs (South Asia, Middle East for women's safety, or even Western markets focusing on general personal safety or teen safety). Adapt to local regulations and integrate with local emergency systems. The Indian success story will help, but we

approach that after solidifying home ground.

- **Metrics and Efficacy:** Work with researchers (maybe criminology or sociology departments) to study the impact of the device on user behavior and outcomes. If we can publish data like “Users of this device had X% faster emergency response and felt Y% safer according to surveys,” it validates the concept and can influence policy (e.g., government grants could be more forthcoming with solid evidence).

Go-to-Market Summary: We start with **focused pilots** to refine the product, then leverage **partnerships** (corporate, institutional, governmental) for an initial rollout to those who need it most and to build credibility. Then we broaden to consumers with strong marketing emphasizing safety empowerment. We incorporate **subsidies and financing options** to overcome cost barriers. All along, we ensure robust customer support and iterate on feedback. A labeled block diagram of the *user journey flow* could be envisioned as an infographic: from **Awareness** (hearing about it via a workshop or ad) ->

Purchase/Onboarding (through a partner or online, setting up contacts) -> **Everyday Use** (wearing it, feeling secure) -> **Emergency** (device triggers, help arrives) -> **Post-event** (data used for action, user shares testimonial). Each step requires proper infrastructure (e.g., training at onboarding, reliability at emergency).

By executing this roadmap, our aim is that in a few years the safety smartwatch becomes as common and essential as carrying a mobile phone for many Indian women – an accessible, trusted companion that, alongside broader societal efforts, contributes to a tangible reduction in harm and an increase in women’s confidence to move freely.

Conclusion & Future Work

In this report, we presented a comprehensive blueprint for a women’s safety smartwatch – detailing the urgent **need**, the envisioned **features**, the underlying **architecture**, and the path to bring it to market responsibly. This device, blending wearable convenience with cutting-edge safety technology, has the potential to fill critical gaps in personal security for women and girls in India. By enabling instant distress signaling, evidence capture, and proactive threat detection (like hidden cameras), it empowers users to protect themselves and seek help in a way not previously possible.

The design is rooted in currently available technology, but we have kept an eye on the **future** throughout. Looking ahead 2–3 years, several emerging technologies and enhancements could further elevate the smartwatch’s capabilities:

1. Satellite-Based SOS Communication: One limitation of any mobile-connected device is cellular coverage. As satellite communication for consumer devices becomes more viable (we’ve already seen early adoption in some smartphones for emergency texting), a future version of our smartwatch could integrate a **satellite SOS feature**. This might involve a small L-band antenna and partnership with a satellite network (e.g., Globalstar or Iridium). In an area with no GSM/LTE signal – say a rural highway or a mountain trek – the watch could still send a basic distress signal via satellite, transmitting GPS coordinates to a central emergency service. By the time this tech is affordable for a wrist form factor (perhaps

through emerging low-earth-orbit constellations offering IoT messaging), we will be ready to integrate it, ensuring the device is not bound by terrestrial network availability.

2. Advanced Anomaly Detection and Predictive Alerts: With more usage data and advancements in AI, the smartwatch could evolve from a reactive device to a more **proactive safety assistant**. For example, using the array of sensors (accelerometer, heart rate, microphone), the device could potentially detect patterns indicative of distress **even if the user hasn't pressed SOS**. A concept for future exploration is a machine-learning model that monitors the user's bio-signals and context – a sudden spike in heart rate and breathing, detected nervousness in voice tone, abrupt irregular motion – and if it matches a profile of an attack or panic, the device could *ask the user if they are okay* (via a vibration and gentle ping). If the user does not respond or cancels, it might trigger an SOS on their behalf. This is delicate (to avoid false positives), but with improved sensors and AI, it could add a layer of protection for cases where the user is incapacitated or unable to press the button. We dub this “**Guardian AI**” – essentially an algorithm that learns a user's normal patterns and flags anomalies that suggest danger.

3. Integration with Public Infrastructure: In future, the device's capabilities (especially location tracking during SOS) could integrate with public safety infrastructure. For instance, cities might allow automatic forwarding of verified SOS alerts to the nearest police patrol vehicle or to CCTV control rooms (to turn cameras to that location). Also, as public panic button projects roll out (some cities put SOS buttons in streets or on public transport), the smartwatch could be integrated so that pressing the watch might also notify nearby public systems (through an API or Bluetooth beacon to a bus's system). This **connected safety net** vision could greatly reduce emergency response times.

4. Hardware Improvements: Anticipated advances in hardware can be incorporated:

- **Battery Technology:** Solid-state batteries or better energy-density cells could double the battery life without increasing size, or allow significant size reduction of the device. Alternatively, advanced energy-harvesting (solar on strap, or kinetic) could supplement battery – perhaps not to fully power it, but to extend time between charges in an eco-friendly way.
- **Miniaturized Sensors:** We might see even smaller and cheaper mmWave radar chips or thermal sensors that could be fit into the watch, enhancing features like hidden camera detection or even detecting approach of people in dark environments.
- **Biometric Authentication:** Perhaps a tiny fingerprint scanner on the watch face or side for quicker secure operations, or even continuous authentication by heartbeat pattern (research exists on using unique heart signatures to verify identity – the watch could ensure it's being worn by the owner).
- **Ruggedization:** Future iterations might tailor versions for different use cases, e.g., a more rugged version for rural/outdoor (with longer battery, satellite, and a built-in loud siren), versus a sleek urban version (with focus on discrete looks and perhaps pepper-spray integration – some concepts have put pepper spray in wearables).

5. Broadening the Safety Ecosystem: As the product gains traction, we foresee complementing it with related offerings:

- A **mobile app** that anyone (even without the watch) can use as a basic SOS (widening our user base) – with the watch offering premium capabilities when paired.
- Partnerships with ride-hailing services or public transit apps so that if a woman is taking a cab late night, her watch data could auto-alert a friend if something goes off-route or if she triggers SOS (integration via APIs).
- Data analytics service for city planners: an anonymized heat map of distress signals or detection of hidden cameras can inform where to increase security measures.
- A community-alert feature: optionally, a user in distress could allow an alert (perhaps without personal details) to be sent to other SafeWatch users in the vicinity, creating a community response (e.g., nearby volunteers coming to help, if deemed safe). This needs careful curation to avoid vigilante issues, but it's a future concept aligning with the idea of **collective safety**.

In conclusion, the development of this safety smartwatch is not just a one-off engineering project but the establishment of a **platform for personal safety** that can continually evolve. The immediate design detailed in this report is feasible with today's technology and addresses a pressing need with a high degree of innovation (integrating video SOS and spycam detection in a wearable). The **impact** of such a device could be significant – from deterring harassment (knowing women might be recording and alerting others) to providing crucial evidence to convict perpetrators, and most importantly, giving women greater confidence and freedom in their daily lives.

As we implement this project, we will remain agile and receptive to emerging tech and user feedback, ensuring that the product roadmap aligns with both technological advancements and the real-world safety landscape. Our ultimate vision is a future where devices like this are *no longer needed* because society has become safer – but until that day, we aim to put the best of technology in the hands of women to safeguard their rights and wellbeing. The journey outlined – from concept to execution and continuous improvement – will require interdisciplinary collaboration, user-centric design, and ethical diligence, all of which we have endeavored to encapsulate in this blueprint.

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